

Quantum Computing - Notes - v0.6.0

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Preface

Every theory section in these notes has been taken from the sources:

- Course slides. [3]

About:

 [GitHub repository](#)



These notes are an unofficial resource and shouldn't replace the course material or any other book on quantum computing. It is not made for commercial purposes. I've made the following notes to help me improve my knowledge and maybe it can be helpful for everyone.

As I have highlighted, a student should choose the teacher's material or a book on the topic. These notes can only be a helpful material.

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1 Introduction

1.1 Complex Numbers

Let's not start from math. Let's start from **physics / computation intuition**. Classical bits are represented by two distinct states: 0 and 1. In quantum computing, we have **qubits**, which are essentially **vector of numbers**. But not just any numbers: a qubit is a vector of **complex numbers**. They are needed because quantum systems have **both magnitude and phase**, and complex numbers can represent both. Real numbers can only represent "*how much*" (magnitude), while complex numbers can represent "*how much*" and "*in which direction / phase*" (magnitude and phase).

🔍 Why complex (and not just real) numbers?

1. In quantum computing, we don't store probabilities directly (e.g., 0.5 for a 50% chance). Instead, we store **probability amplitudes**, which are complex numbers. The probability of an outcome is the square of the magnitude of its amplitude:

$$P = |z|^2$$

2. **Interference.** Quantum states can **combine and cancel**. For example, if we have two states with amplitudes z_1 and z_2 , the combined amplitude is:

$$z_{\text{combined}} = z_1 + z_2$$

If z_1 and z_2 have the same magnitude (*how much*) but opposite phases (*in which direction*), they can cancel each other out, leading to zero probability for that outcome. This is a key feature of quantum mechanics that allows for phenomena like quantum interference, which is essential for quantum algorithms.

In general, complex numbers are needed because **quantum states behave like waves**, not like classical values. Waves have both amplitude and phase, and complex numbers are the natural way to represent them mathematically. This is why quantum mechanics and quantum computing rely heavily on complex numbers.

🔍 What do we actually need?

Not everything. Just a **minimal toolkit**. A complex number is:

$$z = x + iy$$

Where x is the **real part** and y is the **imaginary part**, and i is the **imaginary unit** satisfying $i^2 = -1$. From a geometric interpretation, we can think of z as a point in the 2D plane, where x is the horizontal coordinate (real axis) and y is the vertical coordinate (imaginary axis).

≡ Polar form

Instead of coordinates (x, y) , we can also represent a complex number in **polar form**:

$$z = re^{i\varphi}$$

Where:

- $r = |z| = \sqrt{x^2 + y^2}$ is the **modulus** (or *magnitude*) of the complex number, representing its distance from the origin in the complex plane.
- $\varphi = \arg(z) = \tan^{-1}\left(\frac{y}{x}\right)$ is the **argument** (or *phase angle*), representing the angle between the positive real axis and the line connecting the origin to the point z in the complex plane.
- The **exponential form** $e^{i\varphi}$ means that we can represent the complex number as a point on the unit circle in the complex plane, where φ is the angle from the positive real axis. The modulus r then scales this point to its actual distance from the origin.

Using **Euler's formula**:

$$e^{i\varphi} = \cos \varphi + i \sin \varphi$$

We can rewrite z as:

$$z = r(\cos \varphi + i \sin \varphi)$$

This form is particularly useful in quantum computing because it makes it easy to understand how complex numbers can represent both magnitude and phase, which are essential for describing quantum states and their evolution.

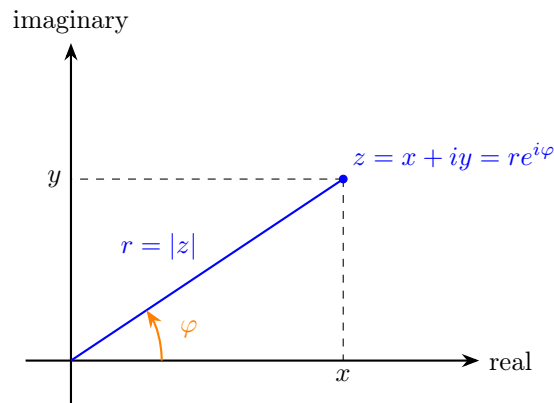


Figure 1: Geometric interpretation of a complex number in the complex plane. The arrow represents the complex number z as a vector from the origin to the point (x, y) . The length of the vector is the modulus r , and the angle it makes with the positive real axis is the argument φ .

Complex Conjugate

The **complex conjugate** of a complex number $z = x + iy$ is denoted as $\bar{z}$ and is defined as:

$$\bar{z} = x - iy$$

Geometrically, the complex conjugate $\bar{z}$ is the **reflection** of z **across the real axis in the complex plane**. This means that if z is represented by the point (x, y) , then $\bar{z}$ is represented by the point $(x, -y)$. The real part remains the same, while the imaginary part changes sign. See Figure 2 for a visual representation of this concept.

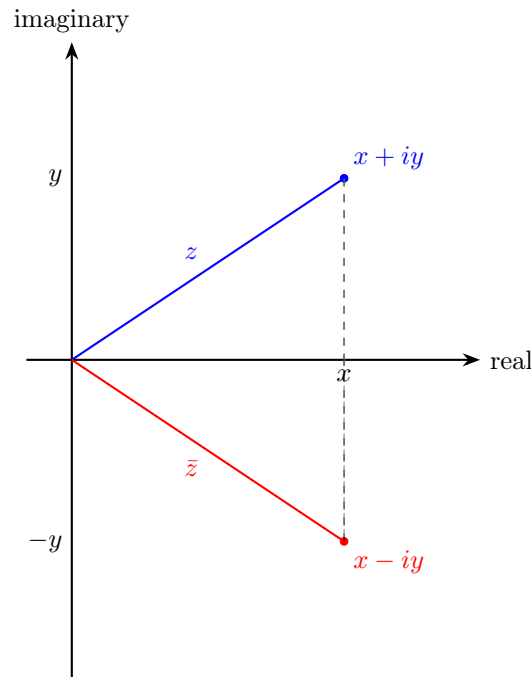


Figure 2: Geometric interpretation of the complex conjugate: reflection across the real axis.

Magnitude

The **magnitude** (or **modulus**) of a complex number $z = x + iy$ is denoted as $|z|$ and is defined as:

$$|z| = \sqrt{x^2 + y^2}$$

And it can also be expressed using the complex conjugate:

$$|z|^2 = z \cdot \bar{z} = (x + iy)(x - iy) = x^2 + y^2$$

The magnitude represents the **distance** of the point z from the origin in the complex plane. It is **always a non-negative real number**. The magnitude is important in quantum mechanics because the probability of an outcome is given by the square of the magnitude of the corresponding probability amplitude. For

example, if a quantum state has an amplitude of z , the probability of measuring that state is $|z|^2$. This is why the magnitude of complex numbers is crucial in quantum computing, as it directly relates to the probabilities of different outcomes when measuring quantum states.

↻ Rotation in the complex plane

Multiplying a complex number by a complex exponential of the form $e^{i\theta}$ corresponds to a **rotation** in the complex plane:

$$z' = z \cdot e^{i\theta} = r e^{i(\varphi+\theta)}$$

Geometrically, this operation:

- **Preserves the magnitude** $|z|$
- **Adds θ to the phase φ**

Therefore, multiplication by $e^{i\theta}$ rotates the complex number by an angle θ around the origin.

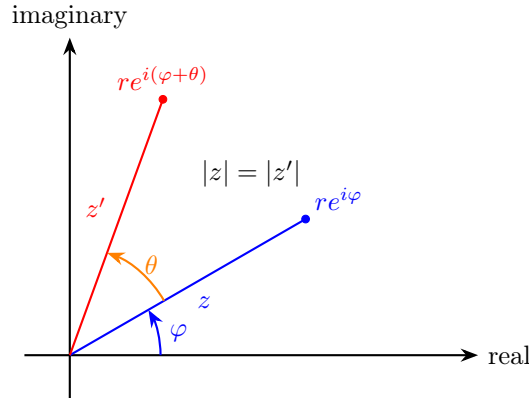


Figure 3: Rotation of a complex number: multiplying by $e^{i\theta}$ rotates the vector by angle θ while preserving its magnitude.

↻ Hermitian (Conjugate Transpose)

Given a complex vector:

$$\mathbf{v} = \begin{bmatrix} a \\ b \end{bmatrix}$$

Its **Hermitian** (or **conjugate transpose**) is defined as:

$$\mathbf{v}^\dagger = [\bar{a} \quad \bar{b}]$$

This operation consists of:

- Taking the **transpose** (column $\rightarrow$ row)
- Taking the **complex conjugate** of each element

1.2 Dirac's Notation

Dirac notation was introduced to make **linear algebra with complex vectors easier, cleaner, and more expressive**, especially for quantum systems.

Let's imagine our work only with standard linear algebra notation, where we have to write out all the vectors and matrices in their full form:

- Vectors:

$$\mathbf{v} = \begin{bmatrix} a \\ b \end{bmatrix}$$

- Conjugate transpose:

$$\mathbf{v}^\dagger = [\bar{a} \quad \bar{b}]$$

- Inner product:

$$\mathbf{v}^\dagger \mathbf{w} = \bar{a}c + \bar{b}d$$

- Matrix-vector multiplication:

$$M\mathbf{v} = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} m_{11}a + m_{12}b \\ m_{21}a + m_{22}b \end{bmatrix}$$

This quickly becomes **cumbersome and hard to read**, especially as we deal with more complex quantum systems. In quantum mechanics we constantly write expressions involving inner products, matrix operations, and state transformations.

✔ **Dirac's idea.** Paul Dirac said “*let's give different roles **different symbols***”. So he introduced:

- **Kets** $|v\rangle$ to represent **column vectors** (quantum states).
- **Bras** $\langle v|$ to represent row vectors (**conjugate transposes**).

Where the operators stay in the middle, and we can write everything in a much cleaner way:

- Vectors (**kets**):

$$|v\rangle = \begin{bmatrix} a \\ b \end{bmatrix} \tag{1}$$

Where the entries are complex numbers and usually we **require** $|v\rangle$ to have unit norm (normalized state).

- Conjugate transpose (**bra**):

$$\langle v| = [\bar{a} \quad \bar{b}] \tag{2}$$

- Inner product:

$$\langle v|w\rangle = \bar{a}c + \bar{b}d \tag{3}$$

The result is a scalar (complex number) that tells us **how much the two states “overlap”**. If $\langle v|w\rangle = 1$ then $|v\rangle$ and $|w\rangle$ are **identical** (up to a global phase, they overlap completely). If $\langle v|w\rangle = 0$ then $|v\rangle$ and

$|w\rangle$ are **orthogonal** (completely different states, no overlap). Otherwise, if $|\langle v|w\rangle| < 1$ then $|v\rangle$ and $|w\rangle$ are partially **similar** (some overlap, but not identical). So the **inner product is basically a notion of “distance” or “similarity” between two quantum states.**

- Matrix-vector multiplication:

$$M|v\rangle = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} m_{11}a + m_{12}b \\ m_{21}a + m_{22}b \end{bmatrix} \quad (4)$$

An operator M acting on a state $|v\rangle$ gives us a new state. The result is a new ket.

Even if we ignore physics for now, we can see that Dirac's notation is more expressive and easier to read than standard linear algebra notation.

❓ What is *Chaining* in Dirac notation?

Chaining means writing multiple operations together in a single expression without needing to write out intermediate steps. For example:

$$\langle x|M|y\rangle$$

It is just a **compact way to combine linear algebra operations** (matrix multiplication and inner products) in a single expression. It allows us to write complex quantum mechanical expressions in a much cleaner and more intuitive way.

Without Dirac notation, we would have to write:

$$\mathbf{x}^\dagger M \mathbf{y}$$

That is a **chain of operations**: take the conjugate transpose of $\mathbf{x}$, multiply it by M , and then multiply the result by $\mathbf{y}$. With Dirac notation, we can write this in a much more compact and readable way as $\langle x|M|y\rangle$.

To read $\langle x|M|y\rangle$, we can think of it as:

1. Start with the ket $|y\rangle$ (a quantum state).
2. Apply the operator M to $|y\rangle$, which gives us a new state $M|y\rangle$.
3. Take the inner product of $\langle x|$ with the new state $M|y\rangle$, which gives us a scalar (complex number) that represents the probability amplitude of transitioning from state $|y\rangle$ to state $|x\rangle$ via the operator M . At first, this might seem abstract, but as we go through more examples and applications in quantum mechanics and quantum computing, it will become much clearer how powerful and intuitive Dirac's notation is for describing quantum systems.

2 Single Qubit States

Before introducing qubits, we briefly describe what we mean by a **quantum system**.

A **Quantum System** is a **physical system** (such as a photon, an electron, or an atom) **whose behavior is described by the laws of quantum mechanics**.

In this course, we will focus on the simplest case: systems that can produce **two possible outcomes** when measured. These are called **two-level systems**, and they form the basis of qubits.

2.1 Building and Measuring Qubits (Intuition)

Before introducing the mathematical formalism of qubits, it is useful to develop an **intuitive understanding** of how a simple quantum system behaves when it is **prepared** and **measured**.

🔗 A Simple Quantum Experiment

Consider a very simple experimental setup:

- A **source** emits a quantum particle (for example, a photon).
- The particle passes through a device.
- A **detector** measures the outcome.

Each time we perform the experiment, the detector returns one of two possible results:

$$0 \quad \text{or} \quad 1$$

This motivates the idea of a **two-level system**, which is the simplest type of quantum system.

💡 Key Observation

If we repeat the same experiment many times under identical conditions, we observe that:

- The outcome is **not deterministic**.
- Instead, it is **probabilistic**.

For example, we might obtain:

- Outcome 0 with probability p
- Outcome 1 with probability $1 - p$

This is fundamentally different from classical systems, where the outcome is determined in advance.

❓ What is a Measurement?

A **Measurement** is the process that:

- Extracts **classical information** from a quantum system;
- Produces a **definite outcome** (e.g., 0 or 1).

An important property of quantum measurement is that it **affects the system itself**. After a measurement is performed, the **system is no longer in its original configuration**.

💡 From Intuition to Formalization

This simple experiment suggests that:

- A quantum system can produce **two possible outcomes**
- The outcomes are governed by **probabilities**
- The system must therefore contain some form of **internal state** that determines these probabilities

In the following example, we will introduce a concrete physical example (photon polarization) that allows us to represent this internal state using **vectors with complex coefficients**.

Example 1: Example of a Two-Level Quantum System

To make the previous discussion more concrete, we introduce a simple physical system that behaves as a two-level quantum system: the **polarization of a photon**.

❓ What is a Photon?

A **Photon** is the elementary particle of light. It carries energy and behaves both as a particle and as a wave. In this course, we will use photons as a simple example of a **quantum system**, since their properties can be easily controlled and measured.

❓ What is Polarization?

Polarization describes the direction in which the electric field of a light wave oscillates. In simple terms, it can be thought of as the **orientation** of the light wave. For a photon, polarization can be visualized as a direction in space. In particular, we consider two basic polarization directions:

- **horizontal** ($\rightarrow$)
- **vertical** ($\uparrow$)

These two directions form a **two-level system**, since a measurement can distinguish between them (i.e., we can determine whether the photon is horizontally or vertically polarized).

💡 Superposition of Polarizations

A photon is not restricted to being only horizontal or vertical. In general, it can be in a **combination** of both states:

$$v = a \rightarrow + b \uparrow$$

Where:

- a and b are **complex coefficients**;
- They determine the behavior of the system when measured.

This is the first example of a **quantum state**: a system described by a linear combination of basic states.

❓ Vector Representation

We can represent these polarization states using vectors:

$$\rightarrow = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \uparrow = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

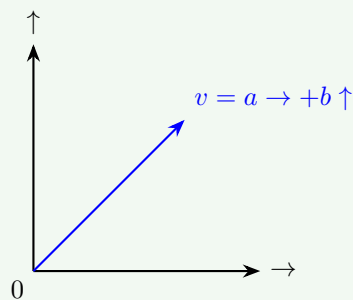
Therefore, a general polarization state becomes:

$$v = \begin{bmatrix} a \\ b \end{bmatrix}$$

This shows that the state of the system can be described as a **vector with complex components**, exactly as introduced in Dirac notation.

📐 Geometric Interpretation

The horizontal and vertical polarization states can be interpreted as two **orthogonal axes** in a two-dimensional space.



The point where the axes intersect represents the origin, and any polarization state can be represented as a **vector starting from the origin**.



In this representation:

- $\rightarrow$ and $\uparrow$ form a **basis**
- Any state is a **linear combination** of these basis states

💡 Measurement with Polarizers

A **Polarizer** is a device that **filters light based on its polarization**. A polarizer acts like a **filter** that allows only one polarization direction to pass and blocks the other. For example:

- A polarizer aligned with the horizontal direction ($\rightarrow$) will allow horizontally polarized light to pass and block vertically polarized light.
- A polarizer aligned with the vertical direction ($\uparrow$) will allow vertically polarized light to pass and block horizontally polarized light.

If the photon is a combination:

$$v = a \rightarrow + b \uparrow$$

And we use a:

- Horizontal polarizer ($\rightarrow$), the photon will pass with probability $|a|^2$ and be absorbed with probability $|b|^2$.
- Vertical polarizer ($\uparrow$), the photon will pass with probability $|b|^2$ and be absorbed with probability $|a|^2$.

This shows that:

- Probabilities are determined by the amplitudes (i.e., the coefficients a and b).
- Measurement **modifies the state of the system** (e.g., if the photon passes through the horizontal polarizer, it will be in the state $\rightarrow$).

2.2 From Polarization to Qubits

The example of photon polarization (page 12) shows that a physical system can be described using two basic states and their linear combinations. We now **abstract** this idea and introduce a **general model that does not depend on a specific physical system**.

🔗 From Physical States to Abstract States

In the polarization example, we used:

$$\rightarrow \quad \text{and} \quad \uparrow$$

As the two basic states of the system. We now replace them with an abstract notation:

$$\rightarrow \rightsquigarrow |0\rangle \quad \uparrow \rightsquigarrow |1\rangle$$

These are called the **computational basis states**. They are just symbols (used in Dirac notation) that represent the two basic states of the system, without referring to any specific physical implementation.

📖 Definition of a Qubit

A **Qubit** is a **two-level quantum system** whose state can be written as:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

It is the **unit of quantum information**, analogous to a classical bit. However, unlike a classical bit that can only be in one of two states (0 or 1), a qubit can exist in **infinitely many possible states**. Indeed, any normalized vector of the form:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

Represents a **valid state** of a qubit. Some **properties of the qubit state** are:

- The coefficients a and b are **complex numbers** (probability amplitudes). They determine the outcome of a measurement:

- $|a|^2$ is the probability of measuring the state $|0\rangle$.
- $|b|^2$ is the probability of measuring the state $|1\rangle$.

Therefore, even though a qubit can be in infinitely many possible states, a measurement (in this basis) can only produce **two possible outcomes**: $|0\rangle$ or $|1\rangle$.

- The state is **normalized**:

$$|a|^2 + |b|^2 = 1$$

Because the probabilities must sum to 1 (i.e., we must get either $|0\rangle$ or $|1\rangle$ when we measure).

Definition 1: Normalized Quantum State

A quantum state $|\psi\rangle$ is said to be **normalized** if its norm is equal to one:

$$\| |\psi\rangle \| = 1$$

Equivalently, using the inner product:

$$\langle \psi | \psi \rangle = 1$$

For a single-qubit state:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

This condition becomes:

$$|a|^2 + |b|^2 = 1 \quad (5)$$

This is required because $|a|^2$ and $|b|^2$ represent measurement probabilities, and the total probability of all possible outcomes must be equal to 1.

- The states $|0\rangle$ and $|1\rangle$ form a **basis** of the vector space describing the qubit.

🔍 What is a *basis*? A **Basis** is a set of vectors that allows us to represent any vector in the space as a linear combination of them. In quantum computing, $|0\rangle$ and $|1\rangle$ are the **basis vectors** and any qubit state is built from them.

In particular, the states $|0\rangle$ and $|1\rangle$ form an **Orthonormal Basis**, meaning that:

- They are **orthogonal**: $\langle 0|1\rangle = 0$.
🔍 Why is orthogonality important? Orthogonality means that the two basis states are completely distinguishable by a measurement (not overlapping). If we measure a qubit in the state $|0\rangle$, we will never get $|1\rangle$, and vice versa.
- They are **normalized**: $\langle 0|0\rangle = \langle 1|1\rangle = 1$.
🔍 Why is normalization important? Normalization ensures that the probabilities of measurement outcomes are well-defined and sum to 1. It also means that the basis states have unit length in the vector space, which is a fundamental property for quantum states.

🔍 Why This Abstraction Matters

This abstraction is fundamental because it allows us to:

- Describe **any** two-level quantum system (not only photons);
- Use a unified mathematical framework based on vectors;
- Apply the tools of linear algebra to quantum systems.

From now on, we will work with qubits using Dirac notation, without referring to a specific physical implementation.

Example 2: Different Choices of Basis

The computational basis $\{|0\rangle, |1\rangle\}$ is not the only possible choice. A qubit can be described using **any orthonormal basis**.

For example, another important basis is the **Hadamard basis**, defined as:

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad |-\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

This shows that the same quantum state can be represented in different bases, depending on the context.

Example 3: Orthogonality of the Hadamard Basis

Show that the Hadamard basis states $|+\rangle$ and $|-\rangle$ are orthogonal.

Proof. Recall:

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad |-\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Orthogonal means that their inner product is zero. So, we need to compute $\langle + | - \rangle$ and show that it equals zero:

$$\langle + | - \rangle = \frac{1}{\sqrt{2}} [1 \quad 1] \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{2} (1 - 1) = 0$$

Therefore, $|+\rangle$ and $|-\rangle$ are orthogonal.

QED

Example 4: Change of Basis from Standard $\leftrightarrow$ Hadamard

Given a qubit state in the standard basis:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

We want to express it in the Hadamard basis:

$$|\psi\rangle = x|+\rangle + y|-\rangle$$

Proof. Using the definitions of $|+\rangle$ and $|-\rangle$:

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad |-\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

We can write $|\psi\rangle$ in terms of $|+\rangle$ and $|-\rangle$:

$$|\psi\rangle = a|0\rangle + b|1\rangle = a \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}$$

Now, we express $|\psi\rangle$ as a linear combination of $|+\rangle$ and $|-\rangle$:

$$|\psi\rangle = x|+\rangle + y|-\rangle$$

Substituting the definitions of $|+\rangle$ and $|-\rangle$:

$$= x \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + y \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Combining the terms:

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} x+y \\ x-y \end{bmatrix}$$

We want this to equal $\begin{bmatrix} a \\ b \end{bmatrix}$, so we set:

$$\frac{1}{\sqrt{2}} \begin{bmatrix} x+y \\ x-y \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}$$

This gives us the system of equations:

$$x+y = \sqrt{2}a, \quad x-y = \sqrt{2}b$$

Adding the two equations:

$$2x = \sqrt{2}(a+b) \implies x = \frac{a+b}{\sqrt{2}}$$

Subtracting the second from the first:

$$2y = \sqrt{2}(a-b) \implies y = \frac{a-b}{\sqrt{2}}$$

It follows that:

$$|\psi\rangle = x|+\rangle + y|-\rangle = \frac{a+b}{\sqrt{2}}|+\rangle + \frac{a-b}{\sqrt{2}}|-\rangle$$

QED

Example 5: Orthogonal State Construction

Given:

$$|v\rangle = a|0\rangle + b|1\rangle$$

Define:

$$|w\rangle = \bar{b}|0\rangle - \bar{a}|1\rangle$$

Show that $|v\rangle$ and $|w\rangle$ are orthogonal.

Proof. To show that $|v\rangle$ and $|w\rangle$ are orthogonal, we need to compute their inner product $\langle w|v\rangle$ and show that it equals zero. First, we write the inner product converting $|w\rangle$ to its corresponding bra:

$$|w\rangle = \bar{b}|0\rangle - \bar{a}|1\rangle \implies \langle w| = b\langle 0| - a\langle 1|$$

Now we compute:

$$\langle w|v\rangle = (b\langle 0| - a\langle 1|) \cdot (a|0\rangle + b|1\rangle)$$

Expanding this expression:

$$= ba\langle 0|0\rangle + bb\langle 0|1\rangle - aa\langle 1|0\rangle - ab\langle 1|1\rangle$$

Using the orthonormality of the basis states:

$$\langle 0|0\rangle = \langle 1|1\rangle = 1, \quad \langle 0|1\rangle = \langle 1|0\rangle = 0$$

We get:

$$\langle w|v\rangle = \overset{1}{ba\langle 0|0\rangle} + \cancel{bb\langle 0|1\rangle} - \cancel{aa\langle 1|0\rangle} - \overset{1}{ab\langle 1|1\rangle} = ba - ab$$

Since a and b are complex numbers, multiplication is commutative, so $ba = ab$. Therefore:

$$\langle w|v\rangle = ba - ab = 0$$

Hence, $|v\rangle$ and $|w\rangle$ are orthogonal.

QED

2.3 Single Qubit Measurement

After introducing the mathematical representation of a qubit, we now describe how information is extracted from it through **measurement**.

🔍 What is *Single Qubit Measurement*?

We have a qubit:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

Where a and b are complex numbers such that $|a|^2 + |b|^2 = 1$. Measurement answers the question: “*which classical value do we observe?*”.

A measurement takes a **quantum state** as input and produces a **classical outcome** as output. In the case of a single qubit, the possible outcomes are 0 and 1.

⚠️ Measurement rule (computational basis)

The probability of an outcome is the **squared magnitude of the coefficient of the basis vector**:

$$P(0) = |a|^2, \quad P(1) = |b|^2 \quad (\text{in the computational basis})$$

But which coefficients? This is where the **basis** comes in. A state can be written in different ways depending on the basis. For example, take:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

- If we measure in basis $\{|0\rangle, |1\rangle\}$:
 - Outcome 0 with probability $|a|^2$
 - Outcome 1 with probability $|b|^2$
- If we measure in basis $\{|+\rangle, |-\rangle\}$:
 - Outcome $+$ with probability $|x|^2$
 - Outcome $-$ with probability $|y|^2$

The **coefficients depend on the chosen basis**. Let’s analyze how a basis change affects the coefficients.

🔍 **What is a basis *really*?** Any measurement device (e.g., a polarizer, page 12) has two preferred states (i.e., the states that it can distinguish). A basis is not just a mathematical construct; it corresponds to **what our measurements device can distinguish**. For example:

- Computational basis, the device distinguishes: *are you $|0\rangle$ or $|1\rangle$?*
- Hadamard basis, the device distinguishes: *are you $|+\rangle$ or $|-\rangle$?*

These are **different physical equations**.

? **The math behind the measurement.** Measurement is projecting the state vector onto basis vectors. Forget qubits for a second and imagine a 2D real vector $\vec{v}$, and choose axes:

- **Standard axes:** we project onto x -axis and y -axis. We get components: v_x and v_y .
- **Rotated axes:** we rotate axes by 45° and project onto new axes the same vector $\vec{v}$. We get new components: v_+ and v_- .

Different numbers, same vector. The components are **not intrinsic properties of the vector**, but depend on the chosen axes. Back to qubits, **measurement projects the vector $|\psi\rangle$ onto the measurement axes** (i.e., the basis vectors). So $\langle u|\psi\rangle$ is the projection of $|\psi\rangle$ onto $|u\rangle$, and $|\langle u|\psi\rangle|^2$ is the probability of observing outcome u .

Example 6: Analogy - Measurement as projecting a vector onto different axes

Consider a classical 2D vector pointing diagonally:

$$\vec{v} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

Case 1: Standard axes (horizontal/vertical). If we describe the vector using the standard basis:

$$\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\},$$

Then the vector has equal components along both axes:

$$\vec{v} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- Projection on horizontal axis: $\frac{1}{\sqrt{2}}$
- Projection on vertical axis: $\frac{1}{\sqrt{2}}$

Case 2: Rotated axes (diagonal basis). Now rotate the coordinate system by 45° and use the basis:

$$\left\{ \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$$

In this basis, the same vector becomes:

$$\vec{v} = 1 \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 0 \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

- Projection on first axis: 1
- Projection on second axis: 0

Connection with qubits. A qubit behaves exactly in the same way:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle = |+\rangle$$

- In basis $\{|0\rangle, |1\rangle\}$: equal projections $\Rightarrow$ probabilistic measurement.
- In basis $\{|+\rangle, |-\rangle\}$: full projection on $|+\rangle \Rightarrow$ deterministic measurement.

💡 The state does not change. What changes is the set of axes (basis) onto which the state is projected. Since measurement probabilities are given by these projections, the outcome depends on the chosen basis.

Given the state:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

- If we measure in the **computational basis**, we get:

- Outcome 0 with probability $\left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$
- Outcome 1 with probability $\left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$

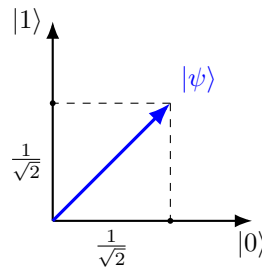


Figure 4: Measurement as projection onto basis vectors.

- If we measure in the **Hadamard basis**, we get:
 - Outcome + with probability $|\langle+|\psi\rangle|^2 = 1$
 - Outcome - with probability $|\langle-|\psi\rangle|^2 = 0$

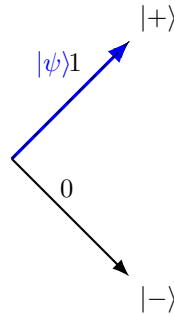


Figure 5: Measurement in the Hadamard basis. The state $|\psi\rangle$ is aligned with $|+\rangle$, so we get outcome $+$ with probability 1. The projection onto $|-\rangle$ is zero, so we get outcome $-$ with probability 0. The axes are rotated by 45° compared to the computational basis.

❓ What happens to the state after measurement? The state **collapses** to the observed outcome. For example, if we measure $|\psi\rangle = a|0\rangle + b|1\rangle$ in the computational basis and observe 0, the state collapses to $|0\rangle$. If we observe 1, the state collapses to $|1\rangle$.

🔗 Two possible situations

As we have seen in page 21, the same state can yield different measurement outcomes depending on the chosen basis. This leads to two important situations:

- **Probabilistic measurement:** If the **state is not aligned with the measurement basis**, we get a probabilistic outcome. For example:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

Measured in the computational basis gives 0 or 1 with equal probability.

- **Deterministic measurement:** If the **state is aligned with one of the basis vectors**, we get a deterministic outcome. For example:

$$|\psi\rangle = |+\rangle$$

Measured in the Hadamard basis gives $+$ with probability 1 and $-$ with probability 0.

❓ When should we prefer one situation to another? It depends on the task. For example, if we want to encode a classical bit, we prefer deterministic measurement (e.g., $|0\rangle$ or $|1\rangle$). If we want to create superposition states for quantum algorithms, we might want probabilistic measurement (e.g., $|+\rangle$).

Now, if the same quantum state can lead to either deterministic or probabilistic measurement outcomes depending on the chosen basis, we can ask: “*how can be explained this phenomenon?*”. It means that a qubit can simultaneously have non-zero components along multiple basis states. This is a fundamental property of quantum mechanics, and it is what allows quantum computers to perform certain computations more efficiently than classical computers. It is called **superposition**, and we will discuss it in the next section.

2.4 Superposition

We have already seen that:

1. **A qubit is represented as a vector:**

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

Where a and b are complex numbers, and $|0\rangle$ and $|1\rangle$ are the basis states of the qubit.

2. **Measurement probabilities depend on the chosen basis:**

- In the computational basis $\{|0\rangle, |1\rangle\}$, the probabilities of measuring $|0\rangle$ and $|1\rangle$ are $|a|^2$ and $|b|^2$, respectively.
- In another basis, such as the Hadamard basis $\{|+\rangle, |-\rangle\}$, the probabilities of measuring $|+\rangle$ and $|-\rangle$ depend on different coefficients, which can be calculated by expressing $|\psi\rangle$ in terms of the new basis states.

3. **The same state can lead to deterministic or probabilistic outcomes depending on the measurement basis.** For example, if $|\psi\rangle = |0\rangle$, measuring in the computational basis always yield $|0\rangle$ (deterministic):

$$|0\rangle = 1|0\rangle + 0|1\rangle = |0\rangle$$

Whereas measuring in the Hadamard basis yields $|+\rangle$ or $|-\rangle$ with equal probability (probabilistic):

$$|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$$

These observations raise an essential question: “*why can a single state exhibit such different behaviors under different measurements?*”. The answer is **superposition**.

🔍 What is Superposition?

Superposition is a fundamental principle of quantum mechanics that **applies to all quantum systems**, not just qubits.

Definition 2: Superposition

Superposition means that a **system can exist in several possible states at the same time until a measurement is made**. It is **basis-dependent**, meaning a state may be a superposition in one basis but not in another, for example:

$$|v\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

Is a superposition with respect to the computational basis $\{|0\rangle, |1\rangle\}$, but it is **not** a superposition in the Hadamard basis $\{|+\rangle, |-\rangle\}$, where it can

be written as:

$$|v\rangle = 1|+\rangle + 0|-\rangle = |+\rangle$$

Formally, let $\{|u\rangle, |u^\perp\rangle\}$ be an orthonormal basis of a single-qubit Hilbert space. A qubit is said to be in **superposition** with respect to this basis if it can be written as:

$$|\psi\rangle = a|u\rangle + b|u^\perp\rangle$$

Where both coefficients a and b are non-zero and satisfy $|a|^2 + |b|^2 = 1$ (normalization condition). If one of the coefficients is zero, the state coincides with one of the basis vectors and the measurements in that basis becomes deterministic, thus it is not a superposition.

More in general, a quantum state $|\psi\rangle$ in a system with multiple possible states can exist in a **linear combination** of these states:

$$|\psi\rangle = c_1|\psi_1\rangle + c_2|\psi_2\rangle + \dots + c_n|\psi_n\rangle$$

Where:

- $|\psi_1\rangle, |\psi_2\rangle, \dots, |\psi_n\rangle$ are **basis states** of the system.
- $c_1, c_2, \dots, c_n$ are **complex probability amplitudes**.
- The system is in **all states at the same time**, with the probability of measuring each state given by $|c_i|^2$.
- The state is **normalized**, meaning:

$$|c_1|^2 + |c_2|^2 + \dots + |c_n|^2 = 1$$

Example 7: Single-Particle Systems (Quantum Mechanics)

In general quantum mechanics, a particle can exist in **multiple positions simultaneously** as a wave function $\Psi(x)$:

$$|\Psi\rangle = \int \Psi(x) |x\rangle dx$$

- The particle is in a **superposition of all possible positions** $|x\rangle$.
- Measurement collapses the wave function to a single position.

❓ How Can a Particle Exist in Multiple States at Once? This question touches the heart of quantum mechanics, where our everyday intuition breaks down. The **key idea** is that a **quantum particle** is not just a tiny ball, it is a **wave function that spreads across multiple possibilities at once**.

1. **Quantum Particles Are Waves, Not Just Points.** In classical physics, we think of particles as tiny, solid objects, like a small ball that always has a precise position and velocity.

In quantum mechanics, however, particles behave more like waves. These waves are described by a wave function $\Psi(x)$, which represents the **probability of finding the particle at different locations**.

The key idea is:

- The **wave function spreads across space**, meaning the **particle does not have a single location** before measurement.
 - Instead, it exists in a superposition of all possible locations.
2. **The Double-Slit Experiment: Proof That a Particle Can Be in Two Places at Once.** The double-slit experiment was first performed by Thomas Young in 1801 to demonstrate the **wave nature of light**. Later, in the 20th century, it was repeated with **electrons, atoms**, and even **large molecules**, revealing that **matter itself exhibits wave-like behavior**. The classical expectation for the double-slit experiment is:

- **Classical Particles** (e.g., tiny balls). Imagine firing small balls at a barrier with two slits:
 - Each ball passes through **either** slit 1 **or** slit 2.
 - On a screen behind the slits, we would observe **two bright bands**, corresponding to the two slits.

The result is a simple pattern of two lines, no interference.

- **Classical Waves** (e.g., Water or Light). If instead we send **waves** (like water waves or classical light):
 - Each wave passes through **both slits simultaneously**.
 - The waves emerging from the slits **interfere: constructive interference** creates bright bands, while **destructive interference** creates dark bands.

The result is an **interference pattern** of alternating bright and dark bands.

The quantum reality is even more surprising:

- **Quantum Particles** (e.g., photons or electrons). When the experiment is performed with **single quantum particles** (photons or electrons), something remarkable happens:
 - (a) **Particles are emitted one at a time.**
 - (b) Each particle is detected as a **localized point** on the screen.
 - (c) After many particles are detected, an **interference pattern gradually emerges**.

This means that each particle behaves as if it passes through **both slits simultaneously**. The particle is described by a **wave function**, representing a **superposition of paths**:

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|\text{slit 1}\rangle + |\text{slit 2}\rangle)$$

The interference pattern arises from the **coherent combination** of these two possibilities.

Suppose we place detectors at the slits to determine **which slit** the particle passes through. The interference pattern **disappears** and the screen now shows **two bands**, just like classical particles. This is because measuring the particle's path **entangles** (i.e., correlates) it with the environment (the detector), leading to **decoherence** (loss of quantum coherence). The superposition:

$$\frac{1}{\sqrt{2}} (|\text{slit 1}\rangle + |\text{slit 2}\rangle)$$

Becomes effectively a **classical mixture**:

$$\frac{1}{2} |\text{slit 1}\rangle \langle \text{slit 1}| + \frac{1}{2} |\text{slit 2}\rangle \langle \text{slit 2}|$$

Without coherence, **interference cannot occur**.

In summary, the double-slit experiment demonstrates the principle of quantum superposition. When quantum particles such as photons or electrons are sent one at a time through two slits, they form an interference pattern on a detection screen. This indicates that each particle behaves as if it travels through both slits simultaneously, corresponding to the superposition:

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|\text{slit 1}\rangle + |\text{slit 2}\rangle)$$

If a measurement is performed to determine which slit the particle passes through, the interference pattern disappears due to decoherence, and the system behaves like a classical probabilistic mixture.

Double Slit Experiment explained! by Jim Al-Khalili



3. **Quantum Superposition: More Than Just Probability**. A common misconception is that a particle in superposition is just **an unknown state**, like a coin that is either heads or tails, but we just don't know which. This is wrong, because quantum superposition is much deeper.

A quantum state is a **combination of all possibilities**. *Until measurement*, the system is in **all possible states at once**.

Mathematically, for an electron in **two locations** x_1 and x_2 :

$$|\psi\rangle = a|x_1\rangle + b|x_2\rangle$$

- The electron is **literally in both places simultaneously**.
- The coefficients a and b are **complex numbers representing the probability amplitudes**.

- **Interference** between these amplitudes **creates quantum effects that cannot be explained by classical probability**.
4. **Superposition in Quantum Computing.** Quantum computing **directly uses** the fact that a particle can be in multiple states at once.
 - A **classical bit** can only be **0 or 1**.
 - A **qubit (quantum bit)** can be in a superposition:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

This means a quantum computer can **perform many calculations simultaneously**. Therefore, **superposition allows quantum computers to process information exponentially faster than classical computers for certain tasks**.

5. **Why Don't We See Superposition in Everyday Life?** In our daily experience, objects are **not in multiple states at once** because of a process called **quantum decoherence**.

Quantum superposition is **fragile**. When a quantum system interacts with the environment (air, light, etc.), the superposition **collapses into one definite state**. This is why large objects (like humans or cars) do not appear in multiple places at once.

However, experiments (like the Double-Slit experiment) confirm that superposition is real at microscopic scales (electrons, photons, atoms, and even molecules).

The key **properties of Superposition** are:

1. **Linearity:** Any combination of valid quantum states is also a valid quantum state.
2. **Interference:** Quantum states in superposition can interfere, leading to constructive or destructive interference.
3. **Measurement Collapse:** When measured, the superposition collapses into a single outcome.
4. **Phase Information:** Unlike classical probabilities, quantum superpositions include complex phases that affect interference patterns.

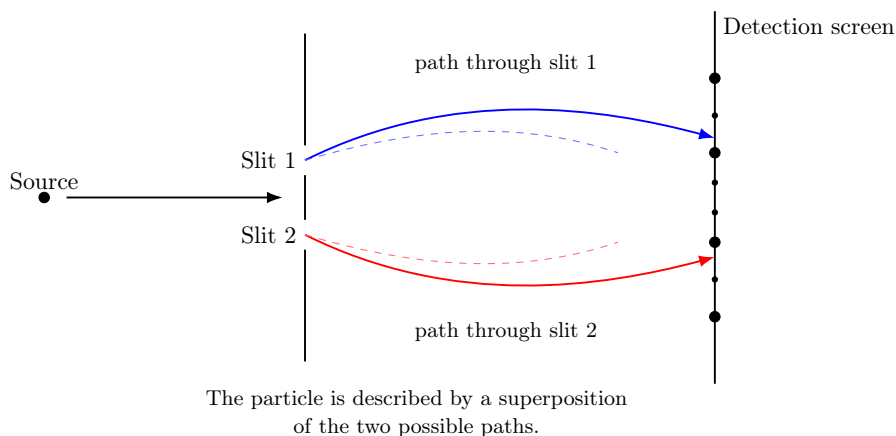


Figure 6: Double-slit experiment, superposition of paths. This figure illustrates the **quantum state of the particle before detection**. Instead of choosing a single path, the particle is described by a **superposition**:

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|\text{slit 1}\rangle + |\text{slit 2}\rangle)$$

The particle is **not** traveling through only one slit, instead, it is described by a **coherent combination** of both possibilities. The diagram emphasizes the **quantum state** rather than the final inference pattern.

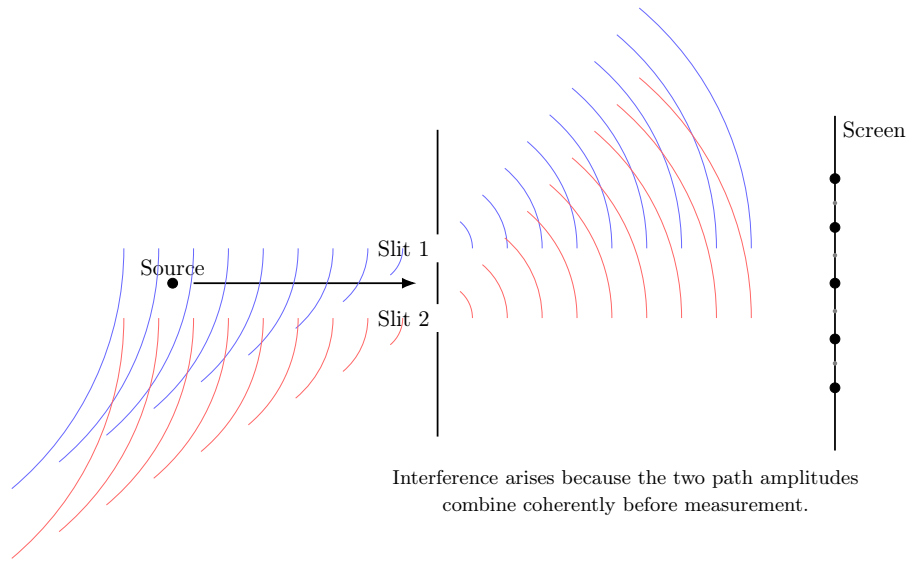


Figure 7: Wave-like representation of the double-slit experiment, interference of amplitudes. This figure focuses on the **wave nature** of the quantum particle. Each slit acts as a **coherent source**, producing wavefronts that overlap and interfere. If $A_1(x)$ and $A_2(x)$ are the probability amplitudes associated with the two slits, the probability of detecting the particle at position x on the screen is:

$$P(x) = |A_1(x) + A_2(x)|^2 = |A_1(x)|^2 + |A_2(x)|^2 + 2\text{Re}(A_1^*(x)A_2(x))$$

The last term is the **inference term**, which can be positive (bright fringes, **constructive interference**) or negative (dark fringes, **destructive interference**). This illustrates how the superposition of paths leads to the characteristic interference pattern observed in the experiment.

2.5 Information Carried by a Single Qubit

This section addresses a fundamental question: *if a qubit can exist in infinitely many states, how much information can it actually convey?* A general single-qubit state is given by:

$$|\psi\rangle = a|0\rangle + b|1\rangle \quad a, b \in \mathbb{C}, \quad |a|^2 + |b|^2 = 1$$

Because a and b are complex numbers, a qubit can be prepared in **infinitely many distinct states**. This might suggest that a qubit can encode an infinite amount of information. However, this is **not** the case. When we perform a **measurement**, we obtain only a **single classical outcome** (either 0 or 1). Therefore, **a single qubit can yield at most one classical bit of information when measured**. This apparent paradox, **infinite possible states but limited measurable information**, is one of the most important conceptual aspects of quantum information theory.

❓ Why can't we determine the Amplitudes a and b of a single qubit state?

Suppose we are given a qubit in an unknown state $|\psi\rangle$. Can we determine a and b from a single measurement? The answer is **no**, because:

- ✗ **Measurement is probabilistic.** A single measurement provides only one outcome (0 or 1), which is insufficient to determine the amplitudes.
- ✗ **Measurement disturbs the state.** After measurement, the qubit collapses to the observed basis state, preventing further measurements on the original state.
- ✗ **State estimation requires many copies.** To estimate a and b , we need **quantum state tomography**, which relies on performing measurements on **many identically prepared qubits**.

This **irreversible collapse** means that the original information encoded in the amplitudes is generally lost after measurement.

⚠ The No-Cloning Theorem

One might think of copying the qubit before measurement to extract more information. However, quantum mechanics forbids this through the **No-Cloning Theorem** (we will see more about this on page 172).

Theorem 1 (No-Cloning). *It is impossible to create an identical copy of an arbitrary unknown quantum state.*

Proof. Assume there exists a unitary operator U that can clone any state $|\psi\rangle$:

$$U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle \quad \forall |\psi\rangle$$

For two arbitrary states $|\psi\rangle$ and $|\phi\rangle$, linearity of quantum mechanics implies:

$$U(\alpha|\psi\rangle + \beta|\phi\rangle)|0\rangle = \alpha|\psi\rangle|\psi\rangle + \beta|\phi\rangle|\phi\rangle$$

Which is **not equal** to:

$$(\alpha |\psi\rangle + \beta |\phi\rangle)(\alpha |\psi\rangle + \beta |\phi\rangle) = \alpha^2 |\psi\rangle |\psi\rangle + \alpha\beta |\psi\rangle |\phi\rangle + \alpha\beta |\phi\rangle |\psi\rangle + \beta^2 |\phi\rangle |\phi\rangle$$

This contradiction proves that perfect cloning is impossible, since the two expressions cannot be equal for all $|\psi\rangle$ and $|\phi\rangle$. QED

Since we cannot clone a qubit, we cannot create multiple copies to determine its exact state from a single instance.

Feature	Classical Bit	Qubit
Possible states	0 or 1	Qubit
Info from measurement	1 bit	At most 1 bit
Measurement disturbance	No	Yes
Ability to copy	Yes	No (No-Cloning)
State determination	Single observation	Requires many identical copies

Table 1: Comparison of classical bits and qubits in terms of information content and measurement properties.

2.6 State Space of a Single-Qubit System

The **State Space** of a physical system is the set of all possible states that the system can occupy. For a **single qubit**, the state space consists of all vectors of the form:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

- $a, b \in \mathbb{C}$ are **complex probability amplitudes**.
- $|0\rangle$ and $|1\rangle$ form an **orthonormal basis** (the computational basis).
- The **normalization condition** must hold: $|a|^2 + |b|^2 = 1$. This ensures that the total probability of all measurement outcomes is equal to 1.

Degrees of Freedom of a Qubit

At first glance, the coefficients a and b appear to provide **4 real parameters**:

$$\begin{aligned} a = a_r + ia_i &\Rightarrow a_r, a_i \quad (2 \text{ parameters}) \\ b = b_r + ib_i &\Rightarrow b_r, b_i \quad (2 \text{ parameters}) \end{aligned}$$

So initially, it seems that a qubit has **four degrees of freedom**. However, not all of these correspond to physically distinguishable states.

- **Normalization constraint.** Quantum states must be normalized, which imposes the condition:

$$|a|^2 + |b|^2 = a_r^2 + a_i^2 + b_r^2 + b_i^2 = 1$$

Geometrically, it defines the surface of a **unit sphere in four dimension** (also known as a 3-sphere, S^3). This constraint effectively removes **one degree of freedom**, since the amplitudes must lie on the surface of a 3-sphere in 4-dimensional space.

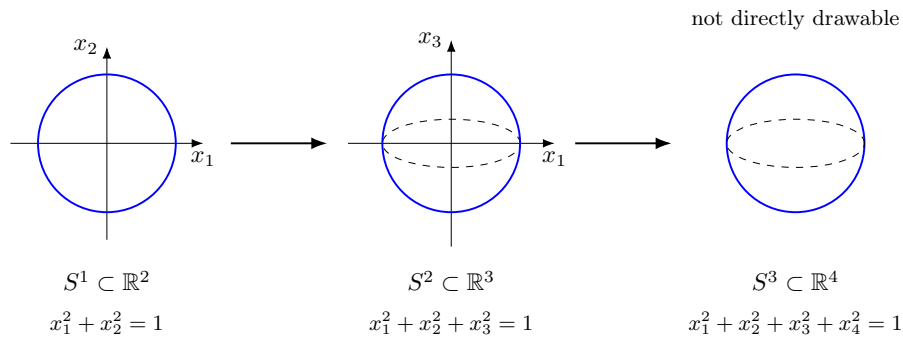


Figure 8: Analogy between the unit circle S^1 , the ordinary sphere S^2 , and the 3-sphere S^3 . The normalized qubit state lives on S^3 because it is specified by four real coordinates constrained by one equation. In this figure, we can only visualize S^1 and S^2 , while S^3 cannot be directly drawn in three-dimensional space, so we represent it with a dashed ellipse to indicate that it is a higher-dimensional object.

- **Global Phase Equivalence.** Two states that differ only by a **global phase** represent the **same physical qubit**:

$$|\psi'\rangle = e^{i\phi} |\psi\rangle$$

For example:

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \quad e^{i\pi/3} |\psi\rangle = \frac{e^{i\pi/3}}{\sqrt{2}} (|0\rangle + |1\rangle)$$

❓ **Why is it physically irrelevant?** Measurement probabilities are given by:

$$P(i) = |\langle i|\psi\rangle|^2$$

If we multiply the state by a global phase:

$$|\langle i|e^{i\gamma}\psi\rangle|^2 = |e^{i\gamma}\langle i|\psi\rangle|^2 = |\langle i|\psi\rangle|^2$$

Because $|e^{i\gamma}| = 1$ for any real γ .

Mathematical proof that $|e^{i\gamma}| = 1$. We can express $e^{i\gamma}$ using Euler's formula:

$$e^{i\gamma} = \cos(\gamma) + i\sin(\gamma)$$

The magnitude of a complex number $z = x + iy$ is given by:

$$|z| = \sqrt{x^2 + y^2}$$

Applying this to $e^{i\gamma}$:

$$\begin{aligned} |e^{i\gamma}| &= \sqrt{(\cos(\gamma))^2 + (\sin(\gamma))^2} \\ &\downarrow \text{using the trigonometric identity} \\ &= \sqrt{\cos^2 \gamma + \sin^2 \gamma} \\ &= \sqrt{1} \\ &= 1 \end{aligned}$$

QED

Therefore, **no experiment can distinguish between $|\psi\rangle$ and $e^{i\phi}|\psi\rangle$** , since **they yield the same measurement probabilities** (due to $|e^{i\phi}| = 1$).

❓ **How does this remove a degree of freedom?** Unlike normalization, the global phase does **not** impose a constrain equation. Instead, it introduces a **redundancy** in our description:

- Many different vectors in S^3 correspond to the **same physical state**.
- All vectors of the norm $e^{i\gamma}|\psi\rangle$ are considered **physically equivalent** to $|\psi\rangle$.

Mathematically, we are **identifying** all these vectors as a single point. This process is called forming a **quotient space**: the set of physically distinct pure qubit states is $S^3 \setminus S^1 \cong S^2$, which is a 2-dimensional surface (the Bloch sphere, see Figure 8, page 33).

After accounting for these constraints, we find that a single qubit state is fully described by **two real parameters** (two degrees of freedom). Therefore, any qubit state can be written in the form:

1. A normalized qubit can be written as:

$$|\psi\rangle = a|0\rangle + b|1\rangle \quad a, b \in \mathbb{C} \quad |a|^2 + |b|^2 = 1$$

2. Any complex number can be written using its **magnitude** and **phase**:

$$a = |a|e^{i\alpha} \quad b = |b|e^{i\beta}$$

Where $|a|$ and $|b|$ are non-negative real numbers, and α and β are real numbers representing the phases. Substituting these into the expression for $|\psi\rangle$ gives:

$$|\psi\rangle = \underbrace{|a|e^{i\alpha}}_a |0\rangle + \underbrace{|b|e^{i\beta}}_b |1\rangle$$

3. We can factor out the common global phase $e^{i\alpha}$:

$$|\psi\rangle = e^{i\alpha} \left(|a| |0\rangle + |b| e^{i(\beta-\alpha)} |1\rangle \right)$$

Since the global phase $e^{i\alpha}$ has **no physical effect**, we can omit it, leading to:

$$|\psi\rangle = |a| |0\rangle + |b| e^{i\phi} |1\rangle$$

Where we define also the **Relative Phase** $\phi = \beta - \alpha$. At this point, the state is characterized by two real magnitudes $|a|$ and $|b|$, and one relative phase ϕ . Mathematically, if:

$$a = |a|e^{i\alpha} \quad b = |b|e^{i\beta}$$

Then:

$$\text{Relative Phase} = \phi = \arg(b) - \arg(a) = \beta - \alpha \quad (6)$$

4. From normalization:

$$|a|^2 + |b|^2 = 1$$

We recognize a familiar trigonometric identity:

$$\cos^2(x) + \sin^2(x) = 1$$

Therefore, we can parametrize the magnitudes using a single angle θ :

$$|a| = \cos\left(\frac{\theta}{2}\right) \quad |b| = \sin\left(\frac{\theta}{2}\right)$$

With $0 \leq \theta \leq \pi$ to ensure that $|a|$ and $|b|$ are non-negative. The reason for using $\frac{\theta}{2}$ instead of θ will become clear when we discuss the Bloch sphere representation (Figure 9, page 37).

5. Substituting the parametrized magnitudes into the state:

$$|\psi\rangle = \underbrace{\cos\left(\frac{\theta}{2}\right)}_{|a|} |0\rangle + e^{i\phi} \underbrace{\sin\left(\frac{\theta}{2}\right)}_{|b|} |1\rangle \quad (7)$$

This is the **canonical parametrization** of a single qubit state.

Regardless of the specific values of θ and ϕ in Equation 7:

- θ : controls the **relative weights of $|0\rangle$ and $|1\rangle$** in the superposition, determining the **probabilities of measuring each state**.
- ϕ : controls the **relative phase between the $|0\rangle$ and $|1\rangle$** components. While it does not affect measurement probabilities, it is crucial for interference effects and the behavior of the qubit under quantum gates (which we will discuss later).

Bloch Sphere Representation of a Qubit

After normalization and removal of the physically irrelevant global phase, a pure single-qubit state depends on only two real parameters. Therefore, it is natural to represent it geometrically as a point on a two-dimensional surface. This surface is the **Bloch sphere**.

Any pure qubit can be written in the form:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle,$$

Where:

- $\theta \in [0, \pi]$ is the **polar angle**,
- $\phi \in [0, 2\pi)$ is the **Azimuthal Angle**, representing the **relative phase**.

This expression gives a one-to-one correspondence between pure qubit states and points on the surface of a unit sphere:

- $|0\rangle$ corresponds to the north pole, since $\theta = 0$:

$$|0\rangle = \cos\left(\frac{0}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{0}{2}\right) |1\rangle = 1 \cdot |0\rangle + 0 \cdot |1\rangle = |0\rangle$$

- $|1\rangle$ corresponds to the south pole, since $\theta = \pi$:

$$|1\rangle = \cos\left(\frac{\pi}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\pi}{2}\right) |1\rangle = 0 \cdot |0\rangle + e^{i\phi} \cdot 1 \cdot |1\rangle = e^{i\phi} |1\rangle \Rightarrow |1\rangle$$

⚠ Attention! $e^{i\phi} |1\rangle$ becomes $|1\rangle$ after **removing the global phase** (physically unobservable), so $|1\rangle$ is indeed at the south pole. In general, all states of the form $e^{i\phi} |1\rangle$ represent the **same physical state** as $|1\rangle$. Hence, $|1\rangle$ corresponds to the **south pole** of the Bloch sphere.

- States on the equator correspond to equal-weight superpositions of $|0\rangle$ and $|1\rangle$.

In particular:

$$\text{Hadamard} \quad |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle),$$

$$\text{Imaginary} \quad |i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle), \quad |-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle),$$

All lie on the equator and differ only by their relative phase.

The Bloch sphere provides an intuitive visualization of a qubit: moving along the vertical direction changes the relative weights of $|0\rangle$ and $|1\rangle$, while rotating around the vertical axis changes the relative phase between them.

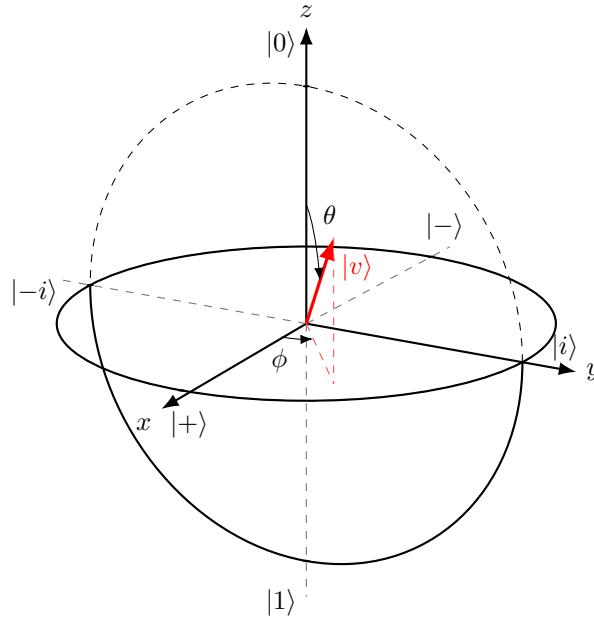


Figure 9: Bloch sphere representation of a pure qubit state. The red vector $|v\rangle$ represents a generic qubit state. Measuring $|v\rangle$ along the z -axis gives probabilities determined by θ , while along x or y -axes reveals information about the relative phase ϕ .

On the Bloch sphere, the axes correspond to **three fundamental measurement bases** for a qubit. Each pair of opposite points represents the **eigenstates of one of the Pauli operators** (we will discuss the Pauli operators in more detail in 3.3.2.2, page 48):

- **z -axis, computational basis** $\{|0\rangle, |1\rangle\}$, eigenstates of σ_z . The states are:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

These states correspond to the classical bit values 0 and 1.

- $|0\rangle$: North pole of the Bloch sphere $(0, 0, 1)$.
- $|1\rangle$: South pole of the Bloch sphere $(0, 0, -1)$.

Measuring along the z -axis answer the question: “*is the qubit $|0\rangle$ or $|1\rangle$?*”.

- **x -axis, Hadamard basis** $\{|+\rangle, |-\rangle\}$, eigenstates of σ_x . The states are:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad \text{and} \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

- $|+\rangle$: $+x$ direction on the equator $(1, 0, 0)$, with $\phi = 0$ (no relative phase).
- $|-\rangle$: $-x$ direction on the equator $(-1, 0, 0)$, with $\phi = \pi$ (relative phase of π between $|0\rangle$ and $|1\rangle$).

These states represent **equal superpositions** of $|0\rangle$ and $|1\rangle$ with **relative phase** 0 and π .

- **y -axis, imaginary basis** $\{|i\rangle, |-i\rangle\}$, eigenstates of σ_y . The states are:

$$|i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle) \quad \text{and} \quad |-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$$

- $|i\rangle$: $+y$ direction on the equator $(0, 1, 0)$, with $\phi = \frac{\pi}{2}$ (relative phase of $\frac{\pi}{2}$ between $|0\rangle$ and $|1\rangle$).
- $|-i\rangle$: $-y$ direction on the equator $(0, -1, 0)$, with $\phi = -\frac{\pi}{2}$ (relative phase of $-\frac{\pi}{2}$ between $|0\rangle$ and $|1\rangle$).

These states also have equal probabilities of measuring $|0\rangle$ or $|1\rangle$ in the computational basis, but they differ by a **relative phase of $\pm\frac{\pi}{2}$** . This phase is crucial for **interference effects** in quantum algorithms.

3 Single Qubit Gates

3.1 Operations on Qubits

Quantum computing relies on three essential operations:

1. **Logic Gates:** transform quantum states.
2. **Measurements:** extract classical information from qubits.
3. **Initialization Procedures:** prepare qubits in known starting states.

Logic Gates: Reversible Transformations

Quantum **Logic Gates** are the building blocks of quantum circuits. They **manipulate the state of qubits in a controlled and deterministic way**, analogous to classical logic gates (like AND or NOT), but with key quantum differences:

- **Operate on Single or Multiple Qubits.** They can affect one qubit (single-qubit gates) or multiple qubits simultaneously (multi-qubit gates):
 - **Single-Qubit Gates:** These gates, such as the Pauli-X, Y, Z, Hadamard (H), and Phase (S) gates, operate on a single qubit to change its state. We will explore these gates in detail in the next sections.
 - **Multi-Qubit Gates:** These gates, such as the CNOT (Controlled-NOT) and Toffoli gates, operate on multiple qubits and can create *entanglement* (we will discuss this in detail later) between them, which is a key resource in quantum computing.
- **Reversible Transformations.** Unlike many classical gates (e.g., AND), **all valid quantum gates are reversible**. Mathematically, this means they are represented by **unitary matrices** U , satisfying:

$$U^\dagger U = U U^\dagger = I$$

Where $U^\dagger$ is the conjugate transpose of U , I is the identity matrix, and U is a unitary matrix¹. Because of this reversibility, quantum gates **do not lose information** and the **original state can be recovered** by applying $U^\dagger = U^{-1}$ to the output state. This is a fundamental difference from classical logic gates, which can be irreversible (e.g., AND gate loses information about the inputs).

Finally, this property is related to fundamental principles such as the **no-cloning theorem** (page 31, page 172) and the **no-deleting theorem**

¹A matrix U is called **unitary** if its **conjugate transpose equals its inverse**:

$$U^\dagger U = U U^\dagger = I \tag{8}$$

We remind the reader that the definition of the inverse of a matrix is:

$$U^{-1}U = U U^{-1} = I$$

So for a unitary matrix, the conjugate transpose $U^\dagger$ plays the role of the inverse. This means that unitary transformations preserve vector norms, orthogonality, and therefore total probability in quantum mechanics.

(we cannot delete quantum information arbitrarily, since it is preserved under unitary transformations, page 181).

- **State Evolution.** If a qubit is in the state:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

Then after applying a gate U , the new state becomes:

$$|\psi'\rangle = U|\psi\rangle = aU|0\rangle + bU|1\rangle$$

This **transformation is deterministic** and **preserves the normalization of the state** (i.e., $|a|^2 + |b|^2 = 1$ remains true after applying U). The specific effect of the gate depends on its matrix representation, which we will explore in the next sections.

We can think of quantum logic gates as **rotations on the Bloch sphere**, smoothly transforming the qubit's state without destroying its quantum nature.

Measurement: Irreversible Collapse of State

Measurement is the process of extracting information from a qubit. Unlike logic gates, measurement is **irreversible** and fundamentally probabilistic.

- **Irreversible Process.** Once a qubit is measured, its original quantum state is **lost**. This distinguishes measurement from unitary gate operations.

What is a unitary gate? A **Unitary Gate** is a quantum gate represented by a **unitary matrix**. More precisely, a quantum gate U is unitary if (see page 39):

$$U^\dagger U = UU^\dagger = I$$

Where $U^\dagger$ is the conjugate transpose of U , and I is the identity matrix. Since a unitary matrix has an inverse equal to its conjugate transpose, then **every quantum gate is reversible**.

Important Property. A fundamental property of **unitary gates** is that they **preserve similarity transformations**. This means that they preserve **geometric relationships** between quantum states, such as lengths (norms), angles, and inner products:

$$\langle\psi|\phi\rangle = \langle U\psi|U\phi\rangle \quad (9)$$

Where $|\psi\rangle$ and $|\phi\rangle$ are any two quantum states. For example, if two states are orthogonal (i.e., $\langle\psi|\phi\rangle = 0$), then after applying a unitary gate U , they remain orthogonal (i.e., $\langle U\psi|U\phi\rangle = 0$). In other words, if two states are “very close”, they remain “very close” after applying a unitary transformation.

- **State Collapse.** Measuring a qubit in the computational basis $\{|0\rangle, |1\rangle\}$ causes the state:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

To **collapse** to:

- $|0\rangle$ with probability $|a|^2$.
- $|1\rangle$ with probability $|b|^2$.

- **Extraction of Classical Information.** The measurement outcome is a **classical bit** (0 or 1). After measurement, subsequent measurements in the same basis yield the same result with probability 1, since the state has collapsed to a definite state.

Measurement can be described using **projection operators**:

$$M_0 = |0\rangle\langle 0| \quad M_1 = |1\rangle\langle 1|$$

The probability of obtaining outcome k (where $k \in \{0, 1\}$) when measuring $|\psi\rangle$ is given by:

$$\begin{aligned}
 p_k &= \langle \psi | M_k | \psi \rangle \\
 &\downarrow \text{substitute } M_k = |k\rangle\langle k| \\
 &= \langle \psi | (|k\rangle\langle k|) | \psi \rangle \\
 &\downarrow \text{using the associativity of inner products} \\
 &= (\langle \psi | k \rangle) (\langle k | \psi \rangle) \\
 &\downarrow \text{since } \langle \psi | k \rangle = (\langle k | \psi \rangle)^* \\
 &\quad \text{where } * \text{ denotes complex conjugation} \\
 &= (\langle k | \psi \rangle)^* (\langle k | \psi \rangle) \\
 &\downarrow \text{which is the definition of squared magnitude, page 7} \\
 &= |\langle k | \psi \rangle|^2
 \end{aligned}$$

In other words, the probability of measuring outcome k is the **squared magnitude of the amplitude** of $|\psi\rangle$ along the direction $|k\rangle$. And the post-measurement state of the qubit, conditioned on obtaining outcome k , becomes:

$$|\psi_k\rangle = \frac{M_k |\psi\rangle}{\sqrt{p_k}}$$

Measurement is like **asking a question** to the qubit: “are you in state $|0\rangle$ or $|1\rangle$?”. The answer is classical (i.e., 0 or 1), but the act of asking **changes the system**.

Example 1: Calculating Measurement Probabilities

Consider a qubit in the state:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

With $|a|^2 + |b|^2 = 1$.

- Probability of measuring 0. Using the projector:

$$M_0 = |0\rangle\langle 0|$$

We compute:

$$\begin{aligned}
 p_0 &= \langle \psi | M_0 | \psi \rangle \\
 &= \langle \psi | 0 \rangle \langle 0 | \psi \rangle \\
 &= (\langle \psi | 0 \rangle) (\langle 0 | \psi \rangle) \\
 &\downarrow \text{since } \langle \psi | 0 \rangle = (\langle 0 | \psi \rangle)^* \\
 &= |\langle 0 | \psi \rangle|^2 \\
 &= |a|^2
 \end{aligned}$$

- Probability of measuring 1. Using the projector:

$$M_1 = |1\rangle \langle 1|$$

We compute:

$$\begin{aligned}
 p_1 &= \langle \psi | M_1 | \psi \rangle \\
 &= \langle \psi | 1 \rangle \langle 1 | \psi \rangle \\
 &= (\langle \psi | 1 \rangle) (\langle 1 | \psi \rangle) \\
 &\downarrow \text{since } \langle \psi | 1 \rangle = (\langle 1 | \psi \rangle)^* \\
 &= |\langle 1 | \psi \rangle|^2 \\
 &= |b|^2
 \end{aligned}$$

Initialization: Preparing Known States

The **Initialization** is the process of preparing a qubit in a **known and controlled state**, typically $|0\rangle$ or $|1\rangle$. This step is essential because quantum algorithms require a well-defined starting point. It is implemented through various methods, including:

- **Physical Preparation.** Many quantum technologies naturally relax to the ground state $|0\rangle$. Additional gates (e.g., Pauli-X) can be applied to obtain $|1\rangle$ if needed.
- **Initialization via Measurement.** Measuring a qubit projects it into a basis state. If the outcome is not the desired one, a corrective gate can be applied. For example, if measurement yields $|1\rangle$ but $|0\rangle$ is needed, apply an X gate to flip it to $|0\rangle$.

Initialization is the **first step** in any quantum computation: (1) **prepare** qubits in known states, (2) **apply** them using quantum gates, and (3) **measure** to obtain classical results. It is analogous to **resetting a classical register** before starting a computation in classical computing.

3.2 Quantum Logic Gates Overview

In quantum computing, quantum logic gates are the primary tools used to **manipulate qubits**. These gates perform **unitary transformations** on the state of one or more qubits, meaning they **preserve the norm** of the quantum state and are **reversible**. The **design of quantum algorithms** and the **execution of quantum circuits** rely entirely on the sequential application of these gates.

≡ Types of Quantum Gates

Quantum Gates can be classified into:

- **Single-Qubit Gates:** These gates operate on **individual qubits**, modifying their state in isolation.
- **Multiple-Qubit Gates:** These gates act on **two or more qubits simultaneously**, allowing for entanglement and complex correlations between qubits.

In this section we will focus on single qubit gates.

✓ Mathematical Framework: Qubit as Vectors, Gates as Matrices

To describe quantum superposition mathematically, we require a framework that supports linear combinations of states, probabilities derived from inner products, complex amplitudes and phases, and linear state transformations. The mathematical structure satisfying all these requirements is a **complex Hilbert space**. Thus, a qubit is naturally represented as a **unit vector** in a two-dimensional complex Hilbert space $\mathbb{C}^2$.

Using the Dirac notation, the basis states $|0\rangle$ and $|1\rangle$ form an orthonormal basis for this space, satisfying $\langle 0|0\rangle = \langle 1|1\rangle = 1$ and $\langle 0|1\rangle = \langle 1|0\rangle = 0$. Any qubit state can be expressed as a linear combination of these basis states:

$$|\psi\rangle = a|0\rangle + b|1\rangle \quad \text{with } |a|^2 + |b|^2 = 1$$

Two state vectors that differ only by a global phase factor $e^{i\phi}$ represent the same physical qubit, since measurement probabilities remain unchanged (see page 34).

Choosing the computational basis $\{|0\rangle, |1\rangle\}$, we can represent these states as column vectors:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Therefore, the qubit state can be written as:

$$|\psi\rangle = a|0\rangle + b|1\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$$

This vector representation allows us to apply linear algebra techniques to analyze and manipulate qubits.

Once qubits are represented as vectors, transformations of these states must be **linear operators** acting on the vector space. In a chosen basis, linear operators are represented by **matrices**. To preserve normalization and probabilities, these matrices must be **unitary**:

$$U^\dagger U = U U^\dagger = I$$

Thus, quantum gates – the **building blocks of quantum circuits** – are represented as unitary 2×2 matrices that linearly transform the state vector:

$$|\psi'\rangle = U |\psi\rangle$$

❓ Why Matrices? Because quantum mechanics is inherently linear, any physical evolution of a closed quantum system must be described by a **linear operator**. When expressed in a chosen basis (such as the computational basis $\{|0\rangle, |1\rangle\}$), this operator takes the form of a matrix.

☆ Unitary: The Fundamental Constraint

A matrix U is **unitary** if it satisfies the condition:

$$U^\dagger U = U U^\dagger = I \quad (10)$$

Where:

- $U^\dagger$ is the **Hermitian transpose** (conjugate transpose) of U .
- I is the **identity matrix**.

This property ensures that $U^{-1} = U^\dagger$, meaning every quantum operation is **reversible**, and the transformation preserves the **norm** of the state vector. So, quantum gate **must be unitary** to ensure that the evolution of quantum states is consistent with the principles of quantum mechanics (i.e., linearity, reversibility, and preservation of probabilities).

A **general single-qubit unitary matrix** can be expressed as:

$$U = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \quad \text{with } \alpha, \beta, \gamma, \delta \in \mathbb{C} \quad \text{and} \quad U^\dagger U = I \quad (11)$$

❓ Why must gates be unitary?

1. **No-Cloning Theorem:** It is impossible to create an identical copy of an unknown qubit. Unitary transformations respect this by not duplicating information.
2. **No-Deleting Theorem:** We cannot delete quantum information arbitrarily. Since unitary gates are invertible, they do not erase information.
3. **Preservation of Probability:** The normalization condition $|a|^2 + |b|^2 = 1$ must hold after any operation, and **only unitary matrices preserve this norm.**

Physical Intuition

In classical circuits, logic gates like AND, OR, NOT process bits deterministically. In quantum circuits:

- Gates **rotate** the quantum state on the **bloch sphere**.
- These rotations correspond to **unitary transformations**.
- The **physical implementation** of quantum gates may vary across hardware (e.g., superconducting qubits, trapped ions), but **mathematically**, the **same unitary matrix describes the gate**.

Quantum gates do not correspond to physical gates in the classical sense. They are **abstract operations** realized by **controlled interactions** in quantum systems.

3.3 Main Single-Qubit Gates

3.3.1 Identity Gate (I)

The **Identity Gate**, denoted I , is the most elementary gate in quantum computing. It performs **no change** to the state of a qubit. Its action on a qubit is trivial, yet it **serves important roles in quantum circuits**, particularly **when managing multi-qubit systems** or **timing synchronization**.

√* Matrix Representation

The Identity gate is represented by the 2×2 **identity matrix**:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (12)$$

Applying the Identity gate to any qubit **leaves the state unchanged**. Formally, for a qubit in state $|\psi\rangle$:

$$I|\psi\rangle = |\psi\rangle$$

To better understand its effect, consider its action on the computational basis:

$$\begin{aligned} I|0\rangle &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle \\ I|1\rangle &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle \end{aligned}$$

Hence, the Identity gate **preserves the state** of the qubit, including any superposition or entangled state it may be in.

📖 Physical Interpretation

Although it may appear useless at first glance, the Identity gate has **conceptual and practical utility**:

- ✓ It acts as a **placeholder** or “*do nothing*” operation when required by **circuit timing**.
- ✓ In **multi-qubit circuits**, it allows one **qubit to remain unchanged** while other qubits are operated on.
- ✓ In **matrix composition** of gates, it serves to **align dimensions** when performing **tensor products** (e.g., $I \otimes H$ applies Hadamard to the second qubit only).

🌀 Bloch Sphere Perspective

In the Bloch sphere (page 37), the **identity gate** represents a **rotation of zero degrees**, meaning the state vector remains exactly where it is. It is the neutral element of quantum transformations.

3.3.2 Pauli Gates

The **Pauli Gates** are a set of fundamental single-qubit quantum gates that play a central role in quantum computing. They are represented by the three **Pauli matrices** X , Y , and Z , and can be interpreted as the quantum analogues of basic operations on a qubit.

Each Pauli gate corresponds to a **rotation of π radians** around one of the principal axes of the Bloch sphere:

- X : rotation around the x -axis (bit flip).
- Z : rotation around the z -axis (phase flip).
- Y : rotation around the y -axis (bit and phase flip).

These operators are **unitary** ($U^\dagger U = U U^\dagger = I$), **Hermitian** ($U = U^\dagger$), and **self-inverse** ($U^2 = I$), and their eigenstates define the three fundamental measurement bases of a qubit. Together, they form the building blocks for describing single-qubit transformations and quantum dynamics. Also, they have eigenvalues ± 1 and satisfy the property:

$$X^2 = Y^2 = Z^2 = I$$

Meaning that applying the same Pauli gate twice returns the qubit to its original state.

3.3.2.1 Rotations Generated by Pauli Operators

In quantum mechanics, rotations of a qubit are generated by the Pauli operators. A rotation around an axis $\hat{n}$ by an angle θ is given by:

$$R_{\hat{n}}(\theta) = e^{-i\frac{\theta}{2}(\hat{n} \cdot \vec{\sigma})} \quad (13)$$

Where $\vec{\sigma} = (X, Y, Z)$ are the Pauli matrices. We will see those matrices in more detail in the next sections, but for now we can just note that they are the generators of rotations around the x , y , and z axes, respectively.

In particular:

$$R_x(\theta) = e^{-i\frac{\theta}{2}X} \quad (14)$$

$$R_y(\theta) = e^{-i\frac{\theta}{2}Y} \quad (15)$$

$$R_z(\theta) = e^{-i\frac{\theta}{2}Z} \quad (16)$$

Using the identity $\sigma^2 = I$, these can be expanded as:

$$R_\alpha(\theta) = \cos\left(\frac{\theta}{2}\right) I - i \sin\left(\frac{\theta}{2}\right) \sigma_\alpha \quad (17)$$

3.3.2.2 Pauli-X (NOT) Gate

The **Pauli-X Gate**, often called the **quantum NOT gate**, is one of the *fundamental* single-qubit quantum gates. It is denoted X and **corresponds to the σ_x Pauli matrix** in quantum mechanics. This gate **flips the state of a qubit, similar to how a classical NOT gate inverts a bit.**

Remark: Pauli Matrix

In quantum mechanics, the **Pauli matrices** are a set of three 2×2 **Hermitian and unitary matrices** that represent **spin operators** for spin- $\frac{1}{2}$ particles and are widely used in quantum computing to describe single-qubit gates. They are:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Key properties:

- Hermitian: $\sigma_i^\dagger = \sigma_i$ (observable-related)
- Unitary: $\sigma_i^\dagger \sigma_i = I$ (reversible transformations)
- Involution: $\sigma_i^2 = I$
- Non-commuting: $[\sigma_i, \sigma_j] = 2i\varepsilon_{ijk}\sigma_k$ (with ε the Levi-Civita symbol)

√ Matrix Representation

$$X = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (18)$$

This **matrix swaps** the $|0\rangle$ and $|1\rangle$ **basis states**:

$$X|0\rangle = |1\rangle \quad \text{and} \quad X|1\rangle = |0\rangle \quad (19)$$

≡ Action on a General Qubit

Let's consider a qubit in the general state:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

Applying the X gate:

$$X|\psi\rangle = aX|0\rangle + bX|1\rangle = a|1\rangle + b|0\rangle = b|0\rangle + a|1\rangle$$

As a result, the **amplitudes** a and b are **swapped**.

≡ Key Properties of X Gate

The Pauli-X gate satisfies several important mathematical properties:

1. **Unitary**. The X gate is unitary, meaning it preserves the inner product and is reversible:

$$XX^\dagger = I$$

2. **Hermitian**. The X gate is also Hermitian, which means it is equal to its own conjugate transpose:

$$X = X^\dagger$$

3. **Self-Inverse**. The X gate is its own inverse, which means that **applying it twice returns the qubit to its original state**:

$$X^{-1} = X \quad \Rightarrow \quad X^2 = I$$

🔍 Geometric Interpretation: Rotation on the Bloch Sphere

The Pauli-X gate corresponds to a **rotation around the x -axis by π radians** (180°) on the Bloch sphere, denoted as $X = R_x(\pi)$.

To understand this, recall the **general qubit state in spherical coordinates**:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

After a π rotation around the x -axis:

- $\theta \rightarrow \pi - \theta$
- $\phi \rightarrow -\phi$

This **rotation mirrors the qubit across the x -axis**, flipping its position between $|0\rangle$ and $|1\rangle$, and adjusting the phase accordingly.

⚙️ Rotation Interpretation

From Equation 13 (page 47) and Equation 17 (page 47), a rotation around the x -axis by an angle θ is given by:

$$R_x(\theta) = \cos\left(\frac{\theta}{2}\right)I - i\sin\left(\frac{\theta}{2}\right)X$$

For $\theta = \pi$, we obtain:

$$R_x(\pi) = -iX$$

Therefore, the Pauli-X gate corresponds to a **rotation of π around the x -axis, up to a global phase**:

$$X \equiv R_x(\pi)$$

Since global phases are physically irrelevant, X and $R_x(\pi)$ represent the same physical transformation.

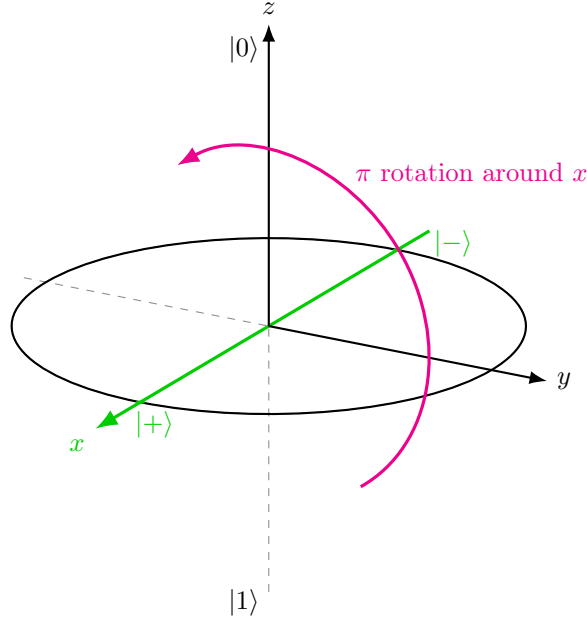


Figure 10: Eigenstates of the Pauli- X gate. Since $|+\rangle$ and $|-\rangle$ lie on the x -axis, they are not moved by a rotation around that axis. The state $|-\rangle$ only acquires a global phase -1 .

Eigenstates

In linear algebra, an **eigenvector** of a matrix (or linear operator) A is a non-zero vector $|v\rangle$ that, when the operator is applied, changes only by a **scalar factor** called the **eigenvalue** λ :

$$A|v\rangle = \lambda|v\rangle$$

In quantum mechanics, we use the term **Eigenstate** instead of eigenvector. So, an eigenstate of an operator U satisfies:

$$U|v\rangle = \lambda|v\rangle$$

An **eigenstate** is a state that remains **unchanged in direction** after the operation is applied, except only its **magnitude** or **phase** may change by a factor of λ . For the Pauli- X gate, solving the eigenvalue equation:

$$X|v\rangle = \lambda|v\rangle$$

Yields two eigenstates:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad \text{with eigenvalue} \quad +1 \Rightarrow X|+\rangle = +1 \cdot |+\rangle = |+\rangle \quad (20)$$

Called the **Plus State**, and:

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} \quad \text{with eigenvalue} \quad -1 \Rightarrow X|-\rangle = -1 \cdot |-\rangle = -|-\rangle \quad (21)$$

Called the **Minus State**. These states are also known as **Hadamard basis states** (page 62), **X-basis states**, or **Eigenstates of the Pauli-X operator**.

Example 2: Measurement in the X-basis

For example, if a qubit is in the state $|x\rangle$ and we measure it in the X-basis, the outcome is 1 with probability 1.

Proof. We want to show that if the qubit is in the state:

$$|x\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

And we measure it in the X-basis $\{|+\rangle, |-\rangle\}$, then the **outcome is +1 with probability 1**.

Measuring in the X-basis means using the eigenstates of the Pauli-X operator:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

These correspond to the new possible outcomes: +1 for $|+\rangle$ and -1 for $|-\rangle$.

If the system is in state $|\psi\rangle$, the probability of obtaining the outcome corresponding to basis state $|u\rangle$ is:

$$P(u) = |\langle u|\psi\rangle|^2$$

In our case, the system is in state $|\psi\rangle = |+\rangle$, so:

- Probability of outcome +1 (measuring $|+\rangle$):

$$P(+1) = |\langle +|+\rangle|^2$$

- Probability of outcome -1 (measuring $|-\rangle$):

$$P(-1) = |\langle -|+\rangle|^2$$

Since every normalized state has inner product 1 with itself, we have:

$$\langle +|+\rangle = 1$$

Therefore:

$$P(+1) = |\langle +|+\rangle|^2 = |1|^2 = 1$$

So the probability of getting +1 is 1.

If we calculate the probability of getting -1, first write the bras and kets:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Changing the ket $|+\rangle$ to a bra:

$$\langle -| = \frac{1}{\sqrt{2}}(\langle 0| - \langle 1|)$$

Now we can compute the inner product:

$$\begin{aligned}\langle -|+\rangle &= \frac{1}{\sqrt{2}} (\langle 0| - \langle 1|) \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \\&\downarrow \text{bring out the factors} \\&= \frac{1}{2} (\langle 0| - \langle 1|) (|0\rangle + |1\rangle) \\&\downarrow \text{expand the product} \\&= \frac{1}{2} (\langle 0|0\rangle + \langle 0|1\rangle - \langle 1|0\rangle - \langle 1|1\rangle) \\&\downarrow \text{use orthonormality } (\langle 0|0\rangle = \langle 1|1\rangle = 1, \langle 0|1\rangle = \langle 1|0\rangle = 0) \\&= \frac{1}{2} (1 + 0 - 0 - 1) \\&= 0\end{aligned}$$

Therefore:

$$P(-1) = |\langle -|+\rangle|^2 = |0|^2 = 0$$

So the probability of getting -1 is 0.

QED

3.3.2.3 Pauli-Z (Phase Flip) Gate

The **Pauli-Z Gate**, also called the **Phase Flip Gate**, is one of the core **Pauli operators** used in quantum computing. It acts **only on the phase** of the qubit, leaving the **$|0\rangle$ amplitude unchanged** but **flipping** the sign of the **$|1\rangle$ amplitude**.

√* Matrix Representation

$$Z = \sigma_Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (22)$$

≡ Action on a General Qubit

For a general state:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

Applying Z :

$$Z|\psi\rangle = aZ|0\rangle + bZ|1\rangle = a|0\rangle - b|1\rangle$$

As a result, Z **flips the sign of the $|1\rangle$ amplitude**, this is why it's called a **phase flip**. Finally, it's trivial to show that:

$$Z|0\rangle = |0\rangle \quad (\text{unchanged}), \quad Z|1\rangle = -|1\rangle \quad (\text{sign flip})$$

So the **magnitudes** of the amplitudes remain unchanged and the **probabilities of measurement outcomes** in the standard basis are unaffected, but the **relative phase** between $|0\rangle$ and $|1\rangle$ is altered (since b changes to $-b$).

Example 3: Effect on Superposition States

The Hadamard (or X-basis) states are:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

These two states differ **only by their relative phase**:

- $|+\rangle$: relative phase 0.
- $|-\rangle$: relative phase π .

Applying Z on $|+\rangle$:

$$\begin{aligned} Z|+\rangle &= Z \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \\ &= \frac{Z|0\rangle + Z|1\rangle}{\sqrt{2}} \\ &= \frac{|0\rangle - |1\rangle}{\sqrt{2}} \\ &= |-\rangle \end{aligned}$$

Applying Z on $|-\rangle$:

$$\begin{aligned} Z|-\rangle &= Z \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \\ &= \frac{Z|0\rangle - Z|1\rangle}{\sqrt{2}} \\ &= \frac{|0\rangle + |1\rangle}{\sqrt{2}} \\ &= |+\rangle \end{aligned}$$

The Pauli-Z gate **does not change the probabilities** of measuring $|0\rangle$ or $|1\rangle$ in the computational basis. However, it **changes the relative phase**, transforming $|+\rangle$ into $|-\rangle$ and vice versa.

🔍 Geometric Interpretation: Rotation around the z -axis

On the Bloch sphere, the Pauli-Z gate performs a π **radians** (180°) **rotation around the z -axis**. In spherical coordinates:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

After Z rotation:

- The angle $\phi \rightarrow \phi + \pi$
- This phase shift **changes the sign of the complex amplitude** for $|1\rangle$

Since **measurement probabilities depend on $|a|^2$ and $|b|^2$** , which **remain unchanged**, the Z gate **does not affect measurement outcomes** in the standard basis. It only affects **interference and future gate interactions**.

⚙️ Rotation Interpretation

From Equation 13 (page 47) and Equation 17 (page 47), a rotation around the z -axis by an angle θ is given by:

$$R_z(\theta) = \cos\left(\frac{\theta}{2}\right)I - i\sin\left(\frac{\theta}{2}\right)Z$$

For $\theta = \pi$, we obtain:

$$R_z(\pi) = -iZ$$

Therefore, the Pauli-Z gate corresponds to a **rotation of π around the z -axis, up to a global phase**:

$$Z \equiv R_z(\pi)$$

Since global phases are physically irrelevant, Z and $R_z(\pi)$ represent the same physical transformation.

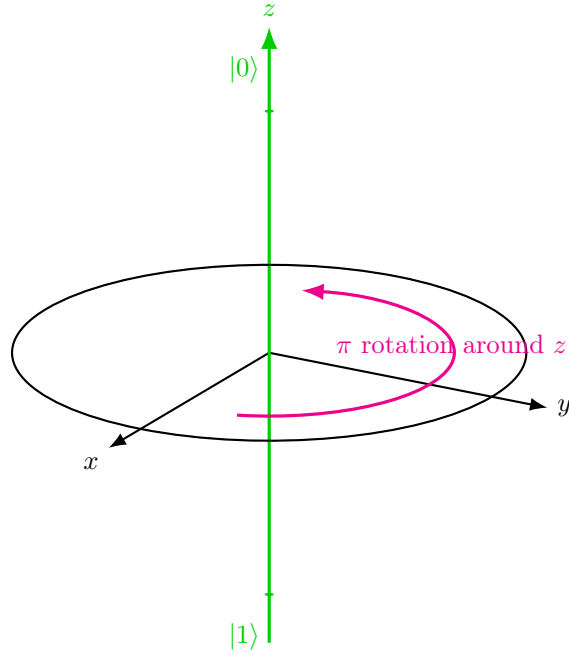


Figure 11: Eigenstates of the Pauli-Z gate. Since $|0\rangle$ and $|1\rangle$ lie on the z -axis, they are not moved by a rotation around that axis. The state $|1\rangle$ only acquires a global phase -1 .

⌵ Key Properties

1. **Unitary:** $Z^\dagger = Z^{-1} = Z$. Ensuring normalization is preserved.
2. **Hermitian:** $Z^\dagger = Z$. Meaning it is equal to its own conjugate transpose.
3. **Self-Inverse:** $Z^{-1} = Z \Rightarrow Z^2 = I$. Applying the Pauli-Z gate twice returns the original state.
4. **Eigenvalues and Eigenstates.** The Pauli-Z operator has *eigenvalues* $+1$ and -1 , with corresponding *eigenstates* $|0\rangle$ and $|1\rangle$:

$$Z|0\rangle = +1 \cdot |0\rangle, \quad Z|1\rangle = -1 \cdot |1\rangle$$

This means that $|0\rangle$ and $|1\rangle$ are **unchanged in direction** by the action of Z (up to a possible phase factor). In particular:

- $|0\rangle$ is completely unchanged.
- $|1\rangle$ acquires only a phase factor $-1 = e^{i\pi}$.

Therefore, these states are **invariant under the action of Z** (up to global phase), and represent the **natural states of the operator**.

💡 **Geometric Insight.** On the Bloch sphere, the Pauli-Z gate corresponds to a rotation around the z -axis. The eigenstates $|0\rangle$ and $|1\rangle$ lie exactly on this axis (north and south poles). As a consequence, they are **not affected by the rotation**, which explains why they are eigenstates of Z .

In contrast, any state that is not aligned with the z -axis (for example $|+\rangle$, Figure 9) is rotated by the action of Z . This highlights the difference between:

- **Eigenstates:** unchanged (up to phase)
- **Superposition states:** geometrically rotated

3.3.2.4 Pauli-Y Gate

The **Pauli-Y Gate**, denoted Y , is one of the three fundamental **Pauli matrices** in quantum mechanics. It is less intuitive than Pauli-X and Z, but equally important, especially for its role in complex rotations and quantum interference.

√* Matrix Representation

$$Y = \sigma_Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

≡ Action on a General Qubit

At first glance, this looks similar to the Pauli-X gate because it exchanges $|0\rangle$ and $|1\rangle$. But there is something extra: X swaps the two basis states (i.e., $|0\rangle \leftrightarrow |1\rangle$), while Y swaps them **and** multiplies them by a phases of $\pm i$. So Pauli-Y is not just a bit flip (like Pauli-X), it is a bit flip together with a phase modification.

Given a qubit in state:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

Apply the Y gate:

$$Y|\psi\rangle = aY|0\rangle + bY|1\rangle = a(-i|1\rangle) + b(i|0\rangle) = ib|0\rangle - ia|1\rangle$$

So the **amplitudes are exchanged** (as with Pauli-X), but each one also acquires a **phase factor of i or $-i$** .

🔍 Geometric Interpretation: Rotation on the Bloch Sphere

The Pauli-Y gate corresponds to a **rotation around the y -axis by π radians** (180°). The effect on Bloch sphere coordinates:

- $\theta \rightarrow \pi - \theta$
- $\psi \rightarrow \pi - \psi$

This rotates the qubit across the y -axis, **altering both amplitudes and their phases**.

⚙️ Rotation Interpretation

From Equation 13 (page 47) and Equation 17 (page 47), a rotation around the y -axis by an angle θ is given by:

$$R_y(\theta) = \cos\left(\frac{\theta}{2}\right) I - i \sin\left(\frac{\theta}{2}\right) Y$$

For $\theta = \pi$, we obtain:

$$R_y(\pi) = -iY$$

Therefore, the Pauli-Y gate corresponds to a **rotation of π around the y -axis, up to a global phase**:

$$Y \equiv R_y(\pi)$$

Since global phases are physically irrelevant, Y and $R_y(\pi)$ represent the same physical transformation.

☐ Pauli-Y as a Composite Gate

A very useful identity is:

$$Y = iXZ \quad (23)$$

This means that Pauli-Y can be understood as the combination of:

1. A Pauli-Z gate (phase flip);
2. Followed by a Pauli-X gate (bit flip);
3. Up to a global phase factor of i .

$$XZ = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

Now, multiply by i :

$$iXZ = i \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = Y$$

This identity confirms that **Pauli-Y can be viewed as a combination of Pauli-X and Pauli-Z with a global phase factor i** , which does not affect measurement outcomes.

☐ Eigenstates of Y Gate

To find the eigenstates, we solve:

$$Y |v\rangle = \lambda |v\rangle$$

Where λ is the eigenvalue. The result is:

$$|i\rangle = \frac{1}{\sqrt{2}} (|0\rangle + i|1\rangle) \quad (24)$$

$$|-i\rangle = \frac{1}{\sqrt{2}} (|0\rangle - i|1\rangle) \quad (25)$$

With eigenvalues:

$$Y |i\rangle = +1 |i\rangle \quad Y |-i\rangle = -1 |-i\rangle \quad (26)$$

⚡ Key Properties of Y Gate

1. **Unitary**: $Y^\dagger Y = I$. So it is a valid quantum gate that preserves normalization.
2. **Hermitian**: $Y^\dagger = Y$. So it is also an observable, meaning it corresponds to a measurable quantity in quantum mechanics.
3. **Self-inverse**: $Y^2 = I$. Applying the Y gate twice returns the qubit to its original state.

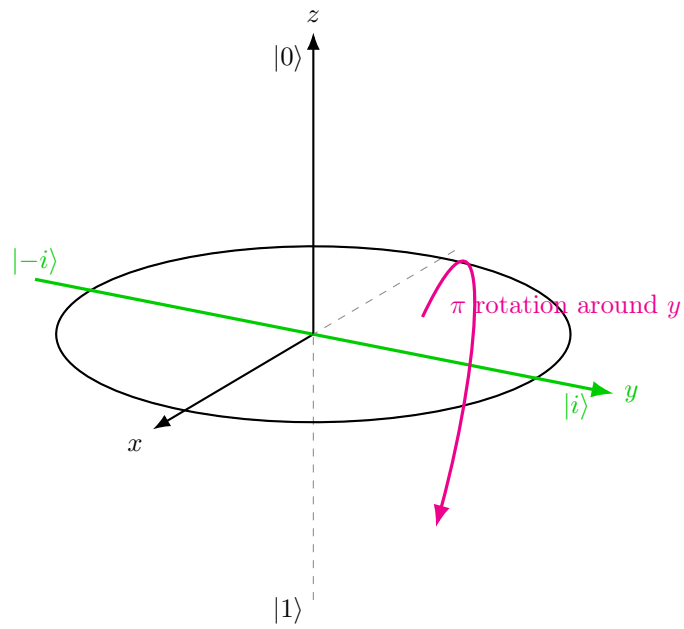


Figure 12: Eigenstates of the Pauli-Y gate. Since $|i\rangle$ and $|-i\rangle$ lie on the y -axis, they are not moved by a rotation around that axis. The state $|-i\rangle$ only acquires a global phase -1 .

3.3.3 Phase Gate (S)

The **Phase Gate (S)** is a single-qubit gate that **adds a phase shift of $\frac{\pi}{2}$ (90°) to the amplitude of the $|1\rangle$ state**. It **leaves $|0\rangle$ unchanged**, like the Pauli-Z gate, but instead of a sign flip (π phase), it **introduces a complex phase factor i** .

√ Matrix Representation

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \quad (27)$$

📖 Action on a General Qubit

For a state:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

Apply S :

$$S|\psi\rangle = a|0\rangle + ib|1\rangle$$

Only the $|1\rangle$ **amplitude gains a phase of i** , affecting interference but **not measurement probabilities** in the computational basis.

$$S|0\rangle = |0\rangle \quad (\text{unchanged}), \quad S|1\rangle = i|1\rangle \quad \left(\text{phase} + \frac{\pi}{2}\right)$$

🌀 Geometric Interpretation: Rotation around z -axis ($\frac{\pi}{2}$)

On the Bloch sphere, S performs a rotation of $\frac{\pi}{2}$ radians around the z -axis. This rotation **shifts the phase angle $\phi \rightarrow \phi + \frac{\pi}{2}$** , altering **how the qubit interferes** in later operations, especially when **Hadamard gates** are involved. In Figure 13, we see how S transforms the state $|+\rangle$ (on the x -axis) into $|i\rangle$ (on the y -axis) through a $\frac{\pi}{2}$ rotation around the z -axis, demonstrating the phase shift's effect on the qubit's state.

↔ Relationship to Pauli-Z

S is often called the **square root of Z** because applying it twice yields the Pauli-Z gate: $S = \sqrt{Z}$. Pauli-Z is a **full phase flip**:

$$e^{i\pi} = -1$$

The S gate is a **partial phase shift**:

$$e^{i\frac{\pi}{2}} = i$$

So:

$$Z = e^{i\pi \cdot |1\rangle\langle 1|} \quad \text{and} \quad S = e^{i\frac{\pi}{2} \cdot |1\rangle\langle 1|}$$

Same transformation, but different phase angles. Let's verify $S^2 = Z$:

$$S \cdot S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & i^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = Z$$

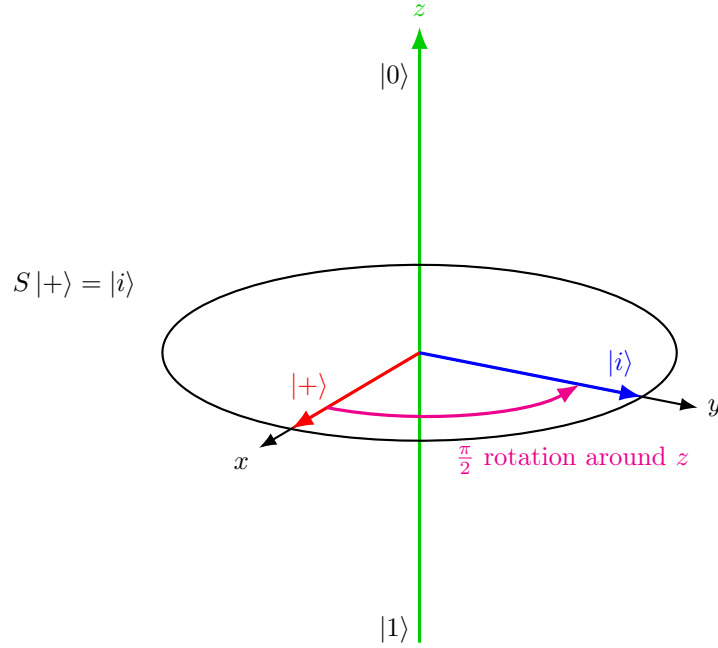


Figure 13: Geometric interpretation of the Phase gate S . The gate performs a rotation of $\frac{\pi}{2}$ around the z -axis, transforming $|+\rangle$ into $|i\rangle$.

Eigenvalues and Eigenstates

The Phase gate S is diagonal in the computational basis, so its eigenstates are $|0\rangle$ and $|1\rangle$:

$$S|0\rangle = 1 \cdot |0\rangle, \quad S|1\rangle = i \cdot |1\rangle$$

Thus, the eigenvalues of S are:

$$\lambda_0 = 1, \quad \lambda_1 = i$$

This means that $|0\rangle$ remains unchanged, while $|1\rangle$ acquires a phase factor $i = e^{i\frac{\pi}{2}}$.

Key Properties

1. **Unitary:** $S^\dagger = S^{-1}$, so normalization is preserved.
2. **Not Hermitian:** $S^\dagger \neq S$, therefore S is not an observable, it is only a transformation.
3. **Inverse:**

$$S^{-1} = S^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix}$$

4. **Relation with Pauli-Z:** $S^2 = Z$, meaning that S is the square root of the Pauli-Z gate.
5. **Periodicity:** $S^4 = I$, since $i^4 = 1$.
6. **Phase Shift:** S applies a relative phase of $\frac{\pi}{2}$ to the $|1\rangle$ component.

3.3.4 Hadamard Gate (H)

The **Hadamard Gate (H)** changes the basis representation of the qubit. More concretely:

$$H : \{|0\rangle, |1\rangle\} \longleftrightarrow \{|+\rangle, |-\rangle\}$$

It maps computational basis states into equal superpositions and vice versa, allowing phase information to be transformed into measurable probabilities.

√* Matrix Representation

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (28)$$

≡ Action on Basis States

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle \quad (29)$$

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle \quad (30)$$

The $|+\rangle$ and $|-\rangle$ states are known as the **Hadamard basis**, or **superposition states**, with **equal probability amplitudes** for $|0\rangle$ and $|1\rangle$.

≡ Superposition and Measurement

After applying H to $|0\rangle$, the resulting qubit is:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

This state is a **superposition** of $|0\rangle$ and $|1\rangle$, meaning that the qubit does not have a definite classical value prior to measurement. Instead, it is described by **probability amplitudes** associated with each basis state.

Upon measurement in the computational basis, the state collapses probabilistically:

• **Probability of 0:**

• **Probability of 1:**

$$\left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} \qquad \frac{1}{2}$$

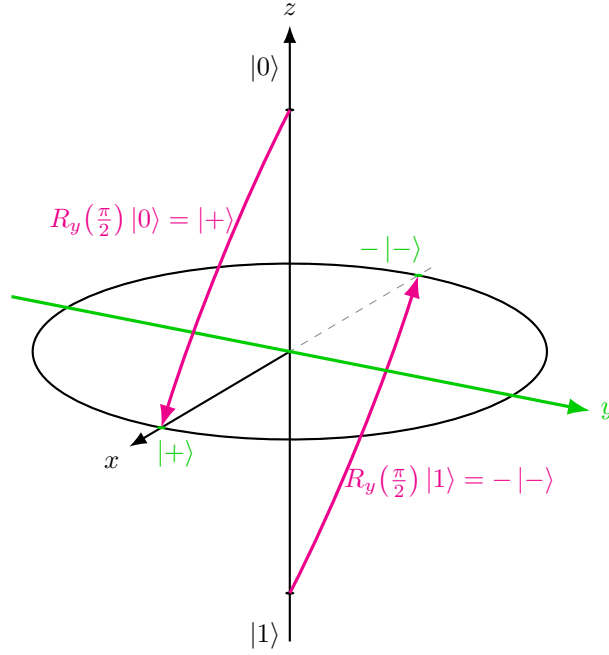
After the measurement:

- The qubit collapses to $|0\rangle$ with probability $\frac{1}{2}$, or to $|1\rangle$ with probability $\frac{1}{2}$.
- The original superposition is **destroyed**, and cannot be recovered.

🔍 Bloch Sphere Interpretation

On the Bloch sphere, the **Hadamard gate** performs two sequential rotations:

1. Rotation around the y -axis by $\frac{\pi}{2}$ radians (90°): $R_y\left(\frac{\pi}{2}\right)$.



The y -axis stays fixed. So the states on the y -axis $|i\rangle$ and $|-i\rangle$ do not move. Instead, the z -axis rotated toward the x -axis: $|0\rangle \rightarrow |+\rangle$ and $|1\rangle \rightarrow -|-\rangle$. So the first step already moves the computational basis from the z -axis to the x axis.

We can verify the matrix representation of $R_y\left(\frac{\pi}{2}\right)$ by using the exponential of the Pauli Y matrix (see page 47):

$$\begin{aligned}
 R_y(\theta) &= e^{-i\frac{\theta}{2}Y} \\
 &= \cos\left(\frac{\theta}{2}\right) I - i \sin\left(\frac{\theta}{2}\right) Y \\
 &= \cos\left(\frac{\theta}{2}\right) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - i \sin\left(\frac{\theta}{2}\right) \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \\
 &= \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix}
 \end{aligned}$$

Now, substituting $\theta = \frac{\pi}{2}$ gives us the expected matrix for the first step of the Hadamard decomposition:

$$R_y\left(\frac{\pi}{2}\right) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

- Apply it to $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$:

$$\begin{aligned}
 R_y\left(\frac{\pi}{2}\right)|0\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\
 &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\
 &= \frac{1}{\sqrt{2}} \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \\
 &= \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \\
 &= \frac{|0\rangle + |1\rangle}{\sqrt{2}} \\
 &= |+\rangle
 \end{aligned}$$

So $|0\rangle$, the north pole of the Bloch sphere, is rotated to the $+x$ -axis, which is the $|+\rangle$ state.

- Apply it to $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$:

$$\begin{aligned}
 R_y\left(\frac{\pi}{2}\right)|1\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\
 &= \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix} \\
 &= \frac{1}{\sqrt{2}} \left(\begin{pmatrix} 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right) \\
 &= \frac{1}{\sqrt{2}} (|1\rangle - |0\rangle) \\
 &= -\frac{|0\rangle - |1\rangle}{\sqrt{2}} \\
 &= -|-\rangle
 \end{aligned}$$

This means that $|1\rangle$, the south pole of the Bloch sphere, moves to the $-x$ -axis, but with a global phase of -1 . Since global phases do not affect measurement outcomes, we can ignore it and say that $|1\rangle$ is effectively rotated to the $-x$ -axis, which is the $|-\rangle$ state. In simple terms: $-|-\rangle \equiv |-\rangle$ up to a global phase.

2. **Rotation around the x -axis by π radians (180°).** This step is represented by the operator $R_x(\pi)$, which can be expressed as:

$$R_x(\pi) = e^{-i\frac{\pi}{2}X} = \cos\left(\frac{\pi}{2}\right)I - i\sin\left(\frac{\pi}{2}\right)X = -iX$$

Applying $R_x(\pi)$ to the states on the x -axis ($|+\rangle$ and $|-\rangle$) does not change their position on the Bloch sphere, but it does introduce a phase factor

(which becomes a global phase only when applied to the entire state):

$$\begin{aligned} R_x(\pi) |+\rangle &= -iX |+\rangle = -i |+\rangle \\ R_x(\pi) |-\rangle &= -iX |-\rangle = -i(-|-\rangle) = i |-\rangle \end{aligned}$$

Applying $R_x(\pi)$ to the superposition state $a|+\rangle - b|-\rangle$ (obtained from the first rotation) gives us:

$$\begin{aligned} &a|+\rangle - b|-\rangle \\ R_x(\pi)(a|+\rangle - b|-\rangle) &= a(-i|+\rangle) - b(i|-\rangle) \\ &= -ia|+\rangle - ib|-\rangle \\ &= -i(a|+\rangle + b|-\rangle) \end{aligned}$$

Since global phases $-i$ do not affect measurement outcomes, we can ignore them and say that $|+\rangle$ and $|-\rangle$ remain on the x -axis. The second rotation does not change the position of the states on the Bloch sphere, but it is necessary to ensure that the transformation matches the Hadamard gate at the operator level, preserving the correct relative phase for all superpositions.

❓ Why is the second rotation necessary? We **cannot** simply change $-|-\rangle$ to $|-\rangle$ inside a superposition, because that would change only one component and therefore change the **relative phase**. The fix comes from applying $R_x(\pi)$. After the first rotation, we have:

$$a|+\rangle - b|-\rangle$$

Now, applying the Pauli-X operator using the rotation $R_x(\pi)$ (page 47):

$$R_x(\pi) = -iX = -i \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{with} \quad X|+\rangle = |+\rangle, \quad X|-\rangle = -|-\rangle$$

Applying $R_x(\pi)$ to the superposition gives us:

$$\begin{aligned} R_x(\pi)(a|+\rangle - b|-\rangle) &= -iX(a|+\rangle - b|-\rangle) \\ &= -i(aX|+\rangle - bX|-\rangle) \\ &= -i(a|+\rangle - b(-|-\rangle)) \\ &= -i(a|+\rangle + b|-\rangle) \end{aligned}$$

Now the factor $-i$ multiplies the **entire state**, so it is a global phase:

$$-i(a|+\rangle + b|-\rangle) \equiv a|+\rangle + b|-\rangle$$

So the important point is:

$$a|+\rangle - b|-\rangle \not\equiv a|+\rangle + b|-\rangle$$

But:

$$R_x(\pi)(a|+\rangle - b|-\rangle) = -i(a|+\rangle + b|-\rangle) \equiv a|+\rangle + b|-\rangle$$

💡 Insight. The second rotation **does not affect the geometric position** of the states, but **corrects the relative phase** structure, ensuring that the transformation is a valid unitary operator acting consistently on the entire Hilbert space.

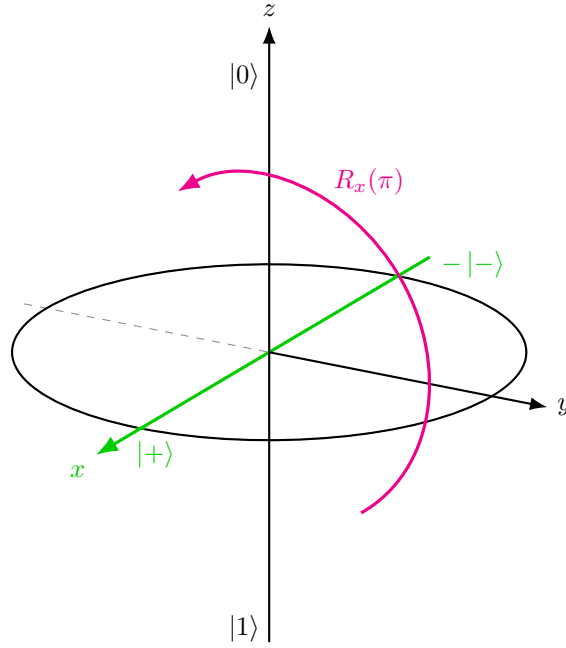


Figure 14: Second step of the Hadamard decomposition: rotation by π around the x -axis. States on the x -axis are not moved on the Bloch sphere; only global phases are introduced.

This brings $|0\rangle$ to the $+x$ -axis ($|+\rangle$) and $|1\rangle$ to the $-x$ -axis ($|-\rangle$). As a result, **qubit moves from pole to equator, entering maximum superposition.**

Eigenvalues and Eigenstates

The Hadamard operator has eigenvalues:

$$\lambda = +1, \quad \lambda = -1$$

with corresponding eigenstates given by the solutions of:

$$H|v\rangle = \lambda|v\rangle$$

These eigenstates are superpositions of $|0\rangle$ and $|1\rangle$, and are aligned along an axis halfway between the x and z directions on the Bloch sphere. Precisely, the eigenvectors are:

$$\begin{aligned} |v_+\rangle &= \cos\left(\frac{\pi}{8}\right)|0\rangle + \sin\left(\frac{\pi}{8}\right)|1\rangle \\ |v_-\rangle &= \sin\left(\frac{\pi}{8}\right)|0\rangle - \cos\left(\frac{\pi}{8}\right)|1\rangle \end{aligned}$$

Where $|v_+\rangle$ corresponds to the eigenvalue $+1$ and $|v_-\rangle$ corresponds to the eigenvalue -1 ; the cosine and sine factors ensure normalization of the eigenstates.

Unlike Pauli gates, the Hadamard eigenstates lie along a **diagonal axis between the computational basis z and the Hadamard basis x** , reflecting its role as a **basis transformation** rather than a simple bit or phase flip.

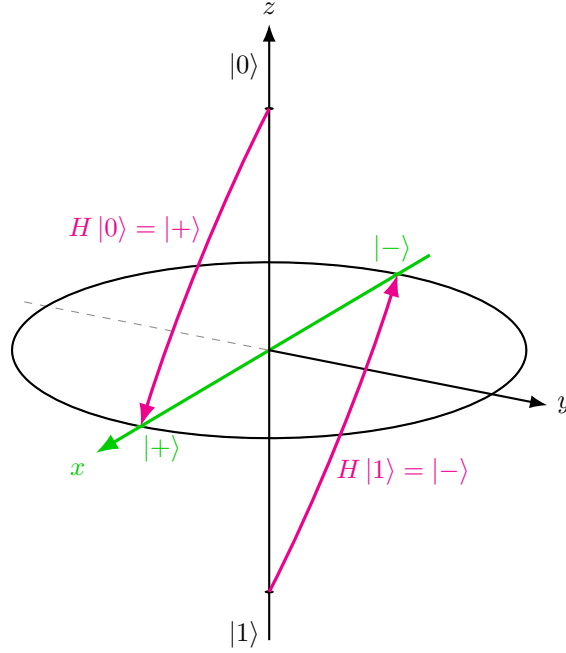


Figure 15: Bloch sphere interpretation of the Hadamard gate. The gate maps $|0\rangle$ to $|+\rangle$ and $|1\rangle$ to $|-\rangle$, moving computational basis states from the z -axis to the x -axis.

⌄ Key Properties

1. **Unitary:** $H^\dagger H = I$, so normalization is preserved.
2. **Hermitian:** $H^\dagger = H$, therefore it corresponds to an observable.
3. **Self-inverse:** $H^2 = I$. Applying the Hadamard gate twice returns the original state.
4. **Basis transformation:** H maps the computational basis $\{|0\rangle, |1\rangle\}$ to the Hadamard basis $\{|+\rangle, |-\rangle\}$:

$$H|0\rangle = |+\rangle, \quad H|1\rangle = |-\rangle$$

5. **Interference enabler:** H converts phase differences into amplitude differences, making them observable through measurement.
6. **Axis mixing:** H transforms Pauli operators as:

$$HZH = X, \quad HXH = Z$$

3.4 Properties

Single-qubit gates exhibit a rich set of mathematical and geometric properties, all of which stem from their nature as unitary transformations on a two-level quantum system. Understanding these properties is key to **predicting gate behavior**, **designing quantum circuits**, and **analyzing quantum algorithms**.

1. All Quantum Gates Are Equivalent to Rotations

A profound insight in quantum computing is that **every single-qubit unitary gate corresponds to a rotation of the qubit's state vector on the Bloch sphere**.

- **Bloch Sphere** is a geometric representation of a qubit's state, where any point on the sphere represents a valid pure state.
- A **unitary operation** on qubit results in **rotating the vector** on this sphere **without changing its length** (preserving probability).

Some examples of rotations:

- **X Gate** → Rotation π around x-axis (section 3.3.2.2, page 48)
- **Z Gate** → Rotation π around z-axis (section 3.3.2.3, page 53)
- **Y Gate** → Rotation π around y-axis (section 3.3.2.4, page 57)
- **S Gate** → Rotation $\frac{\pi}{2}$ around z-axis (section 3.3.3, page 60)
- **H Gate** → Rotation $\frac{\pi}{2}$ around y, then π around x (section 3.3.4, page 62)

🔗 Why is this important? Quantum computation is fundamentally about **state rotations**, understanding gates as rotations helps visualize **interference**, **phase shifts**, and **entanglement creation**.

2. Reversibility: Applying a Gate Twice

Many single-qubit gates, especially the Pauli gates, exhibit the property of **involution**: applying the same gate twice **returns the qubit to its original state**. Mathematically it can be expressed as:

$$X^2 = Y^2 = Z^2 = I$$

This is due to the fact that **two 180° rotations around the same axis bring the vector back** to its original position on the Bloch sphere.

Proof that X Gate has involution.

$$X^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

These identities confirm that **Pauli gates are their own inverses**, and **all unitary gates are reversible**. QED

3. Global Phases Do Not Matter

Quantum gates may **introduce a global phase** (a constant complex factor $e^{i\phi}$) to a state. Physically, **global phases are unobservable**, meaning they do **not affect measurement probabilities**. For example:

- States $|\psi\rangle$ and $e^{i\phi}|\psi\rangle$ are physically indistinguishable.
- The Y gate = iXZ , where i is a global phase that can be ignored in practice.

4. Composite Gates and Commutativity

- Gates can be **combined by matrix multiplication** (note: non commutative in general).
- **Order matters**: $AB \neq BA$ for most gates.
- Exception: **Gate Z and Gate S** are **commuted** because they are both **diagonal phase gates**.

Property	Pauli Gates (X, Y, Z)	Phase Gate (S)	Hadamard (H)
Rotation Axis	x, y, z (π radians)	z ($\frac{\pi}{2}$ radians)	y ($\frac{\pi}{2}$) + x (π)
Involution $G^2 = I$	✓	✗ ($S^2 = Z$)	✓
Self-adjoint $G = G^\dagger$	✓	✗	✓
Unitary	✓	✓	✓
Phase Introduction	Z, S	i for $ 1\rangle$	✓ (interference)
Creates Superposition?	✗ (X, Y, Z alone)	✗	✓

Table 2: Summary of Properties.

3.5 When Does a Gate Create Superposition?

In this section, we will answer the question: *when does a quantum gate create superposition?* More precisely, if we start from a **basis state** $|0\rangle$ or $|1\rangle$, when does a gate produce:

$$a|0\rangle + b|1\rangle \quad \text{with } a, b \neq 0?$$

? What does “*create superposition*” really mean? It means that the output must contain **both basis states**. So:

- $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is not a superposition, it's just $|0\rangle$.
- $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ is not a superposition, it's just $|1\rangle$.
- $\begin{pmatrix} a \\ b \end{pmatrix}$ with $a, b \neq 0$ is a superposition, because it contains both $|0\rangle$ and $|1\rangle$ components. Indeed, we can write it as:

$$a|0\rangle + b|1\rangle = a \begin{pmatrix} 1 \\ 0 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}$$

? How does a matrix act on a basis state?

The main idea is that **each column of the matrix tells us where a basis vector goes**. For example, for a matrix:

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

The first column $\begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix}$ tells us where $|0\rangle$ goes, and the second column $\begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix}$ tells us where $|1\rangle$ goes. In general, a **gate creates superposition if one of its columns has two non-zero entries**.

⚠ Constraint: Unitarity

Not every matrix can be a valid quantum gate. Quantum gates must be **unitary**, meaning:

$$A^\dagger A = I$$

This imposes strong constraints on the matrix entries. In particular, the columns must be:

- **Normalized**, total probability is 1: $\|\psi\|^2 = |a|^2 + |b|^2 = 1$
- **Orthogonal**, inner product is zero: $\langle v|w\rangle = 0$

As a consequence, some matrices that appear to create superposition are **not physically allowed**, because they are not unitary.

💡 Key Insight

Superposition arises from the **mixing of basis states**. Therefore:

- A gate **creates superposition** if at least one of its columns contains **two non-zero entries**.
- A gate **cannot create superposition** if its columns contain only one non-zero entry (as in diagonal or triangular matrices).

In particular:

- Gates like H (Hadamard) create superposition because they mix $|0\rangle$ and $|1\rangle$.
- Gates like X , Y , Z or S do not create superposition because they only modify phases and do not mix basis states.

Proof that A cannot create superposition. Consider the gate:

$$A = \begin{pmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{pmatrix}$$

To create superposition from a basis state, at least one column of A must contain two non-zero entries.

Applying A to $|0\rangle$ gives:

$$A|0\rangle = \begin{pmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a_{11} \cdot 1 + a_{12} \cdot 0 \\ 0 \cdot 1 + a_{22} \cdot 0 \end{pmatrix} = \begin{pmatrix} a_{11} \\ 0 \end{pmatrix} = a_{11} |0\rangle$$

Therefore, $|0\rangle$ is not mapped to a superposition because the second entry is zero.

Applying A to $|1\rangle$ gives:

$$A|1\rangle = \begin{pmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} a_{11} \cdot 0 + a_{12} \cdot 1 \\ 0 \cdot 0 + a_{22} \cdot 1 \end{pmatrix} = \begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix} = a_{12} |0\rangle + a_{22} |1\rangle$$

This would be a superposition only if both $a_{12} \neq 0$ and $a_{22} \neq 0$.

However, since A is a quantum gate, it must be unitary (property of quantum gates):

$$A^\dagger A = I$$

Assuming A is real, we have $A^\dagger = A^T$, so:

$$A^T = \begin{pmatrix} a_{11} & 0 \\ a_{12} & a_{22} \end{pmatrix}$$

Hence:

$$A^T A = \begin{pmatrix} a_{11} & 0 \\ a_{12} & a_{22} \end{pmatrix} \cdot \begin{pmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{pmatrix} = \begin{pmatrix} a_{11}^2 & a_{11}a_{12} \\ a_{11}a_{12} & a_{12}^2 + a_{22}^2 \end{pmatrix}$$

Since A is unitary (i.e., $A^\dagger A = I$), this matrix must equal the identity:

$$\begin{pmatrix} a_{11}^2 & a_{11}a_{12} \\ a_{11}a_{12} & a_{12}^2 + a_{22}^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Therefore, to satisfy this equality, the following equations must hold:

$$\begin{cases} a_{11}^2 = 1 \\ a_{11}a_{12} = 0 \\ a_{12}^2 + a_{22}^2 = 1 \end{cases}$$

From $a_{11}^2 = 1$, we get $a_{11} = \pm 1$, so $a_{11} \neq 0$. Therefore, from $a_{11}a_{12} = 0$, it follows that:

$$a_{12} = 0$$

Substituting this into the last equation gives:

$$a_{22}^2 = 1 \quad \Rightarrow \quad a_{22} = \pm 1$$

Hence the matrix becomes diagonal:

$$A = \begin{pmatrix} \pm 1 & 0 \\ 0 & \pm 1 \end{pmatrix}$$

Since A is diagonal, it only modifies the phases of $|0\rangle$ and $|1\rangle$, without mixing them. Thus, when we apply A to $|0\rangle$ and $|1\rangle$, we get:

$$A|0\rangle = \pm|0\rangle, \quad A|1\rangle = \pm|1\rangle$$

So neither $|0\rangle$ nor $|1\rangle$ is mapped to a superposition, because the output states are just $|0\rangle$ or $|1\rangle$ (up to a global phase). Thus, A cannot create superposition (i.e., it cannot produce a state with both $|0\rangle$ and $|1\rangle$ components). QED

3.6 Single-Qubit Quantum Circuits

A **Quantum Circuit** is a **mathematical model** for quantum computation, **composed of sequential operations applied to qubits**. In other words, a quantum circuit is a **model of computation** that describes **how a quantum state evolves step by step**. The operations in a quantum circuit include initializations, unitary gates, and measurements. In this section, we focus on **circuits involving a single qubit**, which form the foundation for understanding multi-qubit systems.

≡ Circuit Elements

A quantum circuit typically consists of three main elements:

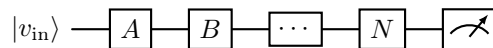
1. **Initialization** (initial state)
 - Qubit is prepared in a **known state**, typically $|0\rangle$ or $|1\rangle$.
 - Essential **starting point** for any computation.
2. **Gates**
 - Single-qubit **unitary operations** (e.g., X, Z, H, S).
 - Transform the qubit's state **reversibly**.
3. **Measurement**
 - **Irreversible process**.
 - Collapses the qubit to $|0\rangle$ or $|1\rangle$ with probabilities dictated by the quantum state's amplitudes.

✂ Quantum Circuit Diagram: Visual Language

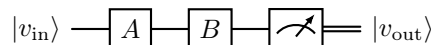
- **Horizontal lines** represent the **timeline of a single qubit**.

$$|v_{\text{in}}\rangle \text{ ————— } |v_{\text{out}}\rangle$$

- **Boxes** on the line represent **gates** (letters) or **measurements** (tachometer icon).



- **Double lines** (if present) represent **classical information** (e.g., measurement result).



- ⚠ Left to right:** The circuit evolves over time from left (input) to right (output).

Gate Sequence: Serial Gates

Consider two gates A and B applied in sequence:

$$|v_{\text{in}}\rangle \longrightarrow \boxed{A} \longrightarrow \boxed{B} \longrightarrow |v_{\text{out}}\rangle$$

1. First apply A, then B.
2. **⚠ Attention:** the **order of matrix multiplication is reversed** when we write the combined effect of the gates as a single matrix. The gate that is applied **last** (B) appears **first** in the product.

$$v_{\text{out}} = BAv_{\text{in}}$$

💡 Why? Because first apply A to the input state v_{in} , then apply B to the result of Av_{in} .

3. The **combined effect** of the two gates is equivalent to a single gate $C = BA$:

$$|v_{\text{in}}\rangle \longrightarrow \boxed{BA} \longrightarrow |v_{\text{out}}\rangle \quad \equiv \quad |v_{\text{in}}\rangle \longrightarrow \boxed{A} \longrightarrow \boxed{B} \longrightarrow |v_{\text{out}}\rangle$$

$$v_{\text{out}} = Cv_{\text{in}} = BAv_{\text{in}}$$

Obviously, if we merge the gates into a single gate C , we must respect the order of operations, which is why $C = BA$ and not AB .

Example 4: Hadamard followed by Hadamard

Let's compute H followed by H to $|0\rangle$:

$$|0\rangle \longrightarrow \boxed{H} \longrightarrow \boxed{H} \longrightarrow |v_{\text{out}}\rangle$$

1. Apply H:

$$\begin{aligned} H|0\rangle &= \frac{1}{\sqrt{2}} \cdot \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} \cdot \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \\ &= \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) = |+\rangle \end{aligned}$$

2. Applying H again, we **restore the original state due to the involution properties** ($H^2 = I$):

$$H(H|0\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$$

3.7 Outer Product of Kets

In quantum mechanics, **Bra-Ket notation** (introduced by Dirac) is a concise and powerful tool for representing quantum states and operations. It uses **kets** $|\psi\rangle$ for **vectors** (states), and **bras** $\langle\psi|$ for their **Hermitian adjoints** (dual vectors).

Given two kets:

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \quad |b\rangle = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$

The **Outer Product** $|a\rangle\langle b|$ is a **matrix**, formed by:

$$|a\rangle\langle b| = |a\rangle \otimes \langle b| = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \begin{pmatrix} \overline{b_1} & \overline{b_2} \end{pmatrix} = \begin{pmatrix} a_1\overline{b_1} & a_1\overline{b_2} \\ a_2\overline{b_1} & a_2\overline{b_2} \end{pmatrix} \quad (31)$$

Where the symbol $\otimes$ is the **tensor product operator**. This operation **maps a vector to a matrix**.

❓ Why is the outer product useful? Because **all quantum gates can be written using outer products of kets**. For example, the **Pauli-X gate** can be expressed as:

$$X = |0\rangle\langle 1| + |1\rangle\langle 0| = \underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix}}_{\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}} + \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix}}_{\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

The **Inner Product** $\langle b|a\rangle$ is a scalar:

$$\langle a|b\rangle = \begin{pmatrix} \overline{a_1} & \overline{a_2} \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \overline{a_1}b_1 + \overline{a_2}b_2 \quad (32)$$

So:

- **Inner product** $\rightarrow$ **projection**, result is number (scalar, see also page 9).
- **Outer product** $\rightarrow$ **matrix**, result is operator.

≡ Associativity between outer and inner products

A key property of the outer product is how it interacts with inner products. Specifically:

$$(|a\rangle\langle b|)|c\rangle = |a\rangle(\langle b|c\rangle) \quad (33)$$

This identity shows that the outer product acts as an operator: it first computes the inner product $\langle b|c\rangle$ (a scalar), and then scales the vector $|a\rangle$.

Proof the Associativity. In this proof, we will demonstrate that applying the outer product $|a\rangle\langle b|$ to a ket $|c\rangle$ is equivalent to scaling $|a\rangle$ by the inner product $\langle b|c\rangle$.

Let's define:

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \quad |b\rangle = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \quad |c\rangle = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

1. Compute **inner product** $\langle b|c\rangle$:

$$\langle b|c\rangle = \overline{b_1}c_1 + \overline{b_2}c_2 = \alpha$$

Where α is a **scalar** value.

2. Multiply **scalar** α with $|a\rangle$ (vector):

$$|a\rangle \alpha = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \alpha = \begin{pmatrix} a_1 \cdot \alpha \\ a_2 \cdot \alpha \end{pmatrix} = \begin{pmatrix} a_1 \cdot (\overline{b_1}c_1 + \overline{b_2}c_2) \\ a_2 \cdot (\overline{b_1}c_1 + \overline{b_2}c_2) \end{pmatrix}$$

3. Compute outer product $|a\rangle\langle b|$ (matrix):

$$|a\rangle\langle b| = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \begin{pmatrix} \overline{b_1} & \overline{b_2} \end{pmatrix} = \begin{pmatrix} a_1\overline{b_1} & a_1\overline{b_2} \\ a_2\overline{b_1} & a_2\overline{b_2} \end{pmatrix}$$

4. Multiply $|a\rangle\langle b|$ by $|c\rangle$:

$$(|a\rangle\langle b|)|c\rangle = \begin{pmatrix} a_1\overline{b_1} & a_1\overline{b_2} \\ a_2\overline{b_1} & a_2\overline{b_2} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} a_1 \cdot (\overline{b_1}c_1 + \overline{b_2}c_2) \\ a_2 \cdot (\overline{b_1}c_1 + \overline{b_2}c_2) \end{pmatrix}$$

The second and fourth steps yield the same result. This proves that **outer product applied to a ket is equivalent to inner product $\langle b|c\rangle$ scaled by $|a\rangle$** . QED

3.8 Measurement

In quantum computing, **measurement** is a special type of operation that differs fundamentally from unitary gates. While gates perform reversible transformations, **measurement is irreversible**. It allows us to **gain information about the state of a qubit**, but at the price of **collapsing the quantum state**.

√* Measurement Operator: Matrix Form

A **measurement operator** is a **non-unitary, non-invertible** matrix. For a measurement **along a specific direction**, represented by a ket $|k\rangle$, the associated **projector** is (outer product):

$$M_k = |k\rangle\langle k| = \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} \begin{pmatrix} \overline{k_1} & \overline{k_2} \end{pmatrix} = \begin{pmatrix} k_1\overline{k_1} & k_1\overline{k_2} \\ k_2\overline{k_1} & k_2\overline{k_2} \end{pmatrix} \quad (34)$$

This is called a **Projection Operator**, it projects any vector **onto the direction of** $|k\rangle$.

Definition 1: Projection Operator

A **Projection Operator** (or simply **Projector**) is a linear operator that **extracts the component of a quantum state along a specific direction** (or subspace).

For a normalized state $|k\rangle$, the projector onto $|k\rangle$ is defined as:

$$P_k = |k\rangle\langle k| \quad (35)$$

When applied to an arbitrary state

$$|v\rangle = \sum_j c_j |j\rangle,$$

The projector returns only the component of $|v\rangle$ parallel to $|k\rangle$:

$$P_k |v\rangle = (|k\rangle\langle k|) |v\rangle = \langle k|v\rangle |k\rangle$$

For example, let's suppose the state:

$$|v\rangle = a |k\rangle + b |k_\perp\rangle$$

Where $|k\rangle$ is the direction we want to keep and $|k_\perp\rangle$ is the direction we want to discard that is orthogonal to $|k\rangle$ (i.e., $\langle k|k_\perp\rangle = 0$). The projector P_k will keep the component $a |k\rangle$ and remove the component $b |k_\perp\rangle$:

$$\begin{aligned} P_k |v\rangle &= |k\rangle\langle k| (a |k\rangle + b |k_\perp\rangle) \\ &= a |k\rangle\langle k|k\rangle + b |k\rangle\langle k|k_\perp\rangle \\ &\downarrow |k\rangle \text{ is normalized, then } \langle k|k\rangle = 1 \\ &\text{and } |k_\perp\rangle \text{ is orthogonal to } |k\rangle, \text{ then } \langle k|k_\perp\rangle = 0 \\ &= a |k\rangle \underbrace{\langle k|k\rangle}_{=1} + b |k\rangle \underbrace{\langle k|k_\perp\rangle}_{=0} \\ &= a |k\rangle + b |k\rangle \cdot 0 \\ &= a |k\rangle \end{aligned}$$

Therefore, a **projector acts as a filter**: it keeps the part of the state aligned with $|k\rangle$ and removes all orthogonal components. For example, suppose:

$$|v\rangle = 0.8|0\rangle + 0.6|1\rangle$$

Project onto $|0\rangle$:

$$|0\rangle\langle 0| |v\rangle = |0\rangle\langle 0| (0.8|0\rangle + 0.6|1\rangle) = 0.8|0\rangle$$

The $|1\rangle$ part disappears because it is orthogonal to $|0\rangle$, while the $|0\rangle$ part is preserved (but scaled by the coefficient 0.8).

Projection operators satisfy the **idempotence property**:

$$P_k^2 = P_k$$

Meaning that applying the same projector twice has the same effect as applying it once.

⚠ Effect of Measurement

Given a qubit $|\psi\rangle$, applying M_k yields the **(unnormalized) projected vector**:

$$|\psi_k\rangle = M_k |\psi\rangle = |k\rangle \langle k|\psi\rangle = (|k\rangle \langle k|) |\psi\rangle \quad (36)$$

To obtain the **new normalized state**, we **divide by the square root of the probability**:

$$|\psi_k^{\text{norm}}\rangle = \frac{M_k \cdot |\psi\rangle}{\sqrt{\langle \psi | M_k | \psi \rangle}} = \frac{\langle k|\psi\rangle \cdot |k\rangle}{\sqrt{|\langle k|\psi\rangle|^2}} = |k\rangle \quad (37)$$

So after measurement, the qubit **collapses** to $|k\rangle$, the **eigenstate corresponding to the measurement outcome**.

✓ Why Normalization is Needed

After applying the measurement operator $M_k = |k\rangle \langle k|$ (Equation 34) to a state $|\psi\rangle$ (Equation 36), we obtain:

$$|\psi_k\rangle = M_k |\psi\rangle = |k\rangle \langle k|\psi\rangle$$

This vector is generally **not normalized**. Its norm is:

$$\| |\psi_k\rangle \|^2 = \langle \psi | M_k | \psi \rangle = |\langle k|\psi\rangle|^2 = p_k$$

Where p_k is the **probability of obtaining outcome k** .

Deepening: Why $\| |\psi_k\rangle \|^2 = p_k$?

We want to show:

$$\| |\psi_k\rangle \|^2 = \langle \psi | M_k | \psi \rangle = |\langle k|\psi\rangle|^2 = p_k$$

1. **Definition of squared norm.** The squared norm of a ket is:

$$\| |\psi_k\rangle \|^2 = \langle \psi_k | \psi_k \rangle$$

Since $|\psi_k\rangle = M_k |\psi\rangle$, we can write:

$$\| |\psi_k\rangle \|^2 = \underbrace{(M_k |\psi\rangle)^\dagger}_{\langle \psi_k |} \underbrace{(M_k |\psi\rangle)}_{|\psi_k \rangle}$$

We have only expanded the definition of the squared norm and substituted $|\psi_k\rangle$.

2. **Take the dagger $\dagger$.** The dagger of a product of operators is the product of the daggers in reverse order:

$$(AB)^\dagger = B^\dagger A^\dagger$$

Applying this to $(M_k |\psi\rangle)^\dagger$:

$$(M_k |\psi\rangle)^\dagger = \langle \psi | M_k^\dagger$$

And substituting back into the expression for the squared norm:

$$\| |\psi_k\rangle \|^2 = \langle \psi | M_k^\dagger M_k |\psi\rangle$$

3. **Use the Hermitian property of M_k** (explained later page 81). Since M_k is Hermitian, we have $M_k^\dagger = M_k$. Therefore:

$$\| |\psi_k\rangle \|^2 = \langle \psi | M_k M_k |\psi\rangle$$

4. **Use the idempotent property of M_k** (explained later page 82). Since M_k is idempotent, we have $M_k^2 = M_k$. Therefore:

$$\| |\psi_k\rangle \|^2 = \langle \psi | M_k |\psi\rangle$$

5. **Substitute the definition of M_k .** Since $M_k = |k\rangle \langle k|$, we can write:

$$\| |\psi_k\rangle \|^2 = \langle \psi | (|k\rangle \langle k|) |\psi\rangle = \langle \psi | k \rangle \langle k | \psi \rangle$$

6. **Use the property of inner products.** The inner product $\langle \psi | k \rangle$ is the complex conjugate of $\langle k | \psi \rangle$:

$$\langle \psi | k \rangle = \overline{\langle k | \psi \rangle}$$

Therefore (page 7):

$$\| |\psi_k\rangle \|^2 = \overline{\langle k | \psi \rangle} \langle k | \psi \rangle = |\langle k | \psi \rangle|^2 = p_k$$

This is exactly the probability p_k of obtaining outcome k when measuring the state $|\psi\rangle$ with the measurement operator M_k .

⚠ Important. The result of a measurement must be a valid quantum state, and therefore it must be **normalized** (i.e., have unit norm).

To restore normalization, we divide the projected vector by its norm:

$$|\psi_k^{\text{norm}}\rangle = \frac{|\psi_k\rangle}{\| |\psi_k\rangle \|} = \frac{|k\rangle \langle k|\psi\rangle}{\sqrt{|\langle k|\psi\rangle|^2}}$$

Observe that:

$$\frac{\langle k|\psi\rangle}{|\langle k|\psi\rangle|}$$

Is a complex number of magnitude 1, i.e., a **global phase**. Since global phases do not affect physical states, we conclude:

$$|\psi_k^{\text{norm}}\rangle = |k\rangle$$

💡 Interpretation Measurement can be understood as a two-step process:

1. **Projection:** extract the component of $|\psi\rangle$ along $|k\rangle$ (this determines the probability p_k).
2. **Renormalization:** rescale the result to obtain a valid quantum state.

📊 Probability of Outcome

The **probability of measuring the qubit in the state $|k\rangle$** is:

$$p_k = \langle \psi | M_k | \psi \rangle = |\langle k | \psi \rangle|^2 = \underbrace{|\bar{k}_1 \cdot \psi_1 + \bar{k}_2 \cdot \psi_2|^2}_{|\langle k | \psi \rangle|^2 \text{ expanded}} \quad (38)$$

This is the **Born rule**, a fundamental postulate of quantum mechanics. It states that the probability of obtaining a specific measurement outcome is given by the **squared magnitude of the inner product between the state being measured and the eigenstate corresponding to that outcome**. In other words, it quantifies how much the state $|\psi\rangle$ (*the state being measured*) overlaps with the state $|k\rangle$ (*the eigenstate corresponding to the outcome*). The more $|\psi\rangle$ is aligned with $|k\rangle$, the higher the probability of measuring k . Conversely, if $|\psi\rangle$ is orthogonal to $|k\rangle$, the probability of measuring k is zero. For example:

$$\begin{aligned} p_0 &= \langle \psi | M_0 | \psi \rangle = |\langle 0 | \psi \rangle|^2 = |1 \cdot \psi_1 + \cancel{0 \cdot \psi_2}| = |\psi_1| \\ p_1 &= \langle \psi | M_1 | \psi \rangle = |\langle 1 | \psi \rangle|^2 = |\cancel{0 \cdot \psi_1} + 1 \cdot \psi_2| = |\psi_2| \end{aligned}$$

These probabilities measure the likelihood of obtaining the outcomes 0 and 1 when measuring the qubit $|\psi\rangle$ in the computational basis.

Remark: $\langle \psi | M_k | \psi \rangle$

This is called a quadratic form in linear algebra, and in quantum mechanics it represents the **probability of the measurement outcome k** when **measuring the qubit $|\psi\rangle$** with **measurement operator M_k** . Suppose a qubit ψ :

$$|\psi\rangle = \begin{pmatrix} a \\ b \end{pmatrix} \quad \text{with } |a|^2 + |b|^2 = 1 \text{ (normalization condition)}$$

If we apply the measurement operator M_0 :

$$M_0 |\psi\rangle = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a \\ 0 \end{pmatrix}$$

And we finally calculate the bra $\langle \psi |$:

$$\langle \psi | \begin{pmatrix} a \\ 0 \end{pmatrix} = (\bar{a} \quad \bar{b}) \begin{pmatrix} a \\ 0 \end{pmatrix} = \bar{a} \cdot a = |a|^2$$

In other words, this is the probability of measuring 0: p_0 .

Standard Basis Measurement

In quantum computing, we usually measure in the **computational basis** $\{|0\rangle, |1\rangle\}$. The **projectors** are:

$$M_0 = |0\rangle \langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad M_1 = |1\rangle \langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad (39)$$

For a **general qubit** $|\psi\rangle = a|0\rangle + b|1\rangle$, the **probabilities** are:

$$p_0 = |a|^2 \quad p_1 = |b|^2 \quad (40)$$

After measurement, the qubit **collapses to $|0\rangle$ with probability $|a|^2$** , or to **$|1\rangle$ with probability $|b|^2$** .

Key Properties of Measurement Operator

- **Hermitian:** $M_k^\dagger = M_k$, which means that M_k is equal to its conjugate transpose. This implies that M_k has real eigenvalues and orthogonal eigenvectors.

$$\begin{aligned} M_k^\dagger &= (|k\rangle \langle k|)^\dagger \\ &\downarrow \text{ use the rule } (AB)^\dagger = B^\dagger A^\dagger \\ &= (\underbrace{|k\rangle}_A \underbrace{\langle k|}_B)^\dagger \\ &= (\langle k|)^\dagger (|k\rangle)^\dagger \\ &= |k\rangle \langle k| \\ &= M_k \end{aligned}$$

- **Idempotent**: once applied, **applying it again does nothing**:

$$M_k^2 = M_k$$

A simple proof:

$$M_k^2 = |k\rangle |k\rangle \langle k| \langle k| = |k\rangle \langle k| = M_k$$

This reflects that **after projection, the qubit is already in the state $|k\rangle$** .

- **Non-unitary** and **Non-reversible**: **measurement destroys superposition and cannot be undone.**

4 Multiple Qubit Gates

4.1 Multiple Qubit States

Now that we have a solid understanding of single qubits and their states, we can transition to discussing many qubits as one joint system, rather than just one vector with one qubit.

With **one qubit**, life is simple because the state of the system can be described by a single vector in a two-dimensional complex vector space:

$$|v\rangle = a|0\rangle + b|1\rangle$$

With **two qubits**, something radical happens. We can no longer describe the system by looking at each qubit separately. Instead, we must describe **one global state** that captures the behavior of both qubits together. For example:

$$|v\rangle = c_0|00\rangle + c_1|01\rangle + c_2|10\rangle + c_3|11\rangle$$

This is not just a simple combination of two single-qubit states.

⚠ Exponential Explosion. The first thing to notice is that the number of amplitudes needed to describe the state of a system grows exponentially with the number of qubits.

- For 1 qubit, we need 2 amplitudes: a and b .
- For 2 qubits, we need 4 amplitudes: c_0 , c_1 , c_2 , and c_3 .
- For 3 qubits, we need 8 amplitudes: d_0 , d_1 , d_2 , d_3 , d_4 , d_5 , d_6 , and d_7 .
- For n qubits, we need 2^n amplitudes to fully describe the state of the system.

This is why quantum computing is so powerful: it can **process many possibilities at the same time** and **manipulate them so that only the correct ones survive**. This will become clear when we introduce entanglement (page 92).

📖 The Key New Phenomenon: Entanglement

Entanglement is the key new phenomenon when dealing with multiple qubits. It refers to quantum states that cannot be expressed as a tensor product of individual qubit states. For example, the state:

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Is an entangled state. In such cases, the system must be described globally, and it is not possible to assign independent quantum states to each qubit.

If one qubit is measured, the outcome is random, but it determines the outcome of the other qubit due to the structure of the global state. This leads to strong

correlations between measurements results. However, this does not imply any form of information transfer between the qubits.

Entanglement is a fundamental resource in quantum computing, enabling correlations that have no classical counterpart, especially when combined with interference effects in quantum algorithms.

- Entanglement does not mean that one qubit sends information to the other. The observed correlations arise from the collapse of a **global quantum state**, not from communication between subsystems.
- Before measurement, individual qubits in an entangled state do **not have a well-defined state**. Only the joint system is well-defined.
- Each qubit individually appears **random**, but measurement outcomes are **correlated**.
- Entanglement alone is not the source of quantum computational power; it becomes powerful when combined with **interference** and quantum operations.

⚙️ New tools we need

To handle this new world of multiple qubits, we need new mathematical tools. We will introduce the **tensor product**, which allows us to construct the joint state of multiple qubits; we will also introduce the **multi-qubit gates**, which are operations that can manipulate the joint state of multiple qubits simultaneously; and we will discuss how to **measure** multiple qubits and understand the resulting probabilities.

4.1.1 Tensor Product

Let's start with something simple. As in previous sections, we have a simple one-qubit state. Now, we want to **add a second qubit to the system** to create a multi-qubit state. For simplicity, we will use only two qubits. *How do we describe the **combined system**?* Simply keeping them separate is not enough because we would miss correlations between the two qubits. **We need a way to combine the two qubits into a single object.** The tool for doing so is the **tensor product**.

Example 1: Dice Analogy

Imagine we have two dice. Each die can be in one of six states (1, 2, 3, 4, 5, or 6). If we want to describe the combined state of both dice, we can't just list their individual states separately. Instead, we need to consider all possible combinations of their states. The tensor product allows us to create a new space that includes all these combinations, giving us a complete description of the system.

The **Tensor Product** is a **mathematical operation** that **takes two vectors** (or more generally, two mathematical objects):

$$|v_A\rangle = a_0 |0\rangle + a_1 |1\rangle \quad |v_B\rangle = b_0 |0\rangle + b_1 |1\rangle$$

And **produces a new vector** that represents the combined state of the two systems:

$$\begin{aligned} |v_A\rangle \otimes |v_B\rangle &= (a_0 |0\rangle + a_1 |1\rangle) \otimes (b_0 |0\rangle + b_1 |1\rangle) \\ &= a_0 b_0 |00\rangle + a_0 b_1 |01\rangle + a_1 b_0 |10\rangle + a_1 b_1 |11\rangle \end{aligned} \quad (41)$$

Here, the resulting state $|v_A\rangle \otimes |v_B\rangle$ is a **superposition of all possible combinations of the states of the two qubits**.

❓ Why amplitudes multiply? For example, why is the amplitude of $|00\rangle$ equal to $a_0 b_0$? A qubit is a vector:

$$|v_A\rangle = \begin{pmatrix} a_0 \\ a_1 \end{pmatrix}, \quad |v_B\rangle = \begin{pmatrix} b_0 \\ b_1 \end{pmatrix}$$

The tensor product is **defined** as:

$$|v_A\rangle \otimes |v_B\rangle = \begin{pmatrix} a_0 b_0 \\ a_0 b_1 \\ a_1 b_0 \\ a_1 b_1 \end{pmatrix} \quad (42)$$

So amplitudes multiply because **that's how the tensor product is defined**. **But why is it defined like that?** Because we want a rule that satisfies two requirements:

1. **Linearity.** Quantum mechanics is linear:

$$(a_0 |0\rangle + a_1 |1\rangle) \otimes |v_B\rangle = a_0 (|0\rangle \otimes |v_B\rangle) + a_1 (|1\rangle \otimes |v_B\rangle)$$

The tensor product must be **bilinear**, i.e., linear in each argument, so that superpositions are preserved when combining quantum states.

2. Basis consistency:

$$|0\rangle \otimes |0\rangle = |00\rangle, \quad |0\rangle \otimes |1\rangle = |01\rangle, \quad |1\rangle \otimes |0\rangle = |10\rangle, \quad |1\rangle \otimes |1\rangle = |11\rangle$$

The tensor product must map basis states of individual systems to a consistent basis of the joint system (e.g., $|i\rangle \otimes |j\rangle = |ij\rangle$), preserving the structure of the state space.

This structure ensures that probabilities factorize correctly for independent systems.

Remark: Inner Product of Tensor Products

For composite quantum systems (i.e., systems composed of multiple qubits), the **inner product between tensor product states factorizes**:

$$(\langle a| \otimes \langle b|) (|c\rangle \otimes |d\rangle) = \langle a|c\rangle \cdot \langle b|d\rangle \quad (43)$$

This means that the “similarity” (overlap) between two composite states is the product of the similarities of their individual subsystems (page 9).

For example:

$$(\langle 0| \otimes \langle +|) (|0\rangle \otimes |- \rangle)$$

Has overlap:

$$\langle 0|0\rangle \langle +|- \rangle = 1 \cdot 0 = 0$$

So even though the first qubit is identical, the total states are orthogonal because the second qubits are orthogonal!

 **Intuition.** A tensor product state combines independent subsystems. Therefore, the total overlap must take into account the overlap of *each* subsystem simultaneously. **If one subsystem is perfectly distinguishable (overlap 0), then the entire composite states become perfectly distinguishable.**

Not all states can be factored

Given two qubits, their joint state lives in a 4-dimensional vector space. However, **not all states** in this 4-dimensional space can be written as a tensor product of two single-qubit states. Precisely, if a multi-qubit state can be written as a **tensor product**, then it is **separable**, and we can recover the individual qubit states. But if it is **not a tensor product**, then it is **entangled**, and we cannot assign well-defined quantum states to each qubit individually.

In general, exists a **necessary (and sufficient) condition** for being a tensor product for two qubits. Given a 2-qubit state:

$$|\psi\rangle = (c_0, c_1, c_2, c_3)$$

Where c_0, c_1, c_2, c_3 are the amplitudes of the basis states $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, respectively. The state $|\psi\rangle$ can be written as a tensor product of two single-qubit states if and only if:

$$c_0 c_3 = c_1 c_2 \quad (44)$$

Equivalently, if we reshape the coefficients into a matrix

$$C = \begin{pmatrix} c_0 & c_1 \\ c_2 & c_3 \end{pmatrix}$$

The state is separable if and only if $\det(C) = 0$. More generally, for n -qubit states, there is **no single simple scalar condition** that characterizes separability. A state is separable **if and only if it can be written as a tensor product of n single-qubit states**, which is a more complex condition to verify. So, the **2-qubit case is simpler to analyze**, but for larger systems, determining whether a state is separable or entangled can be more challenging.

Let's make this more concrete with two examples:

✓ Let's start from a 4D vector:

$$|\psi\rangle = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

We can check if it is a tensor product by verifying the condition in Equation 44:

$$c_0 c_3 = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}, \quad c_1 c_2 = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

Since $c_0 c_3 = c_1 c_2$, the state is a tensor product and thus separable.

Now, we can find the two single-qubit states whose tensor product gives us $|\psi\rangle$. We try:

$$|v_A\rangle = \begin{pmatrix} a_0 \\ a_1 \end{pmatrix}, \quad |v_B\rangle = \begin{pmatrix} b_0 \\ b_1 \end{pmatrix}$$

We want:

$$(a_0 b_0, a_0 b_1, a_1 b_0, a_1 b_1) = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2} \right) \Rightarrow \begin{cases} a_0 b_0 = \frac{1}{2} \\ a_0 b_1 = \frac{1}{2} \\ a_1 b_0 = \frac{1}{2} \\ a_1 b_1 = \frac{1}{2} \end{cases}$$

Solving this system of equations, we find:

$$a_0 = b_0 = \frac{1}{\sqrt{2}}, \quad a_1 = b_1 = \frac{1}{\sqrt{2}}$$

And this works because:

$$\begin{cases} a_0 b_0 = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2} \\ a_0 b_1 = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2} \\ a_1 b_0 = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2} \\ a_1 b_1 = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2} \end{cases}$$

Therefore:

$$|\psi\rangle = \left[\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \right] \otimes \left[\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \right]$$

The global 2-qubit state can be written as **two independent qubit states**, confirming that it is separable.

$$|\psi\rangle = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} = \left[\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \right] \otimes \left[\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \right]$$

✗ Let's start from a different 4D vector:

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

We check the condition for separability:

$$c_0 c_3 = \frac{1}{2}, \quad c_1 c_2 = 0$$

Since $c_0 c_3 \neq c_1 c_2$, the state is not a tensor product and thus is **not separable**. This state is an example of an **entangled state** (**Entanglement** phenomenon), where the two qubits are correlated in such a way that we cannot describe them independently. We will see more about entanglement in the next section, but for now, the key takeaway is that not all multi-qubit states can be factored into tensor products of single-qubit states, and this is a fundamental aspect of quantum mechanics that leads to phenomena like entanglement.

In summary, the tensor product generates only a subset of the full 4-dimensional space of two-qubit states. Separable states have a special multiplicative structure and lie on a lower-dimensional manifold within this space. Most states in the full space do not satisfy this structure and are therefore entangled.

4.1.2 Building a Two-Qubit System

In the previous sections, we have seen how to represent the state of a single qubit using a two-dimensional complex vector. Now, we want to extend this representation to a system of two qubits. To do this, we will use the concept of the tensor product seen in the previous section.

We start from two independent qubits:

$$|v_A\rangle = a_0 |0\rangle + a_1 |1\rangle \quad \text{and} \quad |v_B\rangle = b_0 |0\rangle + b_1 |1\rangle$$

And our **goal is to find a state that encodes all joint possibilities** of the two qubits:

- Both qubits are in state $|0\rangle$: $|00\rangle$
- The first qubit is in state $|0\rangle$ and the second in state $|1\rangle$: $|01\rangle$
- The first qubit is in state $|1\rangle$ and the second in state $|0\rangle$: $|10\rangle$
- Both qubits are in state $|1\rangle$: $|11\rangle$

To represent these joint possibilities, we will use the tensor product of the two qubits' states:

$$|v_A v_B\rangle = |v_A\rangle \otimes |v_B\rangle$$

Expanding this expression, we get:

$$\begin{aligned} |v_A v_B\rangle &= (a_0 |0\rangle + a_1 |1\rangle) \otimes (b_0 |0\rangle + b_1 |1\rangle) \\ &= a_0 b_0 |00\rangle + a_0 b_1 |01\rangle + a_1 b_0 |10\rangle + a_1 b_1 |11\rangle \end{aligned} \quad (45)$$

This state $|v_A v_B\rangle$ is a linear combination of the four possible joint states of the two qubits. Each term in the expansion corresponds to one of the joint possibilities, and the coefficients $a_0 b_0$, $a_0 b_1$, $a_1 b_0$, and $a_1 b_1$ represent the probability amplitudes of each joint state.

We are no longer describing two systems separately, we are describing **one system with 4 possible outcomes**.

General Two-Qubit State

In Equation 45 we built a two-qubit state starting from a **very specific structure**, where the coefficients of the state were products of the coefficients of the individual qubits (i.e., $a_i b_j$). However, we can also consider a more **general two-qubit state**, where the coefficients are not necessarily products of the individual qubits' coefficients:

$$|v\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle \quad (46)$$

Where c_0 , c_1 , c_2 , and c_3 are **arbitrary complex numbers** that represent the probability amplitudes of each joint state.

Obviously, this general two-qubit state **must satisfy the normalization condition**:

$$|c_0|^2 + |c_1|^2 + |c_2|^2 + |c_3|^2 = 1 \quad (47)$$

This means that the sum of the probabilities of all joint states must equal 1, ensuring that the state is properly normalized.

We can also generalize deeper to n qubits, where the state of the **system is a linear combination** of 2^n **basis states**, and the **coefficients** are arbitrary **complex numbers** that satisfy the normalization condition. Mathematically, we can express this as:

$$|v\rangle = \sum_{k=0}^{2^n-1} c_k |k\rangle \quad (48)$$

Where $|k\rangle$ represents the basis states of the n -qubit system, and c_k are the corresponding probability amplitudes. As we have already discussed at page 83, this general form **illustrates the exponential growth of the state space as the number of qubits increases**. However, we can write our one- and two-qubit states as special cases from this general form:

- For $n = 1$ qubit, we have $2^1 = 2$ basis states: $|0\rangle$ and $|1\rangle$, and the state can be written as:

$$|v\rangle = c_0 |0\rangle + c_1 |1\rangle$$

- For $n = 2$ qubits, we have $2^2 = 4$ basis states: $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$, and the state can be written as:

$$|v\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle$$

In summary, a general two-qubit state is a normalized superposition (i.e., the *coefficients satisfy the normalization condition*) of the four basis states (i.e., $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$). Unlike tensor product states, its coefficients are arbitrary and *may not* factorize, leading to entanglement and non-classical correlations between the qubits (like the state in Example at page 88).

4.1.3 Separable vs Entangled States

Let the general state of a two-qubit system be (as defined in Equation 46, page 89):

$$|v\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle$$

The question is: *can we interpret this as two independent qubits?* The answer is **not always**. As we saw on page 85, the answer depends on whether the states are separable or entangled.

☐ Separable States

A state is **separable** if it can be written as a tensor product of two **single-qubit states**. In other words, there exist single-qubit states $|v_A\rangle$ and $|v_B\rangle$ such that:

$$|v\rangle = |v_A\rangle \otimes |v_B\rangle$$

So the system can be split into one state for qubit A and one state for qubit B , and the global state is just the combination of these two states. As consequence, the coefficients factorize as:

$$c_0 = a_0 b_0, \quad c_1 = a_0 b_1, \quad c_2 = a_1 b_0, \quad c_3 = a_1 b_1$$

☐ Entangled States

A state is **entangled** if it cannot be written as a tensor product of two **single-qubit states**. In other words, there do not exist single-qubit states $|v_A\rangle$ and $|v_B\rangle$ such that:

$$|v\rangle \neq |v_A\rangle \otimes |v_B\rangle$$

In this case, the state cannot be separated into two independent qubits, and the coefficients do not factorize as above. *The system is simply one single entity.* For example, the Bell state (we will see in future sections, page 115) is an entangled state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

And the demonstration was written on page 88. In other words, we can only say that the system is in state $|\Phi^+\rangle$, but we cannot say that qubit A is in some state and qubit B is in some state, because the state of the system does not factorize into two independent states.

At this stage, entanglement is introduced as a structural property of quantum states. Its physical meaning and implications become clear only when studying measurement and quantum circuits in later sections.

☐ Difference. With **separable** states, the *information is stored in the individual qubits*. In contrast, with **entangled** states, the *information is stored in the correlations between the qubits*.

4.1.4 Why Entanglement Matters

Let's answer to the question “*why entanglement matters*” step by step. First of all, the **complexity of quantum states grows exponentially** with the number of qubits, as we have introduced at page 83. As consequence, to describe the system, we need 2^n complex numbers, where n is the number of qubits. This means that for a system of 50 qubits, we would need 2^{50} complex numbers to describe its state, which is an astronomically large number (approximately 1.13×10^{15}). It could be a **problem for classical computers**, but **quantum computers can handle this complexity naturally**.

The reason why quantum computers can handle this complexity naturally is not that they explicitly store all these amplitudes as a classical computer would. Rather, the quantum system itself physically evolves in this exponentially large state space.

Entanglement matters because **most quantum states cannot be decomposed into independent single-qubit states**. In fact, the set of separable states, i.e., states that can be written as a tensor product of individual qubit states, is **very small compared to the set of all possible states**. For two qubits, separable states must satisfy:

$$c_0 c_3 = c_1 c_2$$

Which is a strong structural constraint. Therefore, **most two-qubit states do not satisfy this condition and are entangled**.

❓ But why does this matter? Because **entanglement prevents the system from being compressed into simpler independent components**. The **state must be described globally**. This global structure is difficult to simulate classically, but **quantum mechanics allows us to manipulate it directly through quantum operations**. This is one of the reasons why quantum computers can solve certain problems more efficiently than classical computers.

In summary, entanglement is a fundamental feature of quantum mechanics that prevents a multi-qubit system from being decomposed into independent subsystems. As the number of qubits increases, the state space grows exponentially, and entanglement forces a global description of the system. This global structure can be exploited by quantum algorithms to achieve computational advantages over classical approaches.

4.1.5 Exercise: Normalization of Tensor Product States

Given two qubits in the states $|v_A\rangle$ and $|v_B\rangle$ respectively:

$$|v_A\rangle = a_0 |0\rangle + a_1 |1\rangle, \quad |v_B\rangle = b_0 |0\rangle + b_1 |1\rangle$$

And the combined state of the two qubits is given by the tensor product:

$$|v_A v_B\rangle = |v_A\rangle \otimes |v_B\rangle = a_0 b_0 |00\rangle + a_0 b_1 |01\rangle + a_1 b_0 |10\rangle + a_1 b_1 |11\rangle$$

Show that:

$$|a_0 b_0|^2 + |a_0 b_1|^2 + |a_1 b_0|^2 + |a_1 b_1|^2 = 1$$

Proof. To show that the sum of the squares of the coefficients equals 1, we start expanding the expression:

$$|a_0 b_0|^2 + |a_0 b_1|^2 + |a_1 b_0|^2 + |a_1 b_1|^2$$

Using the property:

$$|xy|^2 = |x|^2 |y|^2$$

Applying this property to each term, we get:

$$\begin{aligned} &= |a_0|^2 |b_0|^2 + |a_0|^2 |b_1|^2 + |a_1|^2 |b_0|^2 + |a_1|^2 |b_1|^2 \\ &= |a_0|^2 (|b_0|^2 + |b_1|^2) + |a_1|^2 (|b_0|^2 + |b_1|^2) \end{aligned}$$

Since $|v_B\rangle$ is a valid quantum state and must be normalized, we have:

$$|b_0|^2 + |b_1|^2 = 1$$

Substituting this back into our expression, we get:

$$\begin{aligned} &= |a_0|^2 \cdot 1 + |a_1|^2 \cdot 1 \\ &= |a_0|^2 + |a_1|^2 \end{aligned}$$

Since $|v_A\rangle$ is also a valid quantum state and must be normalized, we have:

$$|a_0|^2 + |a_1|^2 = 1$$

Therefore, we conclude that:

$$|a_0 b_0|^2 + |a_0 b_1|^2 + |a_1 b_0|^2 + |a_1 b_1|^2 = 1$$

The combined state $|v_A v_B\rangle$ is also a valid quantum state and must be normalized, which is consistent with our result. QED

This exercise demonstrates that the **tensor product of two normalized quantum states is also normalized**, because the squared amplitudes factorize and the normalization conditions of the individual states ensure that the total probability remains 1.

4.2 Introduction to Multiple Qubit Gates

Until now, we learned that a quantum state is a vector in a complex vector space, and for 2 qubits, the state is a vector in $\mathbb{C}^4$:

$$|v\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle$$

Now the question is: *how do we change this state?*

 **The key idea.** A **quantum gate** is just a rule that transforms one state into another. For 1 qubit we already know (page 39):

$$|v'\rangle = U |v\rangle$$

Where U is a 2×2 unitary matrix. For example, the Hadamard gate H (page 62) transforms the state $|0\rangle$ into $|+\rangle$:

$$H |0\rangle = |+\rangle$$

 **What about 2 qubits?** For 2 qubits, we need a 4×4 unitary matrix to transform the state:

$$|v'\rangle = U |v\rangle \quad \text{with} \quad |v\rangle = \begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{pmatrix}, \quad U = \begin{pmatrix} u_{00} & u_{01} & u_{02} & u_{03} \\ u_{10} & u_{11} & u_{12} & u_{13} \\ u_{20} & u_{21} & u_{22} & u_{23} \\ u_{30} & u_{31} & u_{32} & u_{33} \end{pmatrix}$$

This means that the gate can change **all amplitudes at once**, not just one at a time as in the single-qubit case. This is the important difference between single and multi-qubit gates: with 1 qubit, we modify only 2 amplitudes, but with 2 qubits, we can modify all 4 amplitudes **simultaneously**. And now, the transformation can mix amplitudes, create correlations, and even generate entanglement between the qubits.

In simple terms, a multi-qubit gate is just a transformation of a larger state vector. Nothing more, nothing less. However, this simple idea has important implications: it allows us to manipulate the global state of a quantum system, including correlations between qubits, which can lead to computational advantages over classical approaches.

4.3 Parallel Gates

Now that we have a way to represent the state of multiple qubits, we can also represent the operations (gates) that act on them. In particular, if we have two qubits $|v_A\rangle$ and $|v_B\rangle$, and two gates A and B , what happens when we want to apply A to $|v_A\rangle$ and B to $|v_B\rangle$ simultaneously?

The answer is given by the **tensor product of quantum gates**:

$$(A \otimes B) (|v_A\rangle \otimes |v_B\rangle) = (A |v_A\rangle) \otimes (B |v_B\rangle) \quad (49)$$

This means that the combined operation of applying A to $|v_A\rangle$ and B to $|v_B\rangle$ is represented by the tensor product of the two gates, $A \otimes B$. So, **applying gates in parallel** corresponds to the tensor product of the individual gates.

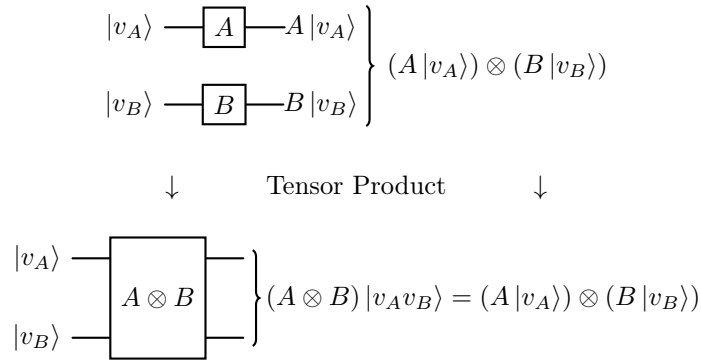


Figure 16: Parallel gates as a tensor product operation.

In simple terms, instead of applying A and B separately, we build one big matrix $A \otimes B$ that acts on the **whole system**. Note that a state is represented as a column vector (e.g., size 4 for two qubits), and a gate is represented as a square matrix (e.g., size 4×4 for two qubits). So *parallel gates* are simply one big transformation of the global state of the system.

❓ What if we want to apply a gate to only one qubit?

Suppose we want to **apply a gate A to the first qubit $|v_A\rangle$ while leaving the second qubit $|v_B\rangle$ unchanged**. We can represent this as:

$$(A \otimes I) (|v_A\rangle \otimes |v_B\rangle) = (A |v_A\rangle) \otimes |v_B\rangle \quad (50)$$

Here, I is the **identity gate** (page 46) that acts on the second qubit, meaning it **leaves it unchanged**. So, to apply a gate to just one qubit in a multi-qubit system, we use the tensor product of that gate with the identity operator on the other qubits.

Example 2: $H \otimes X$

Let's consider the gates H (Hadamard, page 62):

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

And X (Pauli-X, page 48):

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

The tensor product $H \otimes X$ is calculated as follows:

$$\begin{aligned} H \otimes X &= \frac{1}{\sqrt{2}} \left(\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right) \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \cdot X & 1 \cdot X \\ 1 \cdot X & -1 \cdot X \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & 1 \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ 1 \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & -1 \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{pmatrix} \end{aligned}$$

Now, if we apply $H \otimes X$ to a simple two-qubit state, say $|00\rangle$:

$$\begin{aligned} (H \otimes X) |00\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \cdot 1 + 1 \cdot 0 + 0 \cdot 0 + 1 \cdot 0 \\ 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 0 + 0 \cdot 0 \\ 0 \cdot 1 + 1 \cdot 0 + 0 \cdot 0 + -1 \cdot 0 \\ 1 \cdot 1 + 0 \cdot 0 + -1 \cdot 0 + 0 \cdot 0 \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} |01\rangle + \frac{1}{\sqrt{2}} |11\rangle \end{aligned}$$

This can be visually represented as parallel gates acting on the two qubits:

$$\left. \begin{array}{l} |0\rangle \text{ --- } \boxed{H} \text{ --- } H |0\rangle \\ |0\rangle \text{ --- } \boxed{X} \text{ --- } X |0\rangle \end{array} \right\} (H |0\rangle) \otimes (X |0\rangle) = |+\rangle \otimes |1\rangle$$

Or as a single global gate:

$$\left. \begin{array}{l} |0\rangle \text{ --- } \boxed{H \otimes X} \text{ --- } \\ |0\rangle \text{ --- } \boxed{H \otimes X} \text{ --- } \end{array} \right\} (H \otimes X) |00\rangle = |+\rangle$$

Where $|+\rangle$ is $|+\rangle \otimes |1\rangle$. Note that applying the Hadamard gate to the qubit $|0\rangle$ creates the superposition state $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, while applying the X gate to $|0\rangle$ flips it to $|1\rangle$.

Example 3: $I \otimes X$

Now, let's consider applying the identity gate I to the first qubit and the X gate to the second qubit. The tensor product is:

$$\begin{aligned} I \otimes X &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & 0 \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ 0 \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & 1 \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \end{aligned}$$

Applying this to the state $|00\rangle$ gives:

$$\begin{aligned} (I \otimes X) |00\rangle &= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 \cdot 1 + 1 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 \\ 1 \cdot 1 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 \\ 0 \cdot 1 + 0 \cdot 0 + 0 \cdot 0 + 1 \cdot 0 \\ 0 \cdot 1 + 0 \cdot 0 + 1 \cdot 0 + 0 \cdot 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \\ &= |01\rangle \end{aligned}$$

This means that applying I to the first qubit leaves it unchanged as $|0\rangle$, while applying X to the second qubit flips it from $|0\rangle$ to $|1\rangle$, resulting in the combined state $|01\rangle$. Visually, this can be represented as:

$$\left. \begin{array}{l} |0\rangle \text{ --- } \boxed{I} \text{ --- } I|0\rangle \\ |0\rangle \text{ --- } \boxed{X} \text{ --- } X|0\rangle \end{array} \right\} |0\rangle \otimes (X|0\rangle) = |0\rangle \otimes |1\rangle$$

Or as a single global gate:

$$\left. \begin{array}{l} |0\rangle \\ |0\rangle \end{array} \right\} \boxed{I \otimes X} \left. \begin{array}{l} \text{---} \\ \text{---} \end{array} \right\} (I \otimes X) |00\rangle = |01\rangle$$

4.4 Hadamard Transform on Multiple Qubits

Applying the Hadamard gate to every qubit is a useful tool in quantum computing, as it creates a superposition of all possible states. Now, let's see how this is done.

We already know that applying the Hadamard gate to a single qubit in the state $|0\rangle$ gives us:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

This means that H creates a **balanced superposition** of the two basis states $|0\rangle$ and $|1\rangle$.

Now, if we apply this idea to two qubits, we can apply the Hadamard gate to each qubit individually. This is done using the tensor product of the Hadamard gates, for example:

$$(H \otimes H)|00\rangle = H|0\rangle \otimes H|0\rangle$$

So it is equivalent to applying the Hadamard gate to each qubit separately. Expanding this, we get:

$$\begin{aligned} (H \otimes H)|00\rangle &= H|0\rangle \otimes H|0\rangle \\ &= \left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \right) \otimes \left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \right) \\ &= \frac{1}{2}(|0\rangle + |1\rangle) \otimes (|0\rangle + |1\rangle) \\ &= \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) \end{aligned}$$

This means that **applying Hadamard gate in parallel generates all possible combinations of the basis states** for the two qubits, creating a superposition of all four possible states.

In general, for n qubits, **applying the Hadamard gate to each qubit will create a superposition of all 2^n possible states**. The resulting state can be expressed as:

$$(H^{\otimes n})|0\rangle^{\otimes n} = \frac{1}{2^{\frac{n}{2}}} \sum_{k=0}^{2^n-1} |k\rangle = \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} |k\rangle \quad (51)$$

Where $|k\rangle$ represents the basis states of the n -qubit system, and the **sum runs over all possible combinations of the basis states**.

$$\left. \begin{array}{l} |0\rangle \text{---} \boxed{H} \text{---} \\ |0\rangle \text{---} \boxed{H} \text{---} \end{array} \right\} (H \otimes H)|00\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

In summary, **Hadamard on all qubits spreads the state over the entire space of possible states**, creating a uniform superposition that is fundamental in many quantum algorithms.

4.5 Main Multi-Qubit Gates

4.5.1 Controlled NOT (CNOT) Gate

The **Controlled NOT (CNOT) Gate** is a **two-qubit gate** composed of:

- A **control qubit** that **determines** whether the **NOT operation is applied or not**.
- A **target qubit** that is **flipped** (NOT operation) if the control qubit is in the state $|1\rangle$, and remains unchanged if the control qubit is in the state $|0\rangle$.

In simple terms, it is a gate that **performs a NOT operation on the *target* qubit only when the *control* qubit is in the state $|1\rangle$** .

≡ Action on Basis States

The CNOT gate acts on the computational basis states as follows:

Input	Output
$ 00\rangle$	$ 00\rangle$
$ 01\rangle$	$ 01\rangle$
$ 10\rangle$	$ 11\rangle$
$ 11\rangle$	$ 10\rangle$

If the first qubit (control) is $|0\rangle$, the second qubit (target) remains unchanged. If the first qubit is $|1\rangle$, the second qubit is flipped: $|0\rangle \leftrightarrow |1\rangle$.

√* Matrix Representation

The CNOT gate can be represented as a 4×4 unitary matrix in the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$:

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (52)$$

This matrix indicates that the first two basis states $|00\rangle$ and $|01\rangle$ are unchanged, while $|10\rangle$ and $|11\rangle$ are swapped, reflecting the action of the CNOT gate on the target qubit based on the state of the control qubit.

≡ Action on General States

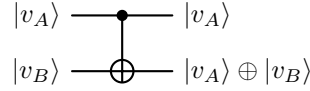
For a general two-qubit state $|v\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle$, the CNOT gate transforms it as follows:

$$\text{CNOT} |v\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |11\rangle + c_3 |10\rangle \quad (53)$$

The amplitudes c_0 and c_1 remain unchanged, while c_2 and c_3 are **swapped**, demonstrating the conditional nature of the CNOT gate based on the state of the control qubit.

✂ Circuit Representation

In quantum circuit diagrams, the CNOT gate is typically represented with a solid dot on the control qubit line and a plus sign $\oplus$ (indicating the NOT operation) on the target qubit line, connected by a vertical line.



Here, $|v_A\rangle$ is the control qubit and $|v_B\rangle$ is the target qubit. The output state of the target qubit is the result of the XOR operation between the control and target qubits, reflecting the conditional flipping of the target qubit based on the state of the control qubit.

In summary, if the control qubit is $|1\rangle$, apply NOT (Pauli-X) to the target qubit; otherwise do nothing.

⚙ Properties

- **Unitary**

$$\text{CNOT}^\dagger \cdot \text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}^\dagger \cdot \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} = I$$

CNOT gate is unitary, so it is a valid quantum gate that preserves normalization of quantum states, and the total probability of all outcomes remains 1.

- **Reversible**

$$\text{CNOT}^{-1} = \text{CNOT}^\dagger = \text{CNOT}$$

The CNOT gate is reversible, meaning that applying the CNOT gate twice will return the system to its original state:

$$\text{CNOT}(\text{CNOT}(|\psi\rangle)) = |\psi\rangle$$

In other words, we can **always undo a CNOT operation without loss of information**.

- **Conditional Operation**

$$\text{CNOT}(|v_A\rangle \otimes |v_B\rangle) \neq \text{CNOT}|v_A\rangle \otimes \text{CNOT}|v_B\rangle$$

The CNOT gate is a conditional operation, meaning that the **action on the target qubit depends on the state of the control qubit**. This is different from parallel gates, where each qubit is acted upon independently:

$$(H \otimes X)|00\rangle = (H|0\rangle) \otimes (X|0\rangle)$$

CNOT **breaks parallelism** because it introduces interaction: the **evolution of one qubit depends on another**, so the **system can no longer be described as independent parts**.

- **Entangling.** CNOT creates *entanglement* **if and only if** the input state has superposition on the control qubit (in the computational basis).

For example:

$$(H \otimes I) |00\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle)$$

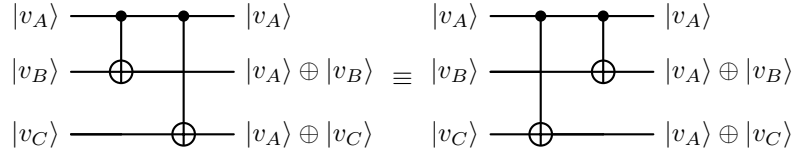
Applying CNOT to this state:

$$\text{CNOT} \left(\frac{1}{\sqrt{2}} (|00\rangle + |10\rangle) \right) = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

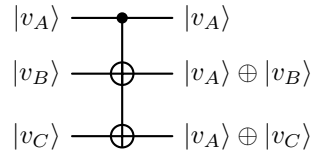
The resulting state is an entangled state (cannot be factored into a tensor product, page 86), demonstrating that CNOT can create entanglement from a superposition state of the control qubit.

✂ Multiple CNOT Gates

Multi-CNOT notation means applying several CNOT gates that share the same control qubit, and these can be represented equivalently as sequential gates or as a single control connected to multiple targets. Visually, this can be represented as:



But it can also be represented as:



The **order of the CNOT gates does not matter** because they **commute when they share the same control qubit**, meaning that the final state of the system will be the same regardless of the order in which the CNOT gates are applied.

4.5.2 Generic Controlled Gate (CU)

The **Generic Controlled Gate (CU)** is a **two-qubit gate** composed of:

- A **control qubit** that determines **whether** the operation is **applied or not** (first qubit).
- A **target qubit** **on which the operation is applied** if the control qubit is in the state $|1\rangle$ (second qubit).
- A **unitary operation** U that is **applied to the target qubit** when the control qubit is in the state $|1\rangle$.

It **generalizes the CNOT gate** (page 100, $U = X$) by allowing **any unitary operation** U to be applied to the target qubit, rather than just the NOT operation.

≡ Action on Basis States

The action of the CU gate on the computational basis states can be described as follows:

Input	Output
$ 00\rangle$	$ 00\rangle$
$ 01\rangle$	$ 01\rangle$
$ 10\rangle$	$ 1\rangle \otimes U 0\rangle$
$ 11\rangle$	$ 1\rangle \otimes U 1\rangle$

If the control qubit is $|0\rangle$, the target qubit remains unchanged. If the control qubit is $|1\rangle$, the unitary operation U is applied to the target qubit, transforming $|0\rangle$ to $U|0\rangle$ and $|1\rangle$ to $U|1\rangle$.

✓✱ Matrix Representation

The CU gate can be represented as a 4×4 unitary matrix in the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$:

$$CU = C_U = \begin{bmatrix} I & 0 \\ 0 & U \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & U_{00} & U_{01} \\ 0 & 0 & U_{10} & U_{11} \end{bmatrix} \quad (54)$$

Where $U_{00}, U_{01}, U_{10}, U_{11}$ are the elements of the 2×2 unitary matrix U that acts on the target qubit when the control qubit is $|1\rangle$. For the first two basis states $|00\rangle$ and $|01\rangle$, the identity operation I is applied, leaving them unchanged. For the last two basis states $|10\rangle$ and $|11\rangle$, the unitary operation U is applied to the target qubit, resulting in the transformation defined by U .

📖 Action on General States

Let's consider two general single-qubit states:

$$|v_A\rangle = a_0 |0\rangle + a_1 |1\rangle \quad |v_B\rangle = b_0 |0\rangle + b_1 |1\rangle$$

Combining these states into a two-qubit state, we have:

$$|v\rangle = |v_A\rangle \otimes |v_B\rangle = a_0 |0\rangle |v_B\rangle + a_1 |1\rangle |v_B\rangle$$

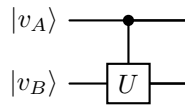
Applying the CU gate to this state, we get:

$$\begin{aligned} \text{CU} |v\rangle &= \text{CU}(a_0 |0\rangle |v_B\rangle + a_1 |1\rangle |v_B\rangle) \\ &= a_0 \text{CU}(|0\rangle |v_B\rangle) + a_1 \text{CU}(|1\rangle |v_B\rangle) \\ &= \underbrace{a_0 |0\rangle |v_B\rangle}_{\text{unchanged}} + \underbrace{a_1 |1\rangle U |v_B\rangle}_{\text{transformed}} \end{aligned}$$

This shows that the CU gate applies the unitary operation U to the target qubit only when the control qubit is in the state $|1\rangle$, while leaving the target qubit unchanged when the control qubit is in the state $|0\rangle$. Each component of the superposition evolves independently according to the value of the control qubit, resulting in a new superposition where different branches have undergone different transformations.

✂ Circuit Representation

In quantum circuit diagrams, the CU gate is typically represented with a **control dot on the control qubit** and a **box labeled with the unitary operation U on the target qubit**. The control dot indicates that the operation is conditional on the state of the control qubit. The circuit representation of the CU gate is as follows:



In summary, if the control qubit is $|1\rangle$, we apply U to the target qubit; otherwise do nothing.

⚙ Properties

The CU gate has the same properties as the CNOT gate because it is a generalization of the CNOT gate. Specifically:

- **Unitary**

$$\text{CU}^\dagger \text{CU} = I$$

- **Reversible**

$$\text{CU}^{-1} = \text{CU}^\dagger$$

- **Conditional Operation**

$$\text{CU}(|v_A\rangle \otimes |v_B\rangle) \neq \text{CU}|v_A\rangle \otimes \text{CU}|v_B\rangle$$

It does not act independently on each qubit, but rather applies a transformation to the target qubit based on the state of the control qubit.

- **Entangling.** The CU gate can create entanglement between the control and target qubits if the unitary operation U and the input state are chosen appropriately. For example, when $U = X$ (i.e., the CNOT gate), we have:

$$\text{CNOT}(|+\rangle |0\rangle) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

4.5.3 SWAP gate

The **SWAP gate** is a **two-qubit gate** that **exchanges the states of two qubits** and has **no control/target distinction**, it is **symmetric**. It simply swaps:

$$|v_A\rangle \otimes |v_B\rangle \xrightarrow{SWAP} |v_B\rangle \otimes |v_A\rangle \quad (55)$$

≡ Action on Basis States

The SWAP gate acts on the basis states as follows:

Input	Output
$ 00\rangle$	$ 00\rangle$
$ 01\rangle$	$ 10\rangle$
$ 10\rangle$	$ 01\rangle$
$ 11\rangle$	$ 11\rangle$

It essentially **swaps the second and third basis states**, while leaving the first and fourth unchanged.

✓* Matrix Representation

The matrix representation of the SWAP gate in the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ is:

$$SWAP = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (56)$$

This matrix clearly shows the **swapping of the second and third basis states**, while the first and fourth remain unchanged.

≡ Action on General States

Let's consider two arbitrary single-qubit states:

$$|v_A\rangle = a_0 |0\rangle + a_1 |1\rangle, \quad |v_B\rangle = b_0 |0\rangle + b_1 |1\rangle$$

The tensor product of these states is:

$$|v_A\rangle \otimes |v_B\rangle = a_0 b_0 |00\rangle + a_0 b_1 |01\rangle + a_1 b_0 |10\rangle + a_1 b_1 |11\rangle$$

Applying the SWAP gate to this state gives:

$$\begin{aligned} SWAP(|v_A\rangle \otimes |v_B\rangle) &= a_0 b_0 |00\rangle + a_1 b_0 |01\rangle + a_0 b_1 |10\rangle + a_1 b_1 |11\rangle \\ &= (b_0 |0\rangle + b_1 |1\rangle) \otimes (a_0 |0\rangle + a_1 |1\rangle) \\ &= |v_B\rangle \otimes |v_A\rangle \end{aligned}$$

This confirms that the SWAP gate indeed **exchanges the states of the two qubits**, as expected.

✂ Circuit Representation

In quantum circuit diagrams, the SWAP gate is typically represented by two crossed lines with $\times$ symbols at the crossing point:

$$\begin{array}{ccc} |v_A\rangle & \text{---} \times & |v_B\rangle \\ |v_B\rangle & \text{---} \times & |v_A\rangle \end{array}$$

This notation visually emphasizes the **swapping of the two qubits' states**.

⚙ Properties

- **Unitary**

$$\text{SWAP}^\dagger \text{SWAP} = I$$

This means that the SWAP gate preserves the inner product and is reversible.

- **Reversible (self-inverse)**

$$\text{SWAP}^2 = I$$

This means that applying the SWAP gate twice will return the system to its original state.

- **Symmetric (no control/target)**. The SWAP gate does not distinguish between the two qubits; it treats them symmetrically. This is in contrast to gates like CNOT, which have a clear control and target qubit.
- **Does NOT create entanglement**. The SWAP gate simply exchanges the states of two qubits and does **not introduce any new correlations** between them. If the input state is separable (not entangled), the output state will also be separable (and vice versa).
- **Can be decomposed into 3 CNOT gates**. The **SWAP gate can be implemented using three CNOT gates**, which is important for practical implementations on quantum hardware that may not have a native SWAP gate.

Proof. We want to transform:

$$|a, b\rangle \rightarrow |b, a\rangle$$

But CNOT only does:

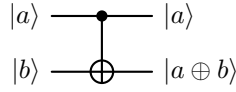
$$|a, b\rangle \rightarrow |a, a \oplus b\rangle$$

So we need to **combine multiple CNOTs** to achieve a SWAP. Let's track what happens to a generic input:

$$|a, b\rangle \quad \text{with } a, b \in \{0, 1\}$$

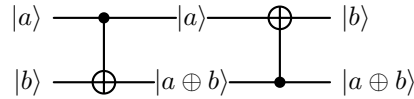
- **First CNOT** (control: first qubit, target: second qubit):

$$|a, b\rangle \xrightarrow{\text{CNOT}} |a, a \oplus b\rangle$$



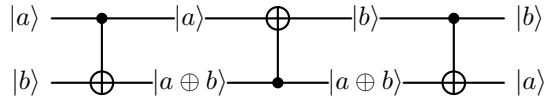
- **Second** CNOT (control: second qubit, target: first qubit):

$$|a, a \oplus b\rangle \xrightarrow{\text{CNOT}} |a \oplus (a \oplus b), a \oplus b\rangle = |b, a \oplus b\rangle$$

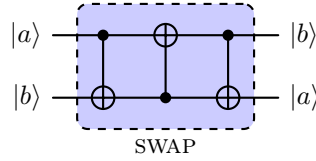


- **Third** CNOT (control: first qubit, target: second qubit):

$$|b, a \oplus b\rangle \xrightarrow{\text{CNOT}} |b, b \oplus (a \oplus b)\rangle = |b, a\rangle$$



In summary, the sequence of operations is:



QED

In general, the SWAP gate **does not create new correlations**; it only **re-orderes the information between qubits**. In other words, it only **changes where the information is stored**.

4.5.4 Controlled-Controlled-NOT (CCNOT) Gate (Toffoli)

The **Controlled-Controlled-NOT (CCNOT) Gate**, also called the **Toffoli Gate**, is a **three-qubit gate** composed of:

- Two **control qubits** that determine whether the gate will be applied.
- One **target qubit** that is flipped (NOT operation) if both control qubits are in the state $|1\rangle$.

It generalizes the CNOT gate by adding an additional control qubit, making it a crucial component in quantum computing for implementing more complex operations and algorithms.

≡ Action on Basis States

The CCNOT gate acts on the basis states as follows:

Input	Output
$ 000\rangle$	$ 000\rangle$
$ 001\rangle$	$ 001\rangle$
$ 010\rangle$	$ 010\rangle$
$ 011\rangle$	$ 011\rangle$
$ 100\rangle$	$ 100\rangle$
$ 101\rangle$	$ 101\rangle$
$ 110\rangle$	$ 111\rangle$
$ 111\rangle$	$ 110\rangle$

The target flips **only if both control qubits are $|1\rangle$** , demonstrating the conditional nature of the gate (last two rows of the table). For all other input combinations, the output remains unchanged.

√* Matrix Representation

The matrix representation of the CCNOT gate in the computational basis is an 8×8 (since it operates on three qubits) unitary matrix given by:

$$\text{CCNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \quad (57)$$

This matrix shows that the first six basis states are unchanged (diagonal entries of 1), while the **last two basis states are swapped** (off-diagonal entries of 1 in the last two rows and columns).

≡ Action on General States

For a general three-qubit state:

$$|\psi\rangle = \sum_{i,j,k \in \{0,1\}} \alpha_{ijk} |ijk\rangle$$

The CCNOT gate acts linearly on each component of the superposition and flips the target qubit (the third qubit) if and only if both control qubits are equal to $|1\rangle$. This can be expressed as:

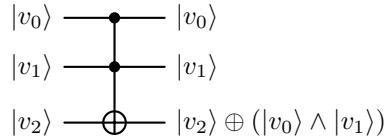
$$\text{CCNOT} |\psi\rangle = \sum_{i,j,k \in \{0,1\}} \alpha_{ijk} |ij(k \oplus (i \cdot j))\rangle \quad (58)$$

Where $\oplus$ denotes the XOR operation, and $i \cdot j = 1$ if and only if $i = j = 1$, and 0 otherwise.

In simple terms, applying CCNOT affects only the c_6 and c_7 coefficients (corresponding to $|110\rangle$ and $|111\rangle$) by swapping them, while all other coefficients remain unchanged.

✂ Circuit Representation

In quantum circuit diagrams, the CCNOT gate is typically represented by a symbol with two control dots connected to a NOT gate (X gate) on the target qubit. The control dots indicate which qubits are the controls, and the NOT gate indicates the operation performed on the target qubit when both controls are active. It is similar to the CNOT gate but with an additional control dot, making it visually distinct and easy to identify in quantum circuits.



In summary, if both control qubits are $|1\rangle$, apply NOT to the target; otherwise do nothing.

⚙ Properties

- **Unitary**

$$\text{CCNOT}^\dagger \text{CCNOT} = I$$

- **Reversible**

$$\text{CCNOT}^{-1} = \text{CCNOT}^\dagger$$

- **Classical universal gate (can simulate AND)**. CCNOT gate is a universal gate for classical computation, meaning that any classical

logic gate can be constructed using CCNOT gates and single-qubit NOT gates. In particular, it can compute the AND operation in a reversible way:

$$|a, b, 0\rangle \xrightarrow{\text{CCNOT}} |a, b, 0 \oplus (a \cdot b)\rangle = |a, b, a \cdot b\rangle$$

Where a and b are the control qubits, and the target qubit is initialized to $|0\rangle$. After applying the CCNOT gate, the target qubit will be in the state $|1\rangle$ if and only if both a and b are $|1\rangle$, effectively computing the AND of the two control qubits.

Since AND and NOT are sufficient to express any Boolean function, the Toffoli gate can reproduce any classical computation.

- **Introduce multi-qubit interaction.** Multi-qubit interaction refers to operations in which the evolution of one qubit depends on the state of one or more other qubits. Unlike parallel gates, which act independently on each qubit, **interacting gates introduce conditional behavior and correlations between qubits.**

Typical examples include controlled gates such as CNOT and CCNOT, where the transformation applied to a target qubit depends on the value of one or more control qubits. These interactions are essential for creating entanglement and for implementing complex quantum algorithms.

- **Can create entanglement.** Similar to the CNOT gate, also the CCNOT gate can create entanglement between the control and target qubits. For example, if the control qubits are in a superposition state, applying the CCNOT gate can result in an entangled state between the control and target qubits.
- **Not decomposable into a single tensor product.** An interacting multi-qubit gate cannot be expressed as a tensor product of single-qubit operators. That is, in general:

$$U \neq U_A \otimes U_B$$

This means that the gate introduces correlations between qubits and cannot be reduced to independent operations.

In summary, the Toffoli gate performs a conditional operation based on two qubits, enabling multi-qubit logic and more complex correlations.

4.6 Quantum Circuit Representation

In the previous sections, we have been using visual representations of quantum gates and their effects on qubits. However, a quantum circuit can be expressed in a more formal way using a **single unitary matrix** that represents the entire operation of the circuit. This matrix acts on the global state of the system and is obtained by composing the individual gates according to the structure of the circuit. This is known as the **quantum circuit representation**.

Definition 1: Quantum Circuit Representation

A **quantum circuit** can be represented as a **unitary matrix** U that acts on the global state vector of the qubits. The overall operation of the circuit is obtained by composing the unitary matrices corresponding to each layer of gates in the circuit.

Mathematically, if a circuit consists of layers $G_1, G_2, \dots, G_n$ with corresponding unitary matrices $U_1, U_2, \dots, U_n$, then the overall unitary matrix U representing the circuit is given by:

$$U = U_n U_{n-1} \cdots U_2 U_1$$

Where the rightmost operator is applied first.

Each U_i is constructed by combining gates acting in parallel using tensor products, and by extending gates with identity operators on qubits where no operation is applied.

In simple terms, a **quantum circuit is just a structured way to build a single unitary matrix acting on the whole system**. So:

$$|\psi_{\text{out}}\rangle = U |\psi_{\text{in}}\rangle$$

Where $|\psi_{\text{in}}\rangle$ is the initial state of the qubits, $|\psi_{\text{out}}\rangle$ is the final state after applying the circuit, and U is the **product of all the gates in the circuit**.

▲ The 3 Fundamental Rules of Quantum Circuit Representation

1. **Parallel composition.** Gates acting on different qubits are combined using the tensor product:

$$U = A \otimes B$$

2. **Sequential composition.** Gates applied in sequence are combined using matrix multiplication, evaluated from right to left:

$$U = U_n \cdots U_2 U_1$$

So the first gate applied is the rightmost one in the product.

3. **Identity extension.** When a gate does not act on a qubit, the identity operator must be used to extend it:

$$U = I \otimes A$$

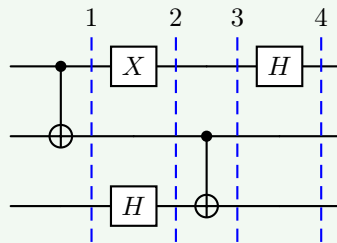
This keeps dimensions consistent when combining gates that act on different subsets of qubits.

Example 4: Quantum Circuit Representation

Let's consider the following quantum circuit representation:

$$\underbrace{(H \otimes I \otimes I)}_4 \cdot \underbrace{(I \otimes \text{CNOT})}_3 \cdot \underbrace{(X \otimes I \otimes H)}_2 \cdot \underbrace{(\text{CNOT} \otimes I)}_1$$

First of all, the order of the gates is from right to left, so we start with the rightmost gate.



4.7 Entanglement

As discussed on page 92, we have already seen why entanglement exists and matters. In short, describing an n -qubit quantum state requires an exponential number of parameters, meaning that most multi-qubit states are entangled.

If qubits were independent (separable), we would only need $2n$ real numbers, since each qubit would require only 2 real parameters:

$$|v\rangle = a_0 |0\rangle + a_1 |1\rangle$$

Each qubit needs two real parameters, so for n qubits, we would need $2n$ real numbers. However, a general quantum state of n qubits is given by (see page 90):

$$|\psi\rangle = \sum_{k=0}^{2^n-1} c_k |k\rangle$$

This representation requires 2^n complex coefficients (subject to normalization and global phase constraints), corresponding to an exponential number of real degrees of freedom.

In conclusion, most multi-qubit states cannot be described as independent qubits. This phenomenon is called **entanglement** and is a fundamental feature of quantum mechanics with no classical analog.

Definition 2: Entanglement

A multi-qubit state is called **Entangled** if it cannot be written as a tensor product of individual qubit states (i.e., it is not separable). Otherwise, it is called **separable** or **unentangled**. Entanglement can be viewed as a “failure of factorization”, where the state of the whole system cannot be factored into states of its parts.

An important consequence of entanglement is that we cannot assign a separate quantum state to each qubit independently.

✂ Analysis of a Real Example: Bell States

To better understand the concept of entanglement, it is useful to analyze a concrete example. In particular, we consider a class of two-qubit states known as **Bell states**, which provide the simplest and most important examples of entangled states. These states cannot be written as a tensor product of individual qubit states, and therefore explicitly demonstrate the impossibility of decomposing certain multi-qubit states into independent subsystems.

Definition 3: Bell States

The **Bell States** are a set of 4 two-qubit states that form an orthonormal basis of maximally entangled states. They are defined as:

$$\begin{aligned} |\Phi^+\rangle &= \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) & |\Phi^-\rangle &= \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle) \\ |\Psi^+\rangle &= \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle) & |\Psi^-\rangle &= \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle) \end{aligned} \quad (59)$$

These states are called **maximally entangled** because each qubit, when considered individually, is in a completely mixed (maximally uncertain) state. In other words, Bell states are characterized by the fact that **each individual qubit is in a maximally mixed (completely random) state**, while the **joint state exhibits perfect correlations between the qubits**.

This means that when we measure a single qubit of a Bell state, the outcome is completely random. In other words, there is a 50% chance of getting a 0 and a 50% chance of getting a 1. Therefore, we cannot say that it is mostly a 0 or a 1. However, once we measure a single qubit, a paradox occurs: even though each qubit alone is random, together they are perfectly correlated. In fact, if we measure the first qubit and get $|0\rangle$, we know with certainty that the second qubit is also $|0\rangle$ (in the case of $|\Phi^+\rangle$ and $|\Phi^-\rangle$), or $|1\rangle$ (in the case of $|\Psi^+\rangle$ and $|\Psi^-\rangle$). Thus, we have **maximum uncertainty individually but maximum correlation jointly**.

Example 5: Proof of Entanglement in Bell States

After introducing the Bell states, we would like to **proof that decomposing them into independent qubits is impossible**. In other words, we want to show that:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Can be written as:

$$|\Phi^+\rangle = |v_A\rangle \otimes |v_B\rangle$$

Proof. Let's assume, for the sake of argument, that $|\Phi^+\rangle$ can be written as a tensor product of two single-qubit states:

$$|\Phi^+\rangle = |v_A\rangle \otimes |v_B\rangle$$

Where:

- $|v_A\rangle = a_0 |0\rangle + a_1 |1\rangle$ is the state of the first qubit.
- $|v_B\rangle = b_0 |0\rangle + b_1 |1\rangle$ is the state of the second qubit.

The tensor product of these two states is:

$$|v_A\rangle \otimes |v_B\rangle = a_0 b_0 |00\rangle + a_0 b_1 |01\rangle + a_1 b_0 |10\rangle + a_1 b_1 |11\rangle$$

For this to equal $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, we must have:

$$a_0b_0 = \frac{1}{\sqrt{2}}, \quad a_1b_1 = \frac{1}{\sqrt{2}}, \quad a_0b_1 = 0, \quad a_1b_0 = 0$$

But here's the problem, if $a_0b_1 = 0$, then either $a_0 = 0$ or $b_1 = 0$:

✗ Case $a_0 = 0$, then $a_0b_0 = 0$, which **contradicts** $a_0b_0 = \frac{1}{\sqrt{2}}$ for $|00\rangle$.

✗ Case $b_1 = 0$, then $a_1b_1 = 0$, which **contradicts** $a_1b_1 = \frac{1}{\sqrt{2}}$ for $|11\rangle$.

Therefore, there is no way to choose a_0, a_1, b_0, b_1 such that the above equations hold simultaneously. This means that $|\Phi^+\rangle$ cannot be written as a tensor product of two single-qubit states, and thus it is an entangled state. All we can say is that these qubits are correlated and cannot be described independently. QED

? Entanglement: Intrinsic or Basis-Dependent?

Having established that Bell states are entangled by showing that they cannot be written as a tensor product of individual qubit states, a natural question arises: **is this property intrinsic to the state, or does it depend on the basis in which the state is expressed?** In other words, could a change of basis make an entangled state appear separable? The answer is no. **Entanglement** is not a property of the representation, but of the **structure of the state itself**.

In contrast to entanglement, **superposition depends on the basis we choose**, for example:

$$|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$$

This shows that the state $|0\rangle$ is a **superposition** of $|+\rangle$ and $|-\rangle$ in the **Hadamard basis**, but it is **not a superposition in the computational basis**, indeed:

$$|0\rangle = 1 \cdot |0\rangle + 0 \cdot |1\rangle$$

It is not a superposition because only one basis vector is present.²

Differently, **entanglement** does **not** depend on basis, because it depends on **factorization**, not representation. So we can change basis, rewrite the state, but we **cannot make an entangled state separable**. This prevents a common misunderstanding: one might wonder whether a suitable change of basis could remove entanglement. This is not the case. Entanglement is preserved under local changes of basis, that is, transformations of the form $U_A \otimes U_B$ acting independently on each qubit. Therefore, **entanglement is an intrinsic property of the quantum state and does not depend on the particular representation of each subsystem**.

²A state is a superposition in a given basis if it is written as a combination of multiple basis vectors with non-zero coefficients.

✂ From States to Operations

So far, we have analyzed entanglement as a property of quantum states, showing that certain states cannot be decomposed into independent subsystems. A natural question now arises: *how can such entangled states be generated in practice?* In other words, **which quantum operations are capable of transforming separable states into entangled ones?**

The answer lies in a special class of quantum gates, known as **entangling gates**, which introduce interactions between qubits and allow the creation of correlations that cannot be reduced to independent components.

Definition 4: Entangling Gates

A multi-qubit gate is called **entangling** (**Entangling Gates**) if it cannot be written as a tensor product of single-qubit gates. Equivalently, a gate is entangling if there exists at least one separable input state that it transforms into an entangled state.

Put simply, an **entangling gate is a gate that makes qubits interact and can create entanglement.**

⚠ Attention: Not every input becomes entangled. There exists at least one input that becomes entangled, but not all inputs necessarily do.

In simple terms, an entangling gate is one that can take two or more qubits that are initially uncorrelated (separable) and produce a state where the qubits become correlated in such a way that they cannot be described independently (entangled). For example, the CNOT gate is an entangling gate because it can create Bell states from separable inputs.

Example 6: CNOT as an Entangling Gate

Consider the separable state:

$$|+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$

This is a separable state, since it is written as a tensor product of two single-qubit states.

Now apply a CNOT gate (with the first qubit as control and the second as target, page 100):

$$\text{CNOT} \left(\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \right) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

The resulting state is the Bell state $|\Phi^+\rangle$, which is entangled. Therefore, the **CNOT gate is an entangling gate.**

Example 7: CNOT Does Not Always Create Entanglement

Not all inputs to an entangling gate produce entangled states. Consider the separable state:

$$|0\rangle \otimes |0\rangle = |00\rangle$$

Applying the CNOT gate:

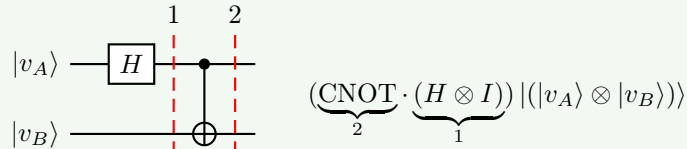
$$\text{CNOT}(|00\rangle) = |00\rangle$$

The state remains separable.

This shows that an entangling gate does not necessarily produce entanglement for every input. However, it is called **entangling** because there exists at least one separable input state that it transforms into an entangled state.

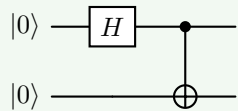
Example 8: Bell-state generator

Exists a simple quantum circuit that produces **different Bell states depending on the input**. So, for the four input states $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$, the circuit creates the four Bell states $|\Phi^+\rangle$, $|\Phi^-\rangle$, $|\Psi^+\rangle$, and $|\Psi^-\rangle$ respectively. This circuit consists of a Hadamard gate followed by a CNOT gate:



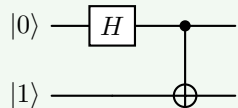
This circuit is a **Bell-state generator** because it can produce any of the four Bell states from the appropriate input.

- Input $|00\rangle$ produces $|\Phi^+\rangle$:



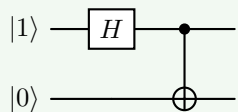
$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

- Input $|01\rangle$ produces $|\Phi^-\rangle$:



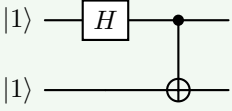
$$|\Phi^-\rangle = \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle)$$

- Input $|10\rangle$ produces $|\Psi^+\rangle$:



$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle)$$

- Input $|11\rangle$ produces $|\Psi^-\rangle$:



$$|\Psi^-\rangle = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle)$$

Example 9: Bell States in the Hadamard Basis

In this exercise, we express the Bell state $|\Phi^+\rangle$ in the Hadamard basis instead of the computational basis.

Bell state $|\Phi^+\rangle$ in the Hadamard basis. Given:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

We aim to re-express this state using the Hadamard basis $\{|+\rangle, |-\rangle\}$, where:

$$|+\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \quad |-\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

1. **Invert the definitions.** To convert from the computational basis to the Hadamard basis, we invert the above equations to express $|0\rangle$ and $|1\rangle$ in terms of $|+\rangle$ and $|-\rangle$:

$$|0\rangle = \frac{1}{\sqrt{2}} (|+\rangle + |-\rangle) \quad |1\rangle = \frac{1}{\sqrt{2}} (|+\rangle - |-\rangle)$$

2. **Substitute into $|00\rangle$ and $|11\rangle$.** We compute the tensor products using the expression above.

- Compute $|00\rangle$:

$$\begin{aligned} |00\rangle &= |0\rangle \otimes |0\rangle \\ &= \left(\frac{1}{\sqrt{2}} (|+\rangle + |-\rangle) \right) \otimes \left(\frac{1}{\sqrt{2}} (|+\rangle + |-\rangle) \right) \\ &= \frac{1}{2} (|++\rangle + |+-\rangle + |-+\rangle + |--\rangle) \end{aligned}$$

- Compute $|11\rangle$:

$$\begin{aligned} |11\rangle &= |1\rangle \otimes |1\rangle \\ &= \left(\frac{1}{\sqrt{2}} (|+\rangle - |-\rangle) \right) \otimes \left(\frac{1}{\sqrt{2}} (|+\rangle - |-\rangle) \right) \\ &= \frac{1}{2} (|++\rangle - |+-\rangle - |-+\rangle + |--\rangle) \end{aligned}$$

3. **Add the two to get $|\Phi^+\rangle$.** Now add the two results:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

Substituting:

$$\begin{aligned} |\Phi^+\rangle &= \frac{1}{\sqrt{2}} \left[\frac{1}{2} (|++\rangle + |+-\rangle + |-+\rangle + |--\rangle) \right. \\ &\quad \left. + \frac{1}{2} (|++\rangle - |+-\rangle - |-+\rangle + |--\rangle) \right] \\ &= \frac{1}{\sqrt{2}} \cdot \frac{1}{2} (2|++\rangle + 0|+-\rangle + 0|-+\rangle + 2|--\rangle) \\ &= \frac{1}{\sqrt{2}} \cdot \frac{1}{2} (2|++\rangle + 2|--\rangle) \\ &= \frac{1}{\sqrt{2}} (|++\rangle + |--\rangle) \end{aligned}$$

Thus, in the Hadamard basis, we have:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|++\rangle + |--\rangle)$$

Which is a Bell-like state expressed in the Hadamard basis instead of the computational basis. QED

❓ Why single-qubit gates cannot create entanglement?

Entanglement means that two qubits become so strongly correlated that they can no longer be described independently. To create this kind of correlation, the qubits must **interact**. Since single-qubit gates act on each qubit **independently**, they cannot introduce any interaction between qubits.

Suppose we have two qubits $|\psi_A\rangle$ and $|\psi_B\rangle$, and the combined state is:

$$|\psi\rangle = |\psi_A\rangle \otimes |\psi_B\rangle$$

The $|\psi\rangle$ is a **separable** state because it can be written as a tensor product of two single-qubit states (see above). In contrast, suppose we apply a gate U_A to the first qubit, and a gate U_B to the second qubit. The overall operation is:

$$U_A \otimes U_B$$

Applying this to the state $|\psi\rangle$ gives:

$$(U_A \otimes U_B)(|\psi_A\rangle \otimes |\psi_B\rangle) = (U_A |\psi_A\rangle) \otimes (U_B |\psi_B\rangle)$$

The result is **still a tensor product of two separate states**. That means the final state is still separable. Therefore, **single-qubit gates preserve separability**, and **cannot create entanglement**.

In general, to create entanglement, qubits must interact. We need a gate that acts on both qubits together, such as the CNOT gate. The CNOT gate is the most common example of an entangling gate because under the hood is very simple, but it can create complex correlations between qubits, including the Bell states. CNOT works because it makes the second qubit (the target) depend on the first qubit (the control), thus creating correlations that cannot be reduced to independent states. Instead, single-qubit gates never do this, because each qubit evolves independently.

Deepening: Physical Meaning of Entanglement

We have introduced entanglement as a mathematical property of quantum states, but it also has profound physical implications. We said that *a state is entangled when it cannot be factorized into tensor products of individual subsystem states*. But physics asks “*what does this impossibility mean physically?*”. And the answer is **entanglement means that the subsystems no longer possess independent physical states**. That is the key idea.

❓ Before Entanglement: Independent Systems. Suppose we have two independent qubits:

$$|\psi_A\rangle = a|0\rangle + b|1\rangle \quad |\psi_B\rangle = c|0\rangle + d|1\rangle$$

The joint system is:

$$|\psi_{AB}\rangle = |\psi_A\rangle \otimes |\psi_B\rangle = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$$

In this case, we can assign a separate state to each qubit, and they evolve independently. The state of the whole system is just the combination of the states of the parts. Physically this means that qubit *A* has its own state, and qubit *B* has its own state, and they do not affect each other.

⚠️ Entanglement: The Global State Exists, Local Ones Do Not.

Now consider the Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

We already know mathematically that this state cannot be written as a tensor product of two single-qubit states (page 115):

$$|\Phi^+\rangle \neq |v_A\rangle \otimes |v_B\rangle$$

But physically this means something deeper:

- There is a state for the **pair**;
- There is NOT a complete independent state for each particle separately.

The **system only has a meaningful description as a whole**. This is the **physical meaning**.

For example, with the Bell state $|\Phi^+\rangle$, the possible outcomes are $|00\rangle$ and $|11\rangle$. So if Alice measures her qubit and gets $|0\rangle$, Bob must get $|0\rangle$ as well. If Alice gets $|1\rangle$, Bob must get $|1\rangle$. The outcomes are perfectly correlated, but neither qubit has a definite state on its own. The state of the whole system is well-defined, but the state of each part is not. This is the essence of entanglement: the **parts do not have independent states; only the whole does**. Entanglement creates strong correlations and the particles behave as parts of a single object.

In summary, entanglement means that multiple quantum particles no longer possess independent physical states; **only the global system has a complete physical description**. Physically this means that the particles are not separate “*objects with separate properties*” anymore, but rather the information is stored in the relationship between them and the measurements become strongly correlated.

4.8 Observables and Measurements

4.8.1 Observables: What Are We Measuring?

Let's start from the basics: *what is an observable?* In quantum mechanics, an **Observable** is a mathematical object representing a measurable physical quantity.

- For example, in classical physics, observables are physical quantities that we can measure, such as position, momentum, energy, etc. (e.g., the position of a particle, the energy of a system).
- In quantum mechanics, the state is no longer a point, but rather a vector (wavefunction) in a complex vector space (Hilbert space). Because of this, observables can no longer be ordinary functions of the state, but instead must be represented as **operators** that act on the state vector.

Definition 5: Observable

In quantum mechanics, an **Observable** is a **Hermitian operator** O associated with a measurable quantity of a quantum system:

$$O = O^\dagger$$

Where $O^\dagger$ denotes the Hermitian adjoint (conjugate transpose) of O .

With *conjugate transpose*, we mean that if O is represented as a matrix, then $O^\dagger$ is obtained by taking the **transpose** of the matrix and then taking the **complex conjugate of each element**. The condition $O = O^\dagger$ ensures that the eigenvalues of O are real, which is necessary for them to correspond to measurable quantities in quantum mechanics.

❓ Why does quantum mechanics specifically choose Hermitian operators for observables?

The answer is simple: **because measurements must produce physically meaningful outcomes**. Specifically, Hermitian operators guarantee: real eigenvalues and orthonormal eigenvectors.

1. **Real eigenvalues**. The eigenvalues of a Hermitian operator are always real numbers, which is **essential because measurement outcomes must be real values**. We cannot measure $3 + 2i$ as a physical quantity, but we can measure 3. Mathematically, if O is a Hermitian operator, then:

$$O |k\rangle = \lambda_k |k\rangle$$

Where λ_k is the possible measurement outcome (eigenvalue) corresponding to the eigenvector $|k\rangle$, and $|k\rangle$ is the state of the system after the measurement.

Remark

When we write:

$$O |k\rangle = \lambda_k |k\rangle$$

We are choosing $|k\rangle$ to be one of the eigenvectors of O , and λ_k is its corresponding eigenvalue. It is the definition of the eigenvector/eigenvalue relationship.

Suppose O is an operator, so a matrix. An eigenvector $|k\rangle$ of O is a special vector such that, when O acts on it, the result is the **same vector direction**, only multiplied by a number. That number is called the eigenvalue λ_k . So:

$$O |k\rangle = \lambda_k |k\rangle$$

Means applying the operator O to $|k\rangle$ does not change its direction; it only scales it by λ_k .

Example 10: Eigenvectors and Eigenvalues of the Pauli-Z Operator

For example, let's take the Pauli-Z operator:

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

And the state $|0\rangle$:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Applying Z to $|0\rangle$:

$$Z |0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1 \cdot |0\rangle = \lambda_0 |0\rangle$$

Therefore, $|0\rangle$ is an eigenvector of Z with eigenvalue $\lambda_0 = 1$. Similarly, applying Z to $|1\rangle$:

$$Z |1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -1 \cdot |1\rangle = \lambda_1 |1\rangle$$

Therefore, $|1\rangle$ is an eigenvector of Z with eigenvalue $\lambda_1 = -1$.

Beyond the mathematical definition, $|0\rangle$ is a special state of the observable Z , measuring Z when the system is in $|0\rangle$ always gives the outcome $+1$. So, when we say that $|0\rangle$ and $|1\rangle$ are eigenvectors of Z , we are saying that these are the states in which the measurement of Z is **perfectly predictable**, yielding the eigenvalues $+1$ and -1 , respectively.

This is a fundamental aspect, because:

- ✓ If the **system is already in an eigenvector**: there is no uncertainty, and the measurement outcome is known with certainty.
- ⚠ In contrast, if the **system is not in an eigenvector**: it is a combination of eigenvectors, and measurement becomes **probabilistic**. For example, suppose the qubit is in:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

This is **not** an eigenvector of Z , because applying Z :

$$\begin{aligned} Z|+\rangle &= \frac{1}{\sqrt{2}} \left(\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) \\ &= \frac{1}{\sqrt{2}} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} \right) \\ &= \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \\ &= |-\rangle \end{aligned}$$

❓ **Why is $|+\rangle$ not an eigenvector of Z ?** If $|+\rangle$ were an eigenvector, we would have by definition:

$$Z|+\rangle = \lambda|+\rangle$$

For some eigenvalue λ . However, as we calculated, applying Z to $|+\rangle$ gives us $|-\rangle$, which is a different state. Therefore, there is no scalar λ such that $Z|+\rangle = \lambda|+\rangle$, confirming that $|+\rangle$ is not an eigenvector of Z . Physically, this means that if the system is in state $|+\rangle$, measuring Z does not yield a definite outcome; instead, it yields $+1$ with probability 0.5 and -1 with probability 0.5, reflecting the superposition of eigenstates.

In summary, if applying an observable O to a state $|\psi\rangle$ produces the same state multiplied by a scalar:

$$O|\psi\rangle = \lambda|\psi\rangle$$

Then $|\psi\rangle$ is an eigenvector of O , and λ is its eigenvalue. In this case, the state already has a definite value for that observable, so measuring O will return λ with certainty.

✓ **Physical consequence.** If the system is in an eigenvector: the **measurement is deterministic**, the **result is known in advance**, the **outcome is the corresponding eigenvalue**.

Why Hermitian operators have real eigenvalues. Let A be a Hermitian operator:

$$A = A^\dagger$$

Assume that $|x\rangle$ is an eigenvector of A with eigenvalue λ :

$$A|x\rangle = \lambda|x\rangle$$

Consider the quantity:

$$\langle x|A|x\rangle$$

Using the eigenvalue equation (i.e., $A|x\rangle = \lambda|x\rangle$) on the right:

$$\langle x|\underbrace{A|x\rangle}_{\lambda|x\rangle} = \langle x|(\lambda|x\rangle) = \lambda\langle x|x\rangle$$

This is possible because λ is just a scalar, so it can be moved outside bra-ket notation.³ Now take the Hermitian conjugate of the eigenvalue equation:

$$A|x\rangle = \lambda|x\rangle \implies \langle x|A^\dagger = \bar{\lambda}\langle x|$$

Since A is Hermitian:

$$A^\dagger = A$$

Therefore:

$$\langle x|A = \bar{\lambda}\langle x|$$

Applying this on the left of $\langle x|A|x\rangle$:

$$\langle x|\underbrace{A|x\rangle}_{\bar{\lambda}\langle x|} = (\bar{\lambda}\langle x|)|x\rangle = \bar{\lambda}\langle x|x\rangle$$

Combining the two expressions:

$$\lambda\langle x|x\rangle = \bar{\lambda}\langle x|x\rangle$$

Since:

$$\langle x|x\rangle > 0$$

Because the inner product of a vector with itself is always positive (unless the vector is the zero vector, which cannot be an eigenvector), we can divide both sides by $\langle x|x\rangle$:

$$\lambda = \bar{\lambda}$$

Therefore, λ is real.

QED

³Remember that bras and kets can be represented using vectors and matrices, so the usual rules of linear algebra apply. In particular, since λ is a scalar, it can be factored out of an inner product:

$$\langle x|(\lambda|x\rangle) = \lambda\langle x|x\rangle$$

More generally, scalar multiplication commutes with matrix products. Therefore, for matrices A and B :

$$\lambda AB = A\lambda B = AB\lambda$$

Provided that the order of the matrices themselves is preserved.

2. **Orthonormal eigenvectors.** A vector is:

- **Orthogonal** if its inner product with another vector is zero (i.e., they are perpendicular in the vector space). For example, in a 2D space, the vectors $|0\rangle$ and $|1\rangle$ are orthogonal because:

$$\langle 0|1\rangle = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0$$

- **Normalized** if its inner product with itself is equal to one (i.e., it has unit length). For example:

$$\langle 0|0\rangle = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1$$

This property is crucial because quantum measurement requires a basis whose states are both perfectly distinguishable (orthogonal) and probabilistically consistent (normalized).

- ✗ **If eigenvectors were not orthogonal**, different measurement states could partially overlap:

$$\langle v_1|v_2\rangle \neq 0$$

In this case, the measurement process would not be able to distinguish the states perfectly, leading to ambiguity in the interpretation of measurement outcomes.

Example 11: Why Orthogonality Matters

Consider the two states:

$$|v_1\rangle = |0\rangle \quad |v_2\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Their inner product is:

$$\langle v_1|v_2\rangle = \langle 0| \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}}$$

Since:

$$\langle v_1|v_2\rangle \neq 0$$

the two states are not orthogonal. This means that the **states partially overlap**, and therefore a **measurement cannot distinguish them perfectly**. For example, if the system is in state $|v_2\rangle$, there is still a non-zero probability of confusing it with $|v_1\rangle$ during a measurement. Therefore, the measurement outcomes would be ambiguous, and we would not be able to assign a clear physical meaning to the measurement results.

- ✗ **If eigenvectors were not normalized:**

$$\langle v|v\rangle \neq 1$$

Then the Born rule (page 80) would not produce properly normalized probabilities, making the probabilistic interpretation of quantum mechanics inconsistent.

Example 12: Why Normalization Matters

Consider the state:

$$|v\rangle = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

Its norm is:

$$\langle v|v\rangle = \begin{bmatrix} 2 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 4$$

Therefore, the state is not normalized.

If we tried to interpret the amplitudes probabilistically using the Born rule, we would obtain:

$$p_0 = |2|^2 = 4$$

Which is **impossible**, since **probabilities must lie between 0 and 1**, and their total sum must equal 1.

To obtain a valid quantum state, we must normalize the vector:

$$|v_{\text{norm}}\rangle = \frac{1}{2} \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Therefore, without normalization, we would end up with an invalid quantum state that cannot be interpreted probabilistically, and the measurement outcomes would not correspond to any physical reality.

Why eigenvectors of Hermitian operators are orthogonal. Let A be Hermitian and let:

$$A|x\rangle = \lambda|x\rangle, \quad A|y\rangle = \mu|y\rangle$$

Where $\lambda \neq \mu$ are the eigenvalues corresponding to the eigenvectors $|x\rangle$ and $|y\rangle$, respectively. Consider the inner product:

$$\langle y|A|x\rangle$$

Because we want to show that $|x\rangle$ and $|y\rangle$ are orthogonal, we need to show that $\langle y|x\rangle = 0$. Using the eigenvalue equation on the right:

$$\langle y|A|x\rangle = \langle y|(\lambda|x\rangle) = \lambda\langle y|x\rangle$$

We moved λ outside the inner product because it is a scalar (see the footnote in the previous proof for more details on this step). Now use the Hermitian property on the left:

$$\underbrace{\langle y|A}_{\langle y|A^\dagger} = \mu\langle y|$$

Therefore:

$$\langle y| A |x\rangle = \mu \langle y|x\rangle$$

Combining the two expressions:

$$\lambda \langle y|x\rangle = \mu \langle y|x\rangle$$

Thus:

$$(\lambda - \mu) \langle y|x\rangle = 0$$

Since $\lambda \neq \mu$ by assumption, we must have:

$$\langle y|x\rangle = 0$$

Therefore, the eigenvectors are orthogonal.

QED

We can think of a **Hermitian operator** as a sort of “*machine*” that **decomposes the quantum state into measurable components** (eigenvectors) **and assigns a real value** (eigenvalue) **to each component**, which corresponds to the possible outcomes of a measurement. The orthonormality of the eigenvectors ensures that these outcomes are distinct and can be interpreted probabilistically in a consistent way. In summary, a Hermitian observable tells us:

- *What outcomes are possible when we measure a quantum system.*
- *With respect to which basis we are measuring the system.*
- *Into which states the system will collapse after the measurement.*

Spectral Theorem

So far, we know that an observable O is a Hermitian operator, it has eigenvectors $|k\rangle$, and each eigenvector has an eigenvalue λ_k . So if the system is in $|k\rangle$, measuring O returns λ_k with certainty. Now, a natural question arises: **what happens if the system is in an arbitrary state $|k\rangle$, not necessarily one of the eigenvectors?** To answer this question, the **Spectral Theorem** comes into play by showing that **every observable can be written as a sum of its eigenvectors and eigenvalues**.

Suppose the observable has a set of eigenvectors $|k\rangle$ with corresponding eigenvalues λ_k . Since the eigenvectors form an orthonormal basis, any quantum state $|v\rangle$ can be expressed as a linear combination of these eigenvectors (page 90):

$$|v\rangle = \sum_k c_k |k\rangle$$

Where c_k are complex coefficients. The **Spectral Theorem** states that the observable O can be expressed as:

$$O = \sum_k \lambda_k |k\rangle\langle k| \quad (60)$$

Where each term $\lambda_k |k\rangle\langle k|$ contains two pieces of information:

- **Projector** $|k\rangle\langle k|$ (page 77) onto the eigenvector $|k\rangle$, which extracts the component of the state along $|k\rangle$ during measurement. It is defined as:

$$|k\rangle\langle k| = |k\rangle \langle k|$$

- **Eigenvalue** λ_k , which assigns the measurement outcome associated with that component.

🔍 **Why the Spectral Theorem is important?** Suppose someone gives us an observable O without telling us its eigenvectors or eigenvalues:

$$O = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Without the Spectral Theorem, the physical meaning of the observable would not be immediately apparent. However, by applying the Spectral Theorem, we can decompose O into its eigenvectors and eigenvalues, allowing us to **understand the measurement process and predict the outcomes based on the state of the system**. For example, we can find the eigenvalues and eigenvectors of O :

- To find the eigenvalues, we solve the characteristic equation:

$$\begin{aligned} \det(O - \lambda I) &= 0 \\ \det\left(\begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix}\right) &= 0 \\ (2-\lambda)(2-\lambda) - 1 \cdot 1 &= 0 \\ 4 - 4\lambda + \lambda^2 - 1 &= 0 \\ \lambda^2 - 4\lambda + 3 &= 0 \\ (\lambda - 3)(\lambda - 1) &= 0 \end{aligned}$$

Thus, the eigenvalues are $\lambda_1 = 3$ and $\lambda_2 = 1$.

- Eigenvectors can be found by solving $(O - \lambda I)|k\rangle = 0$ for each eigenvalue:
 - For $\lambda_1 = 3$:

$$\begin{aligned} (O - 3I)|v_1\rangle &= 0 \\ \left(\begin{bmatrix} 2-3 & 1 \\ 1 & 2-3 \end{bmatrix}\right)|v_1\rangle &= 0 \\ \left(\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\right)|v_1\rangle &= 0 \end{aligned}$$

Now to find the eigenvector, we need to solve the system of equations:

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

That is a homogeneous system of linear equations:

$$\begin{cases} -x + y = 0 \\ x - y = 0 \end{cases} = \begin{cases} y = x \\ x = y \end{cases}$$

So any vector of the form (x, x) is an eigenvector. We choose the simplest one $(1, 1)$, and then we normalize it:

$$\left\| \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

Therefore, the normalized eigenvector is:

$$|v_1\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

– For $\lambda_2 = 1$:

$$\begin{aligned} (O - 1I) |v_2\rangle &= 0 \\ \left(\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \right) |v_2\rangle &= 0 \\ \left(\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \right) |v_2\rangle &= 0 \end{aligned}$$

We need to solve the system of equations:

$$\begin{cases} x + y = 0 \\ x + y = 0 \end{cases} \Rightarrow x + y = 0 \Rightarrow y = -x \xrightarrow{\text{substitute from } \begin{bmatrix} x \\ y \end{bmatrix}} \begin{bmatrix} x \\ -x \end{bmatrix}$$

So any vector of the form $(x, -x)$ is an eigenvector. We choose the simplest one $(1, -1)$, and then we normalize it:

$$\left\| \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$$

Therefore, the normalized eigenvector is:

$$|v_2\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

Then we can express O in terms of its spectral decomposition:

$$O = 3 |v_1\rangle\langle v_1| + 1 |v_2\rangle\langle v_2|$$

From this decomposition, we can immediately read off the possible outcomes from the eigenvalues (3 and 1), the measurement basis from the eigenvectors $(|v_1\rangle$ and $|v_2\rangle)$, and the measurement operators from the projectors $(|v_1\rangle\langle v_1|$ and $|v_2\rangle\langle v_2|)$. It also allows us to calculate the probabilities of obtaining each outcome when measuring O on an arbitrary state $|\psi\rangle$ using the Born rule:

$$p_1 = |\langle v_1 | \psi \rangle|^2 \quad p_2 = |\langle v_2 | \psi \rangle|^2$$

Where p_1 and p_2 are the probabilities of obtaining the outcomes 3 and 1, respectively, when measuring the observable O on a state $|\psi\rangle$. And finally the expectation value of O in the state $|\psi\rangle$ can be calculated as:

$$\langle \psi | O | \psi \rangle = \sum_k \lambda_k p_k$$

In this example, this becomes:

$$\langle \psi | O | \psi \rangle = \sum_{k=1}^2 \lambda_k p_k = 3p_1 + 1p_2$$

In summary, the Spectral Theorem transforms an abstract Hermitian matrix into a complete description of a quantum measurement. It reveals:

- The **possible measurement outcomes** (eigenvalues),
- The **corresponding post-measurement states** (eigenvectors),
- The **projectors used to compute probabilities**,
- And the **expectation value** as a weighted average of the outcomes.

4.8.2 Measurement Operators and Projectors

In this section, we will explain **how measurement is actually performed mathematically**.

Measurement Operators

After introducing the Spectral Theorem, we know that any observable can be written as:

$$O = \sum_k \lambda_k |k\rangle\langle k|$$

Where:

- λ_k are the possible measurement outcomes (eigenvalues);
- $|k\rangle$ are the corresponding eigenvectors;
- $|k\rangle\langle k|$ are projection operators onto the eigenvectors.

At this point, a natural question arises: **what role do these projectors play in the measurement process?**

The answer is that each projector:

$$M_k = |k\rangle\langle k|$$

Is called a **Measurement Operator** (page 77). It represents the mathematical operation associated with the measurement outcome λ_k .

Definition 6: Measurement Operator

Given an observable (page 123):

$$O = \sum_k \lambda_k |k\rangle\langle k|$$

The **Measurement Operator** associated with the outcome λ_k is:

$$M_k = |k\rangle\langle k| = \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} \begin{pmatrix} \overline{k_1} & \overline{k_2} \end{pmatrix} = \begin{pmatrix} k_1 \overline{k_1} & k_1 \overline{k_2} \\ k_2 \overline{k_1} & k_2 \overline{k_2} \end{pmatrix}$$

In other words, **each possible measurement outcome has an associated projector** that:

1. **Extracts the component** of the quantum state along the eigenvector $|k\rangle$;
2. **Determines the probability** of obtaining the corresponding eigenvalue λ_k ;
3. **Produces the post-measurement** state after collapse.

For this reason, a measurement operator can be thought of as a **detector** tuned to a specific eigenstate. If the detector corresponding to $|k\rangle$ “clicks”, then:

- The measurement outcome is λ_k ;
- The system collapses to the state $|k\rangle$.

For example, suppose an observable has two eigenstates $|0\rangle$ and $|1\rangle$. Then there are two measurement operators:

$$M_0 = |0\rangle\langle 0|, \quad M_1 = |1\rangle\langle 1|$$

We can imagine them as two detectors:

- Detector M_0 looks for the component of the state along $|0\rangle$.
- Detector M_1 looks for the component of the state along $|1\rangle$.

Suppose the state is:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

This means that part of the state points along $|0\rangle$, and part of the state points along $|1\rangle$. The detector $M_0 = |0\rangle\langle 0|$ asks “*how much of the state is aligned with $|0\rangle$?*” and extracts $\alpha |0\rangle$. The detector $M_1 = |1\rangle\langle 1|$ asks “*how much of the state is aligned with $|1\rangle$?*” and extracts $\beta |1\rangle$.

Example 13: Example: Measurement Operators for Pauli-Z

The Pauli-Z observable can be written as:

$$Z = |0\rangle\langle 0| - |1\rangle\langle 1|$$

Therefore, the associated measurement operators are:

$$M_0 = |0\rangle\langle 0|, \quad M_1 = |1\rangle\langle 1|$$

These operators correspond to the two possible outcomes:

- +1, associated with the eigenstate $|0\rangle$;
- -1, associated with the eigenstate $|1\rangle$.

Suppose the system is in the state:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

Applying the measurement operator M_0 :

$$M_0 |\psi\rangle = |0\rangle\langle 0| (\alpha |0\rangle + \beta |1\rangle) = \alpha |0\rangle$$

This means that the operator M_0 extracts only the component of the state aligned with $|0\rangle$ and removes the orthogonal component $|1\rangle$.

The probability of obtaining the corresponding outcome is:

$$p_0 = \langle \psi | M_0 | \psi \rangle = |\alpha|^2$$

Similarly:

$$p_1 = \langle \psi | M_1 | \psi \rangle = |\beta|^2$$

Therefore, **measurement operators** provide the complete mathematical description of the measurement process: they **isolate the relevant component** of the quantum state, **determine the probability** of each outcome, and **specify the state** after the measurement.

☰ Completeness of the Basis

In the previous example, we defined two measurement operators M_0 and M_1 corresponding to the two eigenstates of the Pauli-Z observable. But what if we had an observable with more eigenstates? Or what if we had a different observable with a completely different set of eigenvectors? Therefore, **how do we know that the measurement operators we defined account for all possible outcomes of the measurement?**

The answer is that the eigenvectors of a Hermitian observable form a **complete orthonormal basis**. This means that any quantum state can be expressed as a linear combination of these eigenvectors:

$$|\psi\rangle = \sum_k c_k |k\rangle$$

Mathematically, this property is expressed by the **Completeness Relation**:

$$\sum_k |k\rangle\langle k| = I$$

Where I is the identity operator (page 46).

Definition 7: Completeness Relation

If $\{|k\rangle\}$ is a complete orthonormal basis, then:

$$\sum_k |k\rangle\langle k| = I \quad (61)$$

Since the measurement operators are defined as:

$$M_k = |k\rangle\langle k|$$

It follows that:

$$\sum_k M_k = I$$

This means that the **set of measurement operators is complete and accounts for all possible outcomes of the measurement.**

Deepening: When a set of vectors $\{|k\rangle\}$ is a complete orthonormal basis?

A set of vectors $\{|k\rangle\}$ is called a **Complete Orthonormal Basis** when it satisfies **three conditions**:

1. **Normalized**, each vector has length 1: $\langle k|k \rangle = 1$.
2. **Orthogonal**, different vectors are perpendicular: $\langle k|j \rangle = 0$ for $k \neq j$.
3. **Complete**, the vectors span the entire Hilbert space, meaning that every quantum state can be written as a linear combination of them. That is, for any state $|\psi\rangle$, there exist coefficients c_k such that:

$$|\psi\rangle = \sum_k c_k |k\rangle$$

Compactly, the *normalized* and *orthogonal* conditions are written as:

$$\langle k|j \rangle = \delta_{kj} \quad (62)$$

Where δ_{kj} is the **Kronecker delta**, which is 1 if $k = j$ and 0 otherwise:

$$\delta_{kj} = \begin{cases} 1 & k = j \\ 0 & k \neq j \end{cases} \quad (63)$$

And the *completeness* condition is expressed by the completeness relation:

$$\sum_k |k\rangle\langle k| = I$$

Example 14: Single-Qubit Computational Basis

For a single qubit, the computational basis consists of the vectors:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

These vectors are:

- Normalized: $\langle 0|0 \rangle = 1$ and $\langle 1|1 \rangle = 1$.
- Orthogonal: $\langle 0|1 \rangle = 0$.
- Complete: any single-qubit state can be expressed as a linear combination of $|0\rangle$ and $|1\rangle$.

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

This Equation 61 states that the **set of measurement operators is complete**: together, they **describe every possible outcome of the measurement**.

🔍 Why is the completeness relation important? To understand why this relation is important, let us apply both sides to an arbitrary state:

$$|\psi\rangle = \sum_k c_k |k\rangle$$

Where c_k are the coefficients of $|\psi\rangle$ in the basis $\{|k\rangle\}$.

Now take the inner product with $\langle k|$ on both sides:

$$\langle k|\psi\rangle = \langle k|\left(\sum_j c_j |j\rangle\right)$$

Using linearity of the inner product:

$$\langle k|\psi\rangle = \sum_j c_j \langle k|j\rangle$$

Since the basis is orthonormal, $\langle k|j\rangle = \delta_{kj}$, so only the term where $j = k$ survives (Equation 62, page 136):

$$\langle k|j\rangle = \delta_{kj} = \begin{cases} 1 & k = j \\ 0 & k \neq j \end{cases}$$

Thanks to the Kronecker delta, all terms in the sum vanish except the one where $j = k$:

$$\langle k|\psi\rangle = c_k \langle k|k\rangle = c_k \cdot 1 = c_k$$

In other words, the bra $\langle k|$ acts as a coefficient extractor: it selects exactly the amount of $|\psi\rangle$ aligned with the basis vector $|k\rangle$.

Finally, apply the sum of all projectors to $|\psi\rangle$:

$$\left(\sum_k |k\rangle\langle k|\right) |\psi\rangle = \sum_k |k\rangle \langle k|\psi\rangle = \sum_k |k\rangle c_k = \sum_k c_k |k\rangle = |\psi\rangle$$

This shows that the **sum of all projectors reproduces the original state exactly**. In other words, **each projector $|k\rangle\langle k|$ extracts one component of the state, and when all these components are added together, the original state is reconstructed without losing any information.** Therefore:

$$\sum_k |k\rangle\langle k| = I$$

Example 15: Computational Basis

For a single qubit, the computational basis is:

$$|0\rangle, \quad |1\rangle$$

The corresponding projectors are:

$$|0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad |1\rangle\langle 1| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Their sum is:

$$|0\rangle\langle 0| + |1\rangle\langle 1| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

The completeness relation also guarantees that the probabilities of all possible outcomes sum to one. If:

$$p_k = \langle \psi | M_k | \psi \rangle$$

Then:

$$\begin{aligned} \sum_k p_k &= \sum_k \langle \psi | M_k | \psi \rangle \\ &= \langle \psi | \left(\sum_k M_k \right) | \psi \rangle \\ &= \langle \psi | I | \psi \rangle \\ &= \langle \psi | \psi \rangle \\ &= 1 \end{aligned}$$

Therefore, the **completeness relation** ensures that:

- **All possible measurement outcomes** are included;
- **No probability is lost**;
- The **total probability** is always **equal to one**.

4.8.3 Expectation Value

In the previous sections, we have seen how to compute the probability of obtaining a certain measurement outcome when measuring an observable on a quantum state. However, in many cases, we are **interested in the average value of an observable over many measurements**, rather than the probability of specific outcomes. This average value is known as the **expectation value** of the observable.

Definition 8: Expectation Value

Let O be an observable represented by a Hermitian operator, and let $|\psi\rangle$ be a quantum state. The quantity:

$$\langle O \rangle = \langle \psi | O | \psi \rangle = \sum_k \lambda_k p_k \quad (64)$$

Is called the **Expectation Value** of the observable O in the state $|\psi\rangle$.

 **Physical Meaning.** The expectation value $\langle O \rangle$ can be interpreted as the **average result obtained by measuring the observable O many times on identical copies of the state $|\psi\rangle$.**

1. Prepare many identical copies of the state $|\psi\rangle$.
2. Measure the observable O on each copy.
3. Record the outcomes.
4. Compute the arithmetic mean of the recorded outcomes.

As the number of experiments increases, the average converges to the expectation value $\langle \psi | O | \psi \rangle$.

Important Clarification

The expectation value is **not necessarily** one of the possible measurement outcomes. It is the **average (more precisely, the probability-weighted average)** of many outcomes, and can take values that are not eigenvalues of the observable. For example, if possible outcomes are $+1$ with probability 50% and -1 with probability 50%, then the expectation value is:

$$(+1) \cdot \frac{1}{2} + (-1) \cdot \frac{1}{2} = 0$$

Although the actual measurement never returns 0, the expectation value is 0 because it represents the average of many measurements. This means that half of the measurements return $+1$ and the other half return -1 , resulting in an average of 0.

The expectation value tells us the “center of mass” of the measurement distribution. **It summarizes the statistical behavior of the measurement in a single number.**

❓ **Why are we interested in the average value of an observable rather than in the probability of specific outcomes?** Because in many real situations we do not **care about** the result of a single measurement, but about the **overall behavior of the system**. Also, it proves a stable, experimentally meaningful quantity that summarizes the statistical behavior of an observable over many measurements, even when individual outcomes are random.

✂ **How can we compute the expectation value?**

To compute the expectation value $\langle O \rangle$ of an observable O in a state $|\psi\rangle$, we start from the Spectral Theorem (page 129):

$$O = \sum_k \lambda_k |k\rangle\langle k|$$

Substitute this into the expectation value formula (page 139):

$$\begin{aligned} \langle \psi | O | \psi \rangle &= \langle \psi | \left(\sum_k \lambda_k |k\rangle\langle k| \right) | \psi \rangle \\ &\downarrow \text{ use linearity of the inner product} \\ &= \sum_k \lambda_k \langle \psi | k \rangle \langle k | \psi \rangle \end{aligned}$$

Using the property of inner products (page 7):

$$\langle \psi | k \rangle = \overline{\langle k | \psi \rangle} \quad \implies \quad \overline{\langle k | \psi \rangle} \langle k | \psi \rangle = |\langle k | \psi \rangle|^2$$

Here, the Born rule (page 80) states that $|\langle k | \psi \rangle|^2$ is the probability of measuring the outcome λ_k when measuring O on the state $|\psi\rangle$. Therefore, we can rewrite the expectation value as:

$$p_k = |\langle k | \psi \rangle|^2 \quad \implies \quad \langle \psi | O | \psi \rangle = \sum_k \lambda_k p_k \quad (65)$$

This formula shows that the expectation value is the **weighted average** of the possible outcomes λ_k , where each outcome is weighted by its probability p_k . In other words, the **expectation value is the sum of all possible outcomes multiplied by their respective probabilities**, which is consistent with our interpretation of it as an average over many measurements.

4.8.4 Measuring $|+\rangle$ with the Pauli-Z Observable

After introducing observables, projectors, probabilities, and expectation values, it is useful to analyze a concrete example that combines all these concepts.

Consider the quantum state:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

And the Pauli-Z observable:

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Calculate the possible measurement outcomes, their probabilities, the post-measurement states, and the expectation value of Z in the state $|+\rangle$.

From the spectral decomposition (page 129) of Z , we know that:

- The **eigenvalues** of Z are $+1$ and -1 :

$$\begin{aligned} \det(Z - \lambda I) &= 0 \\ \det\left(\begin{bmatrix} 1-\lambda & 0 \\ 0 & -1-\lambda \end{bmatrix}\right) &= 0 \\ (1-\lambda)(-1-\lambda) &= 0 \\ \lambda &= \pm 1 \end{aligned}$$

- The **first eigenvector** $|v_1\rangle$ is $|0\rangle$:

$$\begin{aligned} (Z - 1I)|v_1\rangle &= 0 \\ \begin{bmatrix} 1-1 & 0 \\ 0 & -1-1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= 0 \\ \begin{bmatrix} 0 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= 0 \\ \begin{cases} 0 \cdot x + 0 \cdot y = 0 \\ 0 \cdot x - 2 \cdot y = 0 \end{cases} &\implies y = 0 \end{aligned}$$

So any vector of the form $\begin{pmatrix} x \\ y \end{pmatrix} \xrightarrow{y=0} \begin{bmatrix} x \\ 0 \end{bmatrix}$ is an eigenvector. We choose the simplest one $(1, 0)$, and then we normalize it:

$$\left\| \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\| = \sqrt{1^2 + 0^2} = 1$$

Therefore, the normalized eigenvector is:

$$|v_1\rangle = 1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

- Similarly, the **second eigenvector** $|v_2\rangle$ is $|1\rangle$:

$$\begin{aligned}
 (Z + 1I) |v_2\rangle &= 0 \\
 \begin{bmatrix} 1+1 & 0 \\ 0 & -1+1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= 0 \\
 \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= 0 \\
 \begin{cases} 2 \cdot x + 0 \cdot y = 0 \\ 0 \cdot x + 0 \cdot y = 0 \end{cases} &\implies x = 0
 \end{aligned}$$

So any vector of the form $\begin{pmatrix} x \\ y \end{pmatrix} \xrightarrow{x=0} \begin{bmatrix} 0 \\ y \end{bmatrix}$ is an eigenvector. We choose the simplest one $(0, 1)$, and then we normalize it:

$$\left\| \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\| = \sqrt{0^2 + 1^2} = 1$$

Therefore, the normalized eigenvector is:

$$|v_2\rangle = 1 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

Therefore:

- The possible **measurement outcomes** are **+1** and **-1**;
- The corresponding **eigenstates** are **$|0\rangle$** and **$|1\rangle$** ;
- The **measurement operators** are:

$$M_0 = |0\rangle \langle 0|, \quad M_1 = |1\rangle \langle 1|$$

Measurement Probabilities

The state $|+\rangle$ is an equal superposition of $|0\rangle$ and $|1\rangle$:

$$|+\rangle = \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle$$

Therefore, the amplitudes are:

$$\alpha = \frac{1}{\sqrt{2}}, \quad \beta = \frac{1}{\sqrt{2}}$$

Using the Born rule:

$$p_0 = |\alpha|^2 = \frac{1}{2}, \quad p_1 = |\beta|^2 = \frac{1}{2}$$

Thus:

- There is a 50% probability of obtaining the outcome +1;
- There is a 50% probability of obtaining the outcome -1.

Post-Measurement States

If the measurement outcome is:

- $+1$, the state collapses to $|0\rangle$;
- -1 , the state collapses to $|1\rangle$.

Expectation Value

The expectation value of Z in the state $|+\rangle$ is:

$$\langle + | Z | + \rangle = (+1) \cdot \frac{1}{2} + (-1) \cdot \frac{1}{2} = 0$$

Remark

The expectation value is 0, but this does **not** mean that the measurement can return the value 0. The only possible outcomes are:

$$+1 \quad \text{and} \quad -1$$

The value 0 is simply the probability-weighted average of these outcomes over many repeated measurements.

What if the state to be measured is already an eigenstate of the observable?

Let's assume that now we want to calculate the measurement outcomes, probabilities, post-measurement states, and expectation value when the input state is $|0\rangle$ or $|1\rangle$ instead of $|+\rangle$. How we have seen, $|0\rangle$ and $|1\rangle$ are eigenvectors of Z with eigenvalues $+1$ and -1 , respectively. This means that they are already measurement states of the observable Z . Let's analyze what happens in these cases.

- **Measuring $|0\rangle$:** the measurement probabilities of $|0\rangle$ are:

$$|0\rangle = 1 \cdot |0\rangle + 0 \cdot |1\rangle$$

Therefore:

$$p_0 = |\alpha|^2 = 1, \quad p_1 = |\beta|^2 = 0$$

Using observable Z , there is a 100% probability of obtaining the outcome $+1$ and a 0% probability of obtaining the outcome -1 . Moreover, the (post-)measurement outcome is always $+1$, and the post-measurement state remains $|0\rangle$ because we have 100% probability of collapsing to $|0\rangle$ and 0% probability of collapsing to $|1\rangle$. Finally, the expectation value is:

$$\langle 0 | Z | 0 \rangle = (+1) \cdot 1 + (-1) \cdot 0 = +1$$

And indeed, the expectation value confirms that the measurement outcome is always $+1$.

- **Measuring** $|1\rangle$: the measurement probabilities of $|1\rangle$ are:

$$|1\rangle = 0 \cdot |0\rangle + 1 \cdot |1\rangle$$

Therefore:

$$p_0 = |\alpha|^2 = 0, \quad p_1 = |\beta|^2 = 1$$

Using observable Z , there is a 0% probability of obtaining the outcome $+1$ and a 100% probability of obtaining the outcome -1 . Moreover, the (post-)measurement outcome is always -1 , and the post-measurement state remains $|1\rangle$ because we have 0% probability of collapsing to $|0\rangle$ and 100% probability of collapsing to $|1\rangle$. Finally, the expectation value is:

$$\langle 1|Z|1\rangle = (+1) \cdot 0 + (-1) \cdot 1 = -1$$

And indeed, the expectation value confirms that the measurement outcome is always -1 .

In summary, **when a quantum state is already an eigenvector of the observable being measured:**

- The **measurement** outcome is **deterministic**;
- The **probability** of the corresponding eigenvalue is **1**;
- The **state does not change** after the measurement;
- The **expectation value** is exactly the associated **eigenvalue**.

In contrast, when the state is a superposition of eigenvectors (as in the case of $|+\rangle$), the measurement outcome is probabilistic and the expectation value becomes a weighted average of the possible outcomes.

4.8.5 Recovering Probabilities from Expectation Values

In the previous sections, we have seen that the expectation value of an observable is the weighted average of its possible measurement outcomes. But, is possible to go in the opposite direction? That is, **if we know the expectation value, can we determine the probabilities of the individual outcomes?** Let's explore how to get **Probabilities from the Expectation Value**.

Let's consider the Pauli-Z observable as an example.

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

With (proof at page 141):

- Eigenvalue $+1$ associated with the state $|0\rangle$;
- Eigenvalue -1 associated with the state $|1\rangle$.

Let:

- p_0 be the probability of measuring the state $|0\rangle$;
- p_1 be the probability of measuring the state $|1\rangle$.

Since Pauli-Z is a valid observable, the probabilities must satisfy:

$$p_0 + p_1 = 1$$

Therefore, the expectation value of Z is:

$$\langle Z \rangle = (+1)p_0 + (-1)p_1 = p_0 - p_1$$

Where $+1$ and -1 are the eigenvalues of Z corresponding to the states $|0\rangle$ and $|1\rangle$, respectively. Instead, p_0 and p_1 are the probabilities of obtaining those states upon measurement.

💡 Now, we are interested in the opposite direction: given the expectation value $m = \langle Z \rangle$ (we change the notation just to be general), can we determine p_0 and p_1 ? The answer is yes, and we can derive the formulas for p_0 and p_1 in terms of m (or $\langle Z \rangle$). We have two relationships:

$$\begin{aligned} p_0 + p_1 &= 1 \\ p_0 - p_1 &= m \end{aligned}$$

We can **solve this system of equations to express p_0 and p_1 in terms of m , and thus recover the probabilities from the expectation value.** Let's do that step by step.

Let's try to solve this system of equations to express p_0 and p_1 in terms of m :

$$\begin{aligned}
 \begin{cases} p_0 + p_1 = 1 \\ p_0 - p_1 = m \end{cases} &= \begin{cases} p_0 = 1 - p_1 \\ p_1 = -m + p_0 \end{cases} \\
 &= \begin{cases} p_0 = 1 - p_1 \\ p_1 = -m + (1 - p_1) \end{cases} \\
 &= \begin{cases} p_0 = 1 - p_1 \\ 2p_1 = -m + 1 \end{cases} \\
 &= \begin{cases} p_0 = 1 - p_1 \\ p_1 = \frac{1-m}{2} \end{cases} \\
 &= \begin{cases} p_0 = 1 - \frac{1-m}{2} \\ p_1 = \frac{1-m}{2} \end{cases} \\
 &= \begin{cases} p_0 = \frac{2-(1-m)}{2} \\ p_1 = \frac{1-m}{2} \end{cases} \\
 &= \begin{cases} p_0 = \frac{1+m}{2} \\ p_1 = \frac{1-m}{2} \end{cases}
 \end{aligned}$$

So, we have:

$$p_0 = \frac{1+m}{2} \quad \text{and} \quad p_1 = \frac{1-m}{2} \quad (66)$$

🔍 Why is this important? This result shows that the expectation value of the Pauli-Z observable contains enough information to reconstruct the complete probability distribution of the measurement outcomes. In other words, by measuring only the average value $m = \langle Z \rangle$, we can determine:

- The probability p_0 of obtaining the state $|0\rangle$;
- The probability p_1 of obtaining the state $|1\rangle$.

⚠️ Not always true in general

For a general observable with more than two outcomes, one expectation value is usually **not enough**. In general, to determine n unknown probabilities, we need n independent equations. Normalization provides one equation, so we typically need additional information such as: more expectation values, higher moments (e.g., $\langle O^2 \rangle$), or other constraints to fully determine the probability distribution.

However, if an observable has only two distinct outcomes a and b , then the same idea works. If:

- Outcome a occurs with probability p ,
- Outcome b occurs with probability $1 - p$,

Then the expectation value is:

$$\langle O \rangle = a \cdot p + b \cdot (1 - p) \xrightarrow{\text{solve for } p} p = \frac{\langle O \rangle - b}{a - b} \quad (67)$$

It finds **the probability of the first outcome** a , and the **probability of the second outcome** b can be found by normalization:

$$1 - p = 1 - \frac{\langle O \rangle - b}{a - b} = \frac{a - \langle O \rangle}{a - b} \quad (68)$$

For example, for the Pauli-Z observable, we have $a = +1$ and $b = -1$, so we can recover the probabilities as:

$$p_0 = P(+1) = \frac{\langle Z \rangle - (-1)}{(+1) - (-1)} = \frac{\langle Z \rangle + 1}{2}$$

$$p_1 = P(-1) = \frac{(+1) - \langle Z \rangle}{(+1) - (-1)} = \frac{1 - \langle Z \rangle}{2}$$

4.9 Heisenberg Representation of Quantum Circuits

Up to now, we have described quantum computation using the **Schrödinger representation**. In this approach, the **quantum state evolves over time**, while the observables remain fixed.

However, there is an alternative but completely equivalent description called the **Heisenberg representation**. In this approach, the **quantum state remains fixed**, while the observables evolve instead.

Both descriptions produce exactly the same physical predictions and measurement outcomes. They are just different mathematical viewpoints of the same underlying physics.

≡ Schrödinger Representation

In the **Schrödinger Representation**, a quantum gate U acts directly on the state:

$$|\psi_1\rangle = U |\psi_0\rangle \quad (69)$$

Where:

- $|\psi_0\rangle$ is the initial state;
- U is the unitary operator representing the quantum circuit;
- $|\psi_1\rangle$ is the final state.

The **observable does not change**. For example, if we measure in the computational basis, we use the Pauli-Z observable Z , which **remains the same before and after applying the circuit**. The expectation value after applying the circuit is:

$$\langle \psi_1 | Z | \psi_1 \rangle$$

In this case:

- The **state changes**;
- The **observable stays fixed**.

≡ Heisenberg Representation

In the **Heisenberg Representation**, the state remains unchanged $|\psi_0\rangle$. Instead, the **observable is transformed** according to:

$$Z \longrightarrow U^\dagger Z U \quad (70)$$

Where:

- $U^\dagger$ is the conjugate transpose of the gate U ;
- Z is the original observable;
- U is the gate itself.

The expectation value is then computed as:

$$\langle \psi_0 | U^\dagger Z U | \psi_0 \rangle$$

In this case:

- The **state stays fixed**;
- The **observable changes**.

❓ Why are the two representations equivalent?

Starting from the Schrödinger representation:

$$|\psi_1\rangle = U |\psi_0\rangle$$

The corresponding bra is:

$$\langle \psi_1 | = \langle \psi_0 | U^\dagger$$

Substituting into the expectation value:

$$\begin{aligned} \langle \psi_1 | Z | \psi_1 \rangle &= (\langle \psi_0 | U^\dagger) Z (U | \psi_0 \rangle) \\ &= \langle \psi_0 | U^\dagger Z U | \psi_0 \rangle \end{aligned}$$

Therefore (and this is the key point), the expectation value computed in the Schrödinger representation is exactly equal to the expectation value computed in the Heisenberg representation (see Equation 70):

$$\langle \psi_1 | Z | \psi_1 \rangle = \langle \psi_0 | U^\dagger Z U | \psi_0 \rangle \quad (71)$$

This proves that both representations yield exactly the same measurement result.

🔗 Physical Interpretation

The Schrödinger and Heisenberg representations are two different mathematical viewpoints of the same physical process.

In the **Schrödinger** representation:

- We imagine the quantum state moving through the circuit;
- Gates transform the state;
- Measurements are performed using fixed observables.

In the **Heisenberg** representation:

- We keep the quantum state fixed;
- Gates transform the observables;
- Measurements are performed using transformed observables.

✔ Why is the Heisenberg representation useful?

The Heisenberg representation is useful because it lets us answer a very important question: *what observable is this quantum circuit actually measuring?*

For example, in the Schrödinger representation, suppose we have an initial state $|\psi\rangle$, a quantum circuit H (the Hadamard gate, page 62), and we measure in the computational basis using the observable Z (page 53). *But what does this circuit actually measure?*

To find out, we can switch to the Heisenberg representation and compute the transformed observable:

$$H^\dagger Z H$$

This will give us the new observable that corresponds to the measurement performed by the circuit H . In this case, we find that:

$$H^\dagger Z H = \frac{1}{2} \left(\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \right) = \frac{1}{2} \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = X$$

This means that the circuit H actually measures the observable X (the Pauli- X gate, page 48), not Z . In other words, applying the Hadamard gate before measuring in the computational basis effectively changes our measurement from Z to X .

So, without the Heisenberg representation, we only know the sequence of gates. With the Heisenberg representation, we immediately understand: “*this circuit is actually measuring X !*” This is the real physical meaning of the circuit, and it is crucial for understanding how quantum algorithms work.

Example 16: Analogy with Arithmetic Expressions

Suppose someone gives us:

$$(3 + 5) \times 2 - 16$$

We could compute it step by step. Or we could simplify the whole expression to: 0. The Heisenberg representation does something similar:

- Instead of following every state transformation,
- It simplifies the entire circuit into an equivalent measurement.

Suppose a circuit contains many gates. In the **Schrödinger picture**, we must track the state after each gate. In the **Heisenberg picture**, we ask only: *what observable is measured at the end?* Sometimes the answer is a single operator such as X or Z . Sometimes it is a more complicated operator that involves multiple qubits. But in either case, we get a clear understanding of what the circuit does without having to track the state at every step.

4.10 Clifford Gates

4.10.1 Definition

In the previous section, we introduced the Heisenberg representation of quantum circuits. There, we saw that applying a Hadamard gate H and then measuring an observable Z is equivalent to first transforming the observable Z into X and then measuring X :

$$Z \xrightarrow{H} X$$

This naturally leads to an important question: *what kinds of gates transform simple observables into other simple observables?* A particularly important class of gates with this property is known as the **Clifford gates**.

Definition 9: Clifford Gate

A unitary gate U is called a **Clifford Gate** if, for every Pauli operator P (i.e., $P \in \{I, X, Y, Z\}$), the transformed operator

$$U^\dagger P U = P' \quad (72)$$

Is still a Pauli operator (possibly with an overall sign ± 1).

In other words, **Clifford gates map Pauli operators to other Pauli operators under conjugation.**

For a single qubit, the Pauli operators are:

$$I^4, \quad X, \quad Y, \quad Z.$$

Therefore, a single-qubit Clifford gate must transform each of these operators into another operator from the same set (up to sign).

For example:

$$Z \longrightarrow X$$

Is allowed, because X is still a Pauli operator. However:

$$Z \longrightarrow \frac{X + Y}{\sqrt{2}}$$

Is not allowed, because the **result is not a single Pauli operator**.

Multi-Qubit Extension

For n qubits, Pauli operators are formed by taking tensor products of single-qubit Pauli matrices. For two qubits, examples of Pauli operators include:

$$X \otimes I, \quad I \otimes Z, \quad Y \otimes X, \quad Z \otimes Z$$

⁴Technically, Wolfgang Pauli only defined X , Y , and Z as the Pauli operators because they are the non-trivial ones. However, we include the identity I in the set of Pauli operators for completeness.

For three qubits, examples include:

$$X \otimes I \otimes Z, \quad Y \otimes Y \otimes X$$

In general, an n -qubit Pauli operator has the form:

$$P = P_1 \otimes P_2 \otimes \cdots \otimes P_n$$

Where each P_i is one of the single-qubit Pauli operators $\{I, X, Y, Z\}$.

Example 17: Hadamard Gate is a Clifford Gate

The Hadamard gate H is:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Because H is both Hermitian and unitary, we have $H^\dagger = H$ and $H^{-1} = H$. Therefore, we can compute the conjugation of the Pauli operators by H :

$$HPH = P'$$

In practice, the Hadamard gate transforms the Pauli matrices as follows:

$$\begin{aligned} HXH &= \frac{1}{2} \left(\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \right) \\ &= \frac{1}{2} \left(\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \right) \\ &= \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix} \\ &= Z \end{aligned} \tag{73}$$

$$\begin{aligned} HZH &= \frac{1}{2} \left(\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \right) \\ &= \frac{1}{2} \left(\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \right) \\ &= \frac{1}{2} \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix} \\ &= X \end{aligned} \tag{74}$$

$$\begin{aligned} HYH &= \frac{1}{2} \left(\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \right) \\ &= \frac{1}{2} \left(\begin{bmatrix} i & -i \\ -i & -i \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \right) \\ &= \frac{1}{2} \begin{bmatrix} 0 & 2i \\ -2i & 0 \end{bmatrix} \\ &= -Y \end{aligned} \tag{75}$$

Each result is still a Pauli operator (up to a sign), confirming that the Hadamard gate is indeed a Clifford gate.

Example 18: CNOT Gate is a Clifford Gate

The CNOT gate (page 100) is:

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

As with the Hadamard gate, we can verify that the CNOT gate is a Clifford gate by checking how it transforms the two-qubit Pauli operators. To do this, we compute the tensor products of the single-qubit Pauli operators with the identity (to create two-qubit Pauli operators) and then conjugate them by the CNOT gate (since the CNOT gate is hermitian, we have $\text{CNOT}^\dagger = \text{CNOT}$):

$$\begin{aligned} X \otimes I &= \begin{bmatrix} 0 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} & 1 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ 1 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} & 0 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \\ \text{CNOT}(X \otimes I)\text{CNOT} &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \\ &= X \otimes X \end{aligned} \tag{76}$$

For the Z operator, the CNOT gate does not change the first qubit, so we have $Z \otimes I$:

$$\begin{aligned} Z \otimes I &= \begin{bmatrix} 1 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} & 0 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ 0 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} & -1 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned}
\text{CNOT}(Z \otimes I)\text{CNOT} &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \\
&= Z \otimes I
\end{aligned} \tag{77}$$

Therefore, the CNOT gate transforms $X \otimes I$ into $X \otimes X$ and leaves $Z \otimes I$ unchanged. By performing similar calculations for all other two-qubit Pauli operators, we can confirm that the CNOT gate maps every two-qubit Pauli operator to another two-qubit Pauli operator (up to a sign):

- $\text{CNOT}(I \otimes X)\text{CNOT} = I \otimes X$
- $\text{CNOT}(I \otimes Z)\text{CNOT} = Z \otimes Z$

These results confirm that the CNOT gate is indeed a Clifford gate.

4.10.2 Non-Clifford Gates

While the Hadamard and CNOT gates are examples of Clifford gates, there are many other gates that do not have this property. The most important example of a non-Clifford gate is the T gate.

The **T Gate** (also called the $\pi/8$ gate) is a **single-qubit gate that applies a specific phase shift to the state $|1\rangle$, while leaving $|0\rangle$ unchanged**. Its matrix representation is:

$$T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1+i}{\sqrt{2}} \end{bmatrix} \quad (78)$$

This means:

$$T|0\rangle = |0\rangle \quad T|1\rangle = e^{i\pi/4}|1\rangle \quad (79)$$

On the Bloch sphere, the T gate corresponds to a rotation around the Z -axis by an angle of $\pi/4$ (or 45 degrees). In terms of the Pauli operators, the T gate transforms them as follows:

$$\begin{aligned} TXT^\dagger &= \frac{1}{\sqrt{2}}(X + Y) \\ TYT^\dagger &= \frac{1}{\sqrt{2}}(Y - X) \\ TZT^\dagger &= Z \end{aligned} \quad (80)$$

Since the T gate transforms the X and Y operators into linear combinations of X and Y (which are not single Pauli operators), the T gate is **not a Clifford gate**.

Deepening: The T Gate is the Square Root of the S Gate

The T gate is the square root of the Phase gate S (which is a Clifford gate) because applying T twice gives exactly S .

If we apply the T gate twice, we have:

$$T^2 = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{bmatrix}$$

Where:

$$e^{i\pi/4} \cdot e^{i\pi/4} = \exp\left(\frac{i\pi}{4} + \frac{i\pi}{4}\right) = \exp\left(\frac{2i\pi}{4}\right) = \exp\left(\frac{i\pi}{2}\right) = e^{i\pi/2}$$

Now using Euler's formula, we can express $e^{i\pi/2}$ as:

$$\begin{aligned} e^{i\theta} &= \cos(\theta) + i \sin(\theta) \\ e^{i\pi/2} &= \cos\left(\frac{\pi}{2}\right) + i \sin\left(\frac{\pi}{2}\right) \\ &= 0 + i \cdot 1 \\ &= i \end{aligned}$$

Therefore, we have:

$$T^2 = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} = S$$

Finally, we can also express T as the square root of S :

$$T = \sqrt{S} = \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{i} \end{bmatrix}$$

In summary, applying the T gate twice gives us the S gate, confirming that T is indeed the square root of S .

❓ Why are Non-Clifford Gates important?

The real question is: **why can some quantum circuits be simulated efficiently on a classical computer, while others cannot?** To answer this, we must compare two ways of describing a quantum state.

- The **full state vector representation**, where an n -qubit quantum state has the form (page 90):

$$|\psi\rangle = \sum_{k=0}^{2^n-1} c_k |k\rangle$$

This requires storing 2^n complex amplitudes (e.g., 1 qubit requires 2 amplitudes, 2 qubits require 4 amplitudes, etc.), which **grows exponentially** with the number of qubits.

- The **stabilizer representation**. Certain special states can be described much more compactly. Instead of storing all amplitudes, we store a small set of Pauli operators (stabilizers) that uniquely characterize the state. Only n independent stabilizers are needed for n qubits. For example, 10 qubits can be described by just 10 stabilizers, which is a huge reduction compared to the $2^{10} = 1024$ amplitudes needed in the full state vector representation. This representation **grows only linearly** with the number of qubits, making it much more efficient for certain states.

Clifford gates use the stabilizer representation, because transform Pauli operators into other Pauli operators. Therefore, we **start with n stabilizers** (n Pauli operators) that describe the initial state, and **after applying Clifford gates, we still have n stabilizers** that describe the new state. This allows to **leave the state description compact and efficient**.

❓ **What happens at a Non-Clifford Gate?** Consider the T gate. It transforms, for example, the X operator into a linear combination of X and Y :

$$TXT^\dagger = \frac{1}{\sqrt{2}}(X + Y)$$

The result is not a single Pauli operator, but a sum of two Pauli operators. After another Non-Clifford transformation, each term may split again. So **repeated Non-Clifford gates can cause the number of terms to double repeatedly, leading to exponential growth**.

✂ **Computational Consequence.** Classical simulation cost depends on how much information must be stored.

- Clifford circuits keep the description small.
- Non-Clifford circuits cause the description to grow rapidly.

Therefore, if a circuit contains only Clifford gates, a classical computer can efficiently simulate it. In contrast, if the circuit includes Non-Clifford gates such as T , simulation generally becomes exponentially more difficult.

☆ **So, why are Non-Clifford gates important?** If Clifford circuits are classically simulable, then they cannot provide full quantum computational power. To achieve universal quantum computation, we need gates that escape the stabilizer framework. The T gate does exactly this. By adding the T gate (or any other suitable non-Clifford gate) to the Clifford gate set, we obtain a universal set of quantum gates. This allows us to implement arbitrary quantum algorithms, including computations believed to be infeasible for classical computers.

4.10.3 Stabilizer States

As anticipated in the previous section, a **Stabilizer State** is a **quantum state that can be described** not by listing all of its amplitudes, but by **specifying which Pauli operators leave it unchanged**. This is one of the most elegant ideas in quantum computing and is the foundation of the stabilizer formalism used in quantum error correction and efficient simulation.

Definition 10: Stabilizer State

An n -qubit state $|\psi\rangle$ is a stabilizer state if there exists a Pauli operator P (or, more generally, a set of commuting Pauli operators) such that:

$$P|\psi\rangle = |\psi\rangle \quad (81)$$

This means:

- Applying P does not change the state $|\psi\rangle$.
- The state $|\psi\rangle$ is an eigenvector of P .
- The corresponding eigenvalue is $+1$.

The operator P is called a **Stabilizer** because it “*stabilizes*” the state $|\psi\rangle$.

Meaning of “*stabilize*”. Suppose a transformation leaves an object unchanged. Some examples:

- Rotating a perfect sphere leaves it unchanged;
- Reflecting a symmetric figure may leave it unchanged;
- Applying certain Pauli operators may leave a quantum state unchanged.

In quantum mechanics, if the relation $P|\psi\rangle = |\psi\rangle$ holds, then P is a symmetry of the state $|\psi\rangle$, and we say that P stabilizes $|\psi\rangle$. This means that **the state $|\psi\rangle$ is invariant under the transformation represented by P** .

Example 19: $|0\rangle$ and $|1\rangle$

The Pauli-Z gate is:

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Applying Z to $|0\rangle$:

$$Z|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

Therefore, $|0\rangle$ is a stabilizer state and the Pauli-Z operator is one of its stabilizers.

By contrast, applying Z to $|1\rangle$:

$$Z|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -|1\rangle$$

Shows that $|1\rangle$ is not stabilized by Z (since the eigenvalue is -1), and thus $|1\rangle$ is not a stabilizer state with respect to Z .

Example 20: $|+\rangle$

The state $|+\rangle$ is defined as:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Applying the Pauli-X operator to $|+\rangle$:

$$X|+\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = |+\rangle$$

Therefore, $|+\rangle$ is a stabilizer state and the Pauli-X operator is one of its stabilizers.

Example 21: Bell State

Consider the Bell state (page 115):

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Applying the operator $X \otimes X$ to $|\Phi^+\rangle$:

$$\begin{aligned} (X \otimes X) |\Phi^+\rangle &= \frac{1}{\sqrt{2}} (X|0\rangle \otimes X|0\rangle + X|1\rangle \otimes X|1\rangle) \\ &= \frac{1}{\sqrt{2}} (|11\rangle + |00\rangle) = |\Phi^+\rangle \end{aligned}$$

And also applying $Z \otimes Z$:

$$\begin{aligned} (Z \otimes Z) |\Phi^+\rangle &= \frac{1}{\sqrt{2}} (Z|0\rangle \otimes Z|0\rangle + Z|1\rangle \otimes Z|1\rangle) \\ &= \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) = |\Phi^+\rangle \end{aligned}$$

Therefore, $|\Phi^+\rangle$ is a stabilizer state and both $X \otimes X$ and $Z \otimes Z$ are stabilizers of $|\Phi^+\rangle$.

☰ Multiple Stabilizers

The Equation 81 defines a **stabilizer state with respect to a single Pauli operator** P . However, many quantum states are stabilized by multiple Pauli operators. In fact, for an n -qubit stabilizer state, we typically use n independent commuting stabilizers:

$$P_1, P_2, \dots, P_n$$

Each of these operators satisfies:

$$P_i |\psi\rangle = |\psi\rangle \quad \text{for } i = 1, 2, \dots, n \quad (82)$$

The set of all such stabilizers forms a group under multiplication, known as the **Stabilizer Group**.

✔ Why Stabilizers are powerful

A general n -qubit state requires 2^n complex amplitudes to be fully described. However, a stabilizer state can be completely characterized by its n independent stabilizers, which are much fewer in number. Furthermore, each stabilizer is describable with $O(n)$ symbols (since it is a tensor product of n Pauli operators), leading to an efficient representation of the state. Therefore, the description size grows only polynomially (i.e., $O(n)$) rather than exponentially.

For example, for 100 qubits, a full state-vector simulation requires 2^{100} amplitudes; in contrast a stabilizer formalism requires roughly 100 stabilizers. This is why large Clifford circuits can be simulated efficiently on classical computers.

In summary, instead of describing “*what are all amplitudes of the state?*”, we describe “*which symmetries does the state satisfy?*”. This is often dramatically more efficient.

4.11 Universal set of quantum gates

After studying Clifford and non-Clifford gates, the next question is: *what is the minimum set of gates needed to build **any quantum computation**?* This leads to the concept of a **universal set of quantum gates**, which is a collection of gates that can be combined to approximate any unitary operation on a quantum system to arbitrary precision.

Definition 11: Universal Set of Quantum Gates

A set of quantum gates is called **universal** if any unitary transformation can be approximated to arbitrary accuracy using only gates from that set.

In other words, a set of gates is universal if using only gates from that set, we can approximate any unitary transformation to arbitrary precision.

Universal gate set $\implies \forall U, \exists$ a circuit approximating U

❓ Why “Approximate” and not “Exactly”?

In quantum computing, every quantum gate is represented by a **unitary matrix**. For example the Hadamard gate, the T gate, and the CNOT gate. However, the set of all unitary matrices is **continuous** (i.e., there are infinitely many unitary transformations). For example, a single-qubit rotation may involve any real angle $R_z(\theta)$ with infinitely many possible values of θ :

$$R_z(\theta) = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}$$

Here, the angle θ can be 0.1, 0.12345, $\pi/7$ or any other real number. There are infinitely many real numbers, so there are infinitely many possible rotations and therefore infinitely many possible unitary matrices.

❓ **Why not build hardware for every matrix?** Building a quantum gate for every possible unitary transformation is **impossible**. Imagine trying to create one physical instruction for every possible angle:

- One gate for 0.1 radians
- One gate for 0.12345 radians
- One gate for $\pi/7$ radians
- One gate for 0.00001 radians

And so on forever. This is clearly impractical.

✔ **The solution: a Finite Gate Set.** Instead of trying to build a gate for every possible unitary, we choose a **small set of standard gates**, for example $\{H, \text{CNOT}, T\}$. These are the only **primitive gates the machine needs to implement directly**. All other operations are constructed by combining them.

▲ Which gates are in the universal set?

To keep things simple, we could choose Clifford gates, such as H and CNOT, as our set of gates. Although these gates are powerful, they can still be efficiently simulated on a classical computer (page 156). Therefore, **Clifford gates alone are not sufficient for universal quantum computation**. For this reason, in general, a **universal set requires at least one non-Clifford gate**, and the most common choice is the T gate because of its simplicity and its relationship to the S gate (page 155). A **widely used universal set of quantum gates is:**

$$\{H, \text{CNOT}, T\} \quad (83)$$

? Why these three gates are enough?

- **Hadamard H** **transforms basis states into superpositions**. For example, the $|0\rangle$ state becomes the plus state (page 50):

$$H|0\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

- **CNOT** creates **entanglement between qubits**. For example, applying the CNOT to the state $|+0\rangle$ creates the Bell state $|\Phi^+\rangle$:

$$\text{CNOT}(|+0\rangle) = |\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

- **T gate** adds a **45° phase and breaks the Clifford-only restriction**, allowing us to access a much larger set of unitary transformations. Because T is a non-Clifford gate, it enables us to perform operations that cannot be efficiently simulated classically.

Hadamard changes basis and creates superposition, CNOT creates entanglement between qubits, and T provides the non-Clifford component required for universality. Together, **they can approximate any unitary operation**.

Other universal gate sets exist, such as:

- Toffoli and Hadamard: $\{H, \text{Toffoli}\}$.
- Single-qubit rotations and CNOT: $\{R_x(\theta), R_y(\theta), R_z(\theta), \text{CNOT}\}$.

≡ How to Generate a Universal Set of Quantum Gates

A universal gate set is obtained by combining two ingredients:

1. **A set of gates that can generate arbitrary single-qubit unitary matrices.**
2. **One entangling two-qubit gates**, usually the CNOT gate (page 100), which can create entanglement between qubits.

If both conditions are satisfied, the gate set is universal. Let's see why this is the case:

1. **Choose Gates that Generate Arbitrary Single-Qubit Unitaries** (e.g., H and T gates). A single qubit can undergo any unitary transformation:

$$U \in U^2 = \{U \in \mathbb{C}^{2 \times 2} : U^\dagger U = I\}$$

To obtain universality, we need to approximate any such 2×2 unitary matrix. A common way is to use a Hadamard gate and a T gate. The pair of gates $\{H, T\}$ can approximate any single-qubit unitary to arbitrary precision.

❓ Why universality? While mathematically there are infinitely many possible 2×2 unitary matrices, our hardware can implement only a **finite, discrete set** of gates exactly (similar to how a digital computer uses a finite instruction set). Therefore, the goal of a *universal* gate set is not to implement every unitary matrix exactly. Instead, **for any target unitary matrix U and for any desired accuracy $\varepsilon > 0$, there must exist a finite sequence of gates from the set whose overall operation V satisfies:**⁵

$$\|U - V\| < \varepsilon$$

In other words, we can **make the approximation error as small as we want**. This ability to approximate any unitary operation to arbitrary precision is what makes a gate set universal.

2. **Add an Entangling (two-qubit) Gate.** Single-qubit gates alone cannot create entanglement (page 120). To manipulate multiple qubits, we **add an entangling gate such as CNOT**. Once CNOT is available, qubits can interact and become entangled.

Combining these two ingredients, the resulting set is universal, meaning that any quantum computation can be approximated using just these gates.

⁵The matrix $U - V$ contains the element-by-element differences, but by itself it is still a matrix, not a single number. To obtain one number representing the size of the difference, we apply a **norm**. It plays exactly the same role as absolute value for numbers. In other words, it means that the implemented operation V is within distance ε of the desired operation U .

4.12 Fault-Tolerant Quantum Gates

So far, we have studied quantum computation in an idealized setting, where quantum gates are represented by perfect unitary matrices acting on perfectly isolated qubits. In practice, however, real quantum hardware is extremely fragile. Physical qubits are affected by noise, decoherence, imperfect control operations, and measurement errors, all of which can corrupt the quantum information stored in the system.

To perform reliable quantum computation, it is therefore necessary to protect quantum information against errors. This is the goal of **quantum error correction**, where a single logical qubit is encoded into many physical qubits in such a way that errors can be detected and corrected without destroying the quantum state.

However, protecting quantum information is not sufficient by itself. We must also be able to apply quantum gates directly on encoded logical qubits while preserving the fault-tolerant structure of the error-correcting code. This introduces the notion of **fault-tolerant quantum gates**, namely gates designed so that local physical errors do not spread uncontrollably through the computation.

In this context, Clifford gates play a particularly important role because many of them admit simple fault-tolerant implementations, often through **transversal operations**. Non-Clifford gates, on the other hand, are significantly more difficult to implement fault-tolerantly, even though they are essential for universal quantum computation. This creates one of the central challenges of modern quantum computing: the gates required for computational universality are also the most expensive and difficult to protect against errors.

In the following sections, we introduce the distinction between physical and logical qubits, the concept of transversal gates, and the reasons why non-Clifford gates such as the T gate represent the main obstacle in fault-tolerant quantum computation.

4.12.1 Logical vs Physical Qubits

⚠ Problem. In an ideal mathematical model of quantum computation, qubits are assumed to evolve perfectly under unitary transformations. Real quantum hardware, however, is inherently **noisy**. Physical qubits interact with the surrounding environment and are affected by decoherence, thermal fluctuations, imperfect control pulses, and measurement errors. As a consequence, **quantum information** stored in a single physical qubit is **extremely fragile**.

✅ Solution. To overcome this limitation, quantum computers do not directly perform large computations on individual physical qubits. Instead, they use **quantum error correction**, where the information of a single qubit is encoded across many physical qubits, forming a **logical qubit**.

- **Physical Qubit.** A physical qubit is the actual hardware-level quantum system used to store information. For example superconducting qubits, trapped ions, neutral atoms or photonic qubits⁶. Physical qubits are directly manipulated by the hardware and implement the elementary quantum operations supported by the quantum processor. However, they are noisy and have limited coherence times.
- **Logical Qubit.** A logical qubit is an encoded qubit constructed from many physical qubits in order to protect quantum information against errors.

Instead of storing the quantum state in a single fragile qubit, the information is distributed redundantly across multiple physical qubits:

$$\text{Many physical qubits} \xrightarrow{\text{Encoding}} \text{One logical qubit}$$

This encoding allows the system to detect errors, correct certain errors and preserve the quantum state for a much longer time.

❓ Why do we need logical qubits? Without error correction, errors accumulate extremely rapidly during a quantum computation. Even a small error probability per gate becomes catastrophic when millions or billions of gates are executed.

Logical qubits make large-scale quantum computation possible by continuously protecting the encoded information while the computation is running.

⚠ The Main Challenge

Encoding a logical qubit is only the first step. We must also be able to apply quantum gates, manipulate information, execute algorithms. Directly on logical qubits without destroying the error-correcting structure. This leads to the concept of **fault-tolerant quantum gates**, which are quantum gates specifically designed to prevent local physical errors from spreading uncontrollably through the encoded system.

⁶It's normal don't know the details of these different physical implementations at this stage. The key point is that they are all examples of physical qubits, which are the building blocks of quantum computers.

4.12.2 Transversal Gates

In fault-tolerant quantum computing, one of the main objectives is to apply logical quantum gates without allowing physical errors to spread uncontrollably through the encoded qubits. Let's suppose one logical qubit⁷ is encoded into many physical qubits:

$$\text{Logical Qubit} \rightarrow q_1, q_2, \dots, q_7$$

It means that a single logical qubit is represented by a collection of physical qubits. Now imagine, for example, that q_3 contains an error, and all the other qubits are fine. This situation is still manageable because we can detect and correct the error in q_3 without affecting the other qubits.

⚠ The Dangerous Situation. Now suppose we apply a complicated gate that strongly couples all qubits together. Then, if q_3 has an error, it can spread to all the other qubits, making the entire logical qubit corrupted and uncorrectable.

✅ The core goal of Fault Tolerance. So the main objective becomes: if one physical qubit fails, the failure should remain localized and not spread to other qubits. This is where transversal gates come into play.

Definition 12: Transversal Gates

A **Transversal Gate** is a logical quantum gate implemented by applying independent operations to corresponding physical qubits of an encoded logical qubit.

Suppose a logical qubit is encoded into n physical qubits. A transversal implementation applies the same unitary operation independently to each physical qubit:

$$U_L = U^{\otimes n}$$

Where:

- U is the physical gate;
- U_L is the resulting logical gate;
- n is the number of physical qubits used in the encoding.

In simple terms, a transversal gate applies operations independently to each physical qubit. So no qubit interacts with another qubit inside the same block. Therefore if q_3 was wrong before, only q_3 will be affected by the gate, and the error will *not* spread to the other qubits.

⁷A logical qubit is an encoded qubit constructed from many physical qubits in order to protect quantum information against errors. Instead of storing the quantum state in a single fragile qubit, the information is distributed redundantly across multiple physical qubits.

Error Containment and Fault Tolerance

The **most important property** of transversal gates is that **they prevent error propagation inside the encoded block.**

Suppose one physical qubit contains an error before the gate is applied. Since each qubit is manipulated independently, the error remains localized and does not spread to many other qubits.

Furthermore, transversal gates are **inherently fault-tolerant**. In general, a fault-tolerant operation must ensure that:

- A small physical error remains small;
- A single faulty qubit does not corrupt the entire logical qubit.

Transversal gates naturally satisfy this requirement **because there are no interactions between qubits within the same code block during the gate operation.** Therefore, even if one qubit experiences an error, it does not affect the others.

Example 22: Logical Hadamard Gate

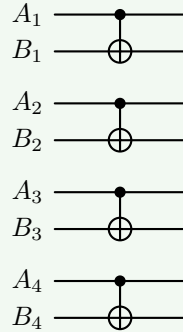
For many quantum error-correcting codes, a **logical Hadamard gate** can be implemented transversally by applying a Hadamard gate independently to every physical qubit:

$$H_L = H^{\otimes n} = \underbrace{H \otimes H \otimes \dots \otimes H}_{n \text{ times}}$$

Thus, the logical operation is realized through many independent physical operations performed in parallel.

Example 23: Logical CNOT Gate

A logical CNOT gate between two encoded logical qubits is often implemented by applying **physical CNOT gates pairwise between corresponding qubits of the two blocks**:



Each pair of physical qubits interacts independently of the others.

4.12.3 Why Non-Clifford Gates Are Difficult

One of the central challenges of fault-tolerant quantum computing is that **most non-Clifford gates cannot be implemented transversally**. While many Clifford gates admit simple fault-tolerant implementations (we saw some examples, like the Hadamard gate and CNOT, in the last section on page 167), non-Clifford gates generally require much more complex procedures.

✓ Clifford Gates and Transversality

As discussed previously, transversal gates are highly desirable because they prevent physical errors from spreading across the encoded logical qubit (page 166).

For many quantum error-correcting codes, **Clifford gates** such as Hadamard gate, Phase gate S , and CNOT gate, can **often be implemented transversally**. This means that the logical operation can be realized by applying independent physical operations to the corresponding physical qubits of the encoded blocks.

⚠ The Problem with Non-Clifford Gates

However, **non-Clifford gates**, such as the T gate, generally **do not admit transversal implementations** that preserve the structure of the error-correcting code. In fact, **if we attempt** to apply a non-Clifford gate transversally, **two problematic situations** may occur:

- ✗ The resulting operation does not correspond to the correct logical gate;
- ✗ Or the encoded structure of the code is broken, allowing errors to propagate uncontrollably through the logical qubit.

The **difficulty of implementing non-Clifford gates in a fault-tolerant manner** is not just a technological limitation of current quantum hardware; it **is also a fundamental theoretical constraint** on the structure of quantum error-correcting codes. The **Eastin-Knill theorem** formalizes this limitation by proving that **no quantum error-correcting code (i.e., no encoding scheme) can realize a universal set of logical gates using only transversal operations**.

Consequently, transversal gates are extremely useful for fault tolerance, but they are not sufficient to implement universal quantum computation. At least one non-Clifford gate must be implemented through more advanced fault-tolerant techniques.

Deepening: Why non-Clifford gates do not admit transversal implementations?

Briefly, because **non-Clifford gates are “too powerful” to fit inside the simple independent structure of transversal operations**. But let us build the intuition behind this statement more carefully.

? What a Transversal Gate really does. A transversal gate acts

independently on each physical qubit: $U_L = U^{\otimes n}$, where U is a single-qubit operation. Each physical qubit is treated separately, and this is **extremely restrictive** since it **cannot create** any **entanglement** between the physical qubits (since there are no multi-qubit interactions).

? Why this restriction is good. Because **independent operations prevent error spreading**. If one physical qubit is wrong, the gate acts independently, then only that qubit remains wrong. So **transversal gates are naturally fault-tolerant**.

⚠ But there is a problem. A universal quantum computer must be able to perform extremely rich transformations. In particular, it must be able to do arbitrary rotations on the Bloch sphere, complicated entangling dynamics, and non-stabilizer transformations. And transversal operations are simply too “*rigid*” to generate all of that while preserving the code structure.

⚠ In fact, there is a no-go theorem. The **Eastin-Knill theorem** [1] states that **no quantum error-correcting code can have a universal set of transversal gates**. This means that while we can have some transversal Clifford gates, we cannot have a transversal implementation of a non-Clifford gate like T that would allow us to achieve universality. Therefore, we must look for more complex methods to implement non-Clifford gates in a fault-tolerant manner, such as magic state distillation or code switching, which are significantly more resource-intensive than transversal implementations.

In summary, transversal gates are simple, local and safe. Non-Clifford gates require transformations that are too complex to be realized by purely independent local operations while still preserving the error-correcting code. That is the real reason why non-Clifford gates generally cannot be implemented transversally.

✓ Solutions to the Non-Clifford Problem

To overcome the limitations imposed by the Eastin-Knill theorem, researchers have developed several techniques to implement non-Clifford gates in a fault-tolerant manner. Two standard approaches are: **magic state distillation** and **magic state injection**. The main idea is to **avoid applying the non-Clifford gate directly to the encoded logical qubit**. Instead, the **quantum computer prepares a special ancillary quantum state**, called a **magic state**, which can **later be consumed** to indirectly implement the desired logical gate.

However, **this course does not cover these techniques in detail**, as they are quite complex and require a deep understanding of quantum error correction and fault tolerance. The key takeaway is that **while non-Clifford gates are essential for universal quantum computation, their implementation in a fault-tolerant manner is significantly more challenging than that of Clifford gates, and it requires sophisticated methods that go beyond simple transversal operations**.

4.12.4 Physical vs Logical Gate Sets

In quantum computing, it is important to distinguish between:

- **Physical Gate sets**, which correspond to the elementary operations directly implemented by the hardware;
- **Logical Gate sets**, which are the fault-tolerant operations acting on encoded logical qubits (e.g., the Clifford gates).

Although the two descriptions correspond to the same underlying quantum computation, they differ significantly in terms of implementation complexity and fault-tolerant requirements.

Physical (Hardware-Level) Gate Sets

At the hardware level, quantum gates act directly on physical qubits. A typical universal physical gate set consists of:

- **Single-Qubit Gates:** These include rotations around the X, Y, and Z axes (e.g., $R_x(\theta)$, $R_y(\theta)$, $R_z(\theta)$).

In general, single-qubit rotations are **easier to implement** than multi-qubit gates because they **act locally on only one physical qubit**. In particular, many hardware platforms can implement $R_z(\theta)$ very efficiently, sometimes even virtually through software phase updates rather than physical operations.

At the physical level, also the T gate is simply a rotation around the Z-axis by $\pi/4$, therefore it is not more difficult to implement than other single-qubit gates.

- **Two-Qubit Gates:** The most common two-qubit gate is the CNOT gate, which is essential for creating entanglement between qubits.

Two-qubit gates such as CNOT are **significantly more difficult** because they require **interactions between qubits**. In general, the **larger the number of qubits involved in a gate, the more complex and error-prone its physical implementation becomes**.

These gates are **implemented through direct hardware control**, such as laser pulses in ion traps or microwave pulses in superconducting qubits.

Logical (Fault-Tolerant) Gate Sets

When quantum information is encoded into logical qubits using quantum error correction, the relevant gate set changes. The focus is no longer only on mathematical universality, but **also on fault-tolerant** implementability. A typical logical fault-tolerant gate set includes (page 161):

- **Clifford gates:**

$$\{H, S, \text{CNOT}\}$$

- **Non-Clifford gates** (e.g., the T gate):

$$\{T\}$$

Key Distinction

The crucial observation is that the **complexity of a gate depends strongly on whether it acts on**: physical qubits or encoded logical qubits. For example:

- On physical qubits, $T = R_z(\pi/4)$ is a simple single-qubit rotation.
- On logical qubits, T is a non-Clifford logical gate not usually transversally implementable, and therefore it requires magic state distillation and fault-tolerantly complex procedures to implement.

This distinction is one of the main practical challenges of large-scale quantum computing. At the mathematical level, the T gate appears to be a simple phase rotation. However, once quantum information is encoded into logical qubits, implementing the same operation fault-tolerantly becomes extremely resource-intensive.

As a consequence, the **cost of fault-tolerant quantum algorithms is often dominated by**:

- The number of logical T gates required;
- Magic state preparation;
- Error-correction overheads associated with non-Clifford operations.

In summary, the same gate may be easy at the hardware level, but extremely difficult at the logical fault-tolerant level.

5 Limits of Quantum Information

5.1 No-Cloning Principle

5.1.1 Why Cloning Matters

Briefly, the **no-cloning principle** states that it is **impossible to create a perfect copy of an arbitrary unknown quantum state**.

At first sight this may seem like a strange technical limitation, because in classical computing copying information is *trivial*: we can easily copy files, duplicate RAM, clone hard disks, send packets over the network, etc. However, **in quantum computing, copying an arbitrary qubit is fundamentally impossible**. This is **not a technological limitation**, instead it is a direct **consequence of the mathematical structure** of quantum mechanics.

In classical computing, a bit can be in one of two states: 0 or 1. In quantum computing, a qubit instead can be in infinitely many states, because it can be in any superposition of 0 and 1:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

With complex amplitudes a and b such that $|a|^2 + |b|^2 = 1$. So a qubit is not just “0” or “1”, but it represents a point on the Bloch sphere, meaning there are infinitely many possible states. **If arbitrary cloning were possible**, we could take an unknown state $|\psi\rangle$ and create as many copies of it as we want. This would allow us to perform measurements on the copies to learn about the original state $|\psi\rangle$ without disturbing it, **violating the fundamental principles of quantum mechanics**.

Relation with Unitary

The no-cloning principle theorem is deeply connected to the fact that quantum evolution is **unitary**. Quantum gates are represented by unitary matrices:

$$U^\dagger U = U U^\dagger = I$$

Unitary operations preserve norms, probabilities and inner products between states. **Cloning would violate this preservation property**. Because, intuitively, if two partially similar states must remain partially similar after evolution (i.e., their inner product must be preserved), then cloning would instead make them “more distinguishable” (i.e., their inner product would change), which is a contradiction. We will see the formal proof of the no-cloning principle in the next section.

Relation with No-Signaling

The no-cloning principle is also related to the no-signaling principle, because **breaking the no-cloning principle would violate the no-signaling principle**.

Suppose Alice and Bob share an entangled Bell pair:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Normally, Bob cannot determine whether Alice has measured her qubit or not, because his local measurements always appear random (50% probability of measuring 0 and 50% probability of measuring 1). But *imagine* Bob could clone his qubit infinitely many times. Then, he could:

1. Create many identical copies of his qubit.
2. Perform measurements/statistics on all copies.
3. Infer whether the global quantum state collapsed (i.e., whether Alice measured her qubit). Using the principle of *reductio ad absurdum*, if Alice does not measure, then Bob will measure random results (0, 1, 0, 1, etc.). When Alice measures and collapses the state to $|0\rangle$, then Bob's qubit will instantly become $|0\rangle$, and therefore, all of his copies will be $|0\rangle$.

This would allow instantaneous communication between Alice and Bob, faster than the speed of light, which is forbidden by the no-signaling principle (violation of relativity, no information can travel faster than the speed of light). Therefore, the **impossibility of cloning helps preserve the no-signaling principle**.

Relation with Quantum State Tomography

Another consequence of breaking the no-cloning principle would be that we would be able to store an infinite amount of classical information inside one qubit.

The idea is the following. Because a qubit can assume infinitely many states, one could theoretically encode a very long classical string into amplitudes (e.g., the binary expansion of π):

$$|\psi_{\text{message}}\rangle$$

But there is a problem: we **cannot directly read the amplitudes** from a single qubit, because measurement collapses the state. If we measure the qubit, we only get one classical outcome (e.g., 0 or 1), and not the full amplitudes a and b . So normally, the information is not accessible.

However, exists a technique called **quantum state tomography** that:

1. Prepares many identical copies⁸ of the same state $|\psi_{\text{message}}\rangle$.
2. Measures them in different bases (e.g., X, Y, Z).

⁸Quantum State Tomography does not violate the no-cloning principle, because it does not create perfect copies of an unknown state. Instead, it prepares many identical copies of a **known** state $|\psi_{\text{message}}\rangle$, which is allowed by quantum mechanics. For example, if we have a quantum circuit that prepares the state $|\psi_{\text{message}}\rangle$, we can run that circuit many times to get many copies of $|\psi_{\text{message}}\rangle$, which is perfectly fine. The no-cloning principle only forbids creating perfect copies of an **unknown** state, not preparing many copies of a **known** state.

3. Reconstructs the amplitudes of the original state $|\psi_{\text{message}}\rangle$ statistically.

In simple terms, given many copies of the same unknown state, estimate what the state was.

Now, imagine there were **no no-cloning theorem**. Then we could do the following:

1. Take the single qubit $|\psi_{\text{message}}\rangle$ that encodes the message.
2. Clone it infinitely many times to get many copies of $|\psi_{\text{message}}\rangle$. This would be possible because we are assuming that the no-cloning principle does not hold.
3. Perform measurements/statistics on all copies to reconstruct the amplitudes of the original state $|\psi_{\text{message}}\rangle$, and therefore read the message encoded in the amplitudes.

So cloning would transform ONE unknown qubit into enough data for tomography, which would allow us to read the message encoded in the amplitudes. This would be a huge problem, because we could encode an infinite amount of classical information into one qubit and then read it using cloning and tomography.

Example 1: Why Cloning Matters

✓ **Normal world (no-cloning theorem exists).** Alice wants to send Bob one qubit:

$$|\psi_{\text{message}}\rangle = \sqrt{0.73148291} |0\rangle + \sqrt{0.26851709} |1\rangle$$

Suppose these decimals secretly encode a super secret message. Bob receives the qubit. Now Bob asks: “*what are the exact amplitudes of the state $|\psi_{\text{message}}\rangle$?*” Bob measures the qubit, but measurement only gives him one classical bit (0 or 1). For example, if Bob measures 0, then the state collapses to $|0\rangle$, and the original amplitudes are lost. So from ONE measurement, Bob cannot reconstruct 0.73148291 exactly. So the hidden information is inaccessible to Bob, and the message is safe.

⚠ **Hypothetical world (no-cloning theorem does not exist).**

Now suppose cloning is allowed. Bob receives the qubit $|\psi_{\text{message}}\rangle$. Now Bob **can clone** the qubit, so he runs a mysterious cloning machine that creates many identical clones of $|\psi_{\text{message}}\rangle$:

$$|\psi\rangle \rightarrow |\psi\rangle |\psi\rangle |\psi\rangle |\psi\rangle |\psi\rangle \dots$$

Now he has millions of copies of $|\psi_{\text{message}}\rangle$. Now Bob can decrypt the super secret message. How? Simple:

1. **Measure many copies in the computational basis.** Bob has millions of copies. Suppose he observes (after many measurements) that 731482 times he gets 0 and 268517 times he gets 1. So Bob can estimate the probabilities of measuring 0 and 1 as 0.731482 and 0.268517. Bob is already very close to the original amplitudes, but he can do better.

2. **Measure in other bases.** Bob is smart, and he knows that measuring in the computational basis only gives him the magnitudes of the amplitudes, but not their phases. So Bob also measures in the X and Y bases to get more information about the state. By combining all the measurement results, Bob can reconstruct the original state $|\psi_{\text{message}}\rangle$ with very high precision, and therefore read the super secret message encoded in the amplitudes.

⚠ The Problem. Alice effectively transmitted thousands, millions, theoretically infinite classical digits using only one qubit, that should not be physically possible (because in physics, the amount of extractable classical information from a physical system must be finite).

🔗 Relation with Quantum Error Correction

In classical systems, redundancy is simple: we can copy bits to create redundancy for error correction. For example, we can encode a bit as three bits:

$$0 \rightarrow 000 \quad 1 \rightarrow 111$$

Then, if one bit gets flipped, we can still recover the original bit by majority voting:

$$000 \rightarrow 000 \quad 001 \rightarrow 000 \quad 010 \rightarrow 000 \quad 100 \rightarrow 000$$

Then majority voting allows us to correct single bit errors. However, in **quantum computing we cannot simply duplicate qubits** and we cannot create copies of arbitrary quantum states. So **quantum error correction cannot protect information by simply copying qubits**. Instead, it uses more complex techniques (e.g., entanglement, syndrome measurements, etc.) to protect quantum information without violating the no-cloning principle.

In summary, the no-cloning theorem is not an isolated curiosity. It is one of the foundational constraints that shape all of quantum information theory. It is connected to reversibility, unitarity, causality, measurement, entanglement, quantum cryptography, quantum error correction. **Without the no-cloning theorem, the entire structure of quantum information would collapse.**

5.1.2 Formal Statement

Definition 1: No-Cloning Theorem

The **No-Cloning Theorem** states that there **does not exist a universal quantum gate capable of perfectly copying an arbitrary unknown quantum state**.

Formally, suppose we have:

- A qubit in an unknown state: $|x\rangle$.
- And a second qubit initialized to: $|0\rangle$.

We would like a quantum gate U such that:

$$U |x\rangle |0\rangle = |x\rangle |x\rangle \quad (84)$$

For every possible quantum state $|x\rangle$. The **No-Cloning Theorem states that such a gate U cannot exist.**

Proof. We now prove that the cloning gate U :

$$U |x\rangle |0\rangle = |x\rangle |x\rangle$$

Cannot exist for an arbitrary quantum state $|x\rangle$.

1. **Assume that cloning is possible.** Suppose there exists a unitary gate U such that:

$$U |x\rangle |0\rangle = |x\rangle |x\rangle$$

And also:

$$U |y\rangle |0\rangle = |y\rangle |y\rangle$$

So U should clone both $|x\rangle$ and $|y\rangle$.

2. **Take the inner product before cloning.** Before cloning, the two input states are:

$$|x\rangle |0\rangle \quad \text{and} \quad |y\rangle |0\rangle$$

Their inner product is:

$$\langle x|y\rangle \langle 0|0\rangle$$

Since $|0\rangle$ is a normalized state, we have $\langle 0|0\rangle = 1$. Thus, the inner product before cloning is:

$$\langle x|y\rangle$$

3. **Take the inner product after cloning.** After cloning, the states become:

$$|x\rangle |x\rangle \quad \text{and} \quad |y\rangle |y\rangle$$

Their inner product is:

$$\langle x|y\rangle \langle x|y\rangle = (\langle x|y\rangle)^2$$

Now we use the tensor product property of inner products, which states that (page 86):

$$(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$$

Applying this to our inner product, we get:

$$(\langle x| \otimes \langle x|)(|y\rangle \otimes |y\rangle) = (\langle x|y\rangle) \otimes (\langle x|y\rangle)$$

Which simplifies to:

$$(\langle x|y\rangle)^2$$

4. **Use unitary.** Valid quantum gates must be unitary, which means they preserve inner products. Therefore, the inner product before and after applying U must be the same:

$$\langle x|y\rangle = (\langle x|y\rangle)^2$$

This is only possible if $\langle x|y\rangle = 0$ or $\langle x|y\rangle = 1$. But for arbitrary states, this is not generally true. Thus, the assumption that cloning is possible leads to a contradiction.

Let's try to understand why this is a contradiction. Suppose two states are $|x\rangle$ and $|y\rangle$, and suppose $|\langle x|y\rangle|$ (i.e., the magnitude of the inner product) is 0.8. It means that the states are not identical, but they are still “quite similar”. They are partially distinguishable, but not perfectly. Now, if cloning were possible, the inner product after cloning would be still 0.8. However, we found that the inner product after cloning would be $(0.8)^2 = 0.64$. This means that the states have become more distinguishable after cloning, which is a contradiction because unitary operations cannot change the inner product between states:

$$|\langle x|y\rangle| = 0.8 \xrightarrow{\text{clone}} |\langle x|y\rangle|^2 = 0.64 \xrightarrow{\text{clone}} |\langle x|y\rangle|^3 = 0.512 \dots$$

QED

Deepening: Alternative Proof of No-Cloning Theorem

We can also prove the no-cloning theorem using a different approach, by directly applying the unitarity of U to the cloning operation. This alternative proof is often more concise and elegant.

Proof. Suppose, by contradiction, that there exists a cloning gate U such that:

$$U |x\rangle |0\rangle = |x\rangle |x\rangle$$

For an arbitrary state $|x\rangle$.

Since U is supposed to be universal, it must also clone another arbitrary state $|y\rangle$:

$$U |y\rangle |0\rangle = |y\rangle |y\rangle$$

At this point, we have two equations describing the action of U on two different input states.

Now, the **only thing we know about unitary operators is that they preserve inner products** (page 40). So if we want to derive a **contradiction from unitarity**, we must compute an inner product. If the inner product before cloning is different from the inner product after cloning, then we have a contradiction, which means that U cannot be unitary, and therefore cannot be a valid quantum gate.

Let's compute the inner product before and after cloning.

- **Before cloning**, the inner product between the two input states is:

$$(U|x\rangle|0\rangle) \cdot (U|y\rangle|0\rangle)$$

But the inner product between two states is written as (page 75):

$$\begin{aligned} (U|x\rangle|0\rangle) \cdot (U|y\rangle|0\rangle) &= (U|x\rangle|0\rangle)^\dagger (U|y\rangle|0\rangle) \\ &= (\langle 0|\langle x|U^\dagger)(U|y\rangle|0\rangle) \\ &\downarrow \text{Remove parentheses} \\ &= \langle 0|\langle x|U^\dagger U|y\rangle|0\rangle \\ &\downarrow \text{Use unitarity: } U^\dagger U = I \\ &= \langle 0|\langle x|y\rangle|0\rangle \end{aligned}$$

- **After cloning**, the inner product between the two output states is:

$$(|x\rangle|x\rangle) \cdot (|y\rangle|y\rangle)$$

Using the tensor product property of inner products:

$$\begin{aligned} (|x\rangle|x\rangle) \cdot (|y\rangle|y\rangle) &= (|x\rangle|x\rangle)^\dagger (|y\rangle|y\rangle) \\ &= (\langle x|\langle x|)(|y\rangle|y\rangle) \\ &= \langle x|y\rangle \langle x|y\rangle \\ &= (\langle x|y\rangle)^2 \end{aligned}$$

Thus, **if cloning were possible**, we would have:

$$\langle 0|\langle x|y\rangle|0\rangle = (\langle x|y\rangle)^2$$

Using the tensor product property of inner products again:

$$\langle \psi|\langle \phi|\chi\rangle|\psi\rangle = \langle \phi|\chi\rangle \langle \psi|\psi\rangle$$

Remarkably, the inner product $\langle \phi|\chi\rangle$ is just a complex number (page 75), so we can pull it out of the inner product. So, we can simplify the left side:

$$\underbrace{\langle 0|\langle x|y\rangle|0\rangle}_{=\langle x|y\rangle \langle 0|0\rangle} = (\langle x|y\rangle)^2$$

Since $\langle 0|0\rangle = 1$, we have:

$$\langle x|y\rangle = (\langle x|y\rangle)^2 \iff \begin{cases} \langle x|y\rangle = 0 \\ \langle x|y\rangle = 1 \end{cases}$$

So unitarity forces the inner product to satisfy this equation. However, this equation is not true for all possible values of $\langle x|y\rangle$. It only holds if $\langle x|y\rangle = 0$ or $\langle x|y\rangle = 1$. But for arbitrary states, $\langle x|y\rangle$ can take any value between 0 and 1, so this is a contradiction. Therefore, a universal cloning gate U cannot exist. QED

❓ Why CNOT Does NOT Clone Qubits

A common confusion is that the CNOT gate seems to copy the control qubit into the target qubit (page 100). This is true only when the control qubit is a computational basis state, i.e. $|0\rangle$ or $|1\rangle$. It is not true for a generic superposition.

Classically, CNOT is often written as:

$$\text{CNOT}(v_A, v_B) = (v_A, v_A \oplus v_B)$$

Where $\oplus$ denotes XOR. ▲ So, if the second qubit is initialized to $|0\rangle$, one *may* be tempted to write:

$$\text{CNOT}(v_A, |0\rangle) = (v_A, v_A \oplus 0) = (v_A, v_A)$$

This is true because the XOR of any bit with 0 is the bit itself:

v_A	$ 0\rangle$	$v_A \oplus 0$
$ 0\rangle$	$ 0\rangle$	$ 0\rangle$
$ 1\rangle$	$ 0\rangle$	$ 1\rangle$

For example, if we start with $v_A = |1\rangle$ and the second qubit is $|0\rangle$, applying CNOT gives:

$$\text{CNOT}(|1\rangle, |0\rangle) = (|1\rangle, |1\rangle \oplus |0\rangle) = (|1\rangle, |1\rangle)$$

This looks **like cloning**: the first qubit v_A appears to be copied into the second qubit.

However, the mistake is that this notation is valid only for computational basis states $|0\rangle$ and $|1\rangle$, not for arbitrary superpositions. The informal notation:

$$\text{CNOT}(v_A, v_B) = (v_A, v_A \oplus v_B)$$

Can be used only when the inputs are basis states. For a generic superposition, we must use the full linear algebra representation of CNOT, which does not clone the state.

✔ **Correct Quantum Computation.** Let the control qubit be a generic superposition:

$$v_A = a_0 |0\rangle + a_1 |1\rangle$$

And let the target qubit be initialized to $|0\rangle$. The joint input state is:

$$v_A |0\rangle = (a_0 |0\rangle + a_1 |1\rangle) |0\rangle$$

Using the tensor product:

$$v_A |0\rangle = a_0 |00\rangle + a_1 |10\rangle$$

Now apply CNOT linearly. CNOT leaves $|00\rangle$ unchanged and maps $|10\rangle$ to $|11\rangle$:

$$\text{CNOT} (a_0 |00\rangle + a_1 |10\rangle) = a_0 |00\rangle + a_1 |11\rangle$$

So the real output is:

$$a_0 |00\rangle + a_1 |11\rangle$$

If CNOT had truly cloned the qubit, the output should instead be:

$$v_A v_A = (a_0 |0\rangle + a_1 |1\rangle)(a_0 |0\rangle + a_1 |1\rangle)$$

Expanding:

$$v_A v_A = a_0^2 |00\rangle + a_0 a_1 |01\rangle + a_1 a_0 |10\rangle + a_1^2 |11\rangle$$

However, these two states are not the same:

$$a_0 |00\rangle + a_1 |11\rangle \neq a_0^2 |00\rangle + a_0 a_1 |01\rangle + a_1 a_0 |10\rangle + a_1^2 |11\rangle$$

Therefore, CNOT does not clone a superposition. More fundamentally, a true universal cloning operation cannot exist because it would violate the unitary structure of quantum mechanics. Indeed, in general:

$$a_0 |00\rangle + a_1 |11\rangle$$

Cannot be factorized as a tensor product of two independent single-qubit states. The second state $v_A v_A$ would be two independent qubits in the same state. This is the key distinction: CNOT can create correlation or entanglement, but it does not create an independent copy of an unknown qubit. In summary, the CNOT output is entangled, while the true cloned state would be two qubits in the same state.

In other words, CNOT preserves quantum coherence between basis components, while true cloning would require duplicating the amplitudes themselves, which is forbidden by unitary quantum evolution.

5.2 No-Deleting Principle

The **No-Deleting Principle** is the dual counterpart of the no-cloning theorem. While the no-cloning theorem states that it is impossible to perfectly copy an arbitrary unknown quantum state, the no-deleting principle states that it is also **impossible to perfectly delete quantum information through a unitary operation**.

In other words:

- Quantum information **cannot** be freely **copied**;
- Quantum information **cannot** be freely **destroyed**.

Like the no-cloning theorem, the no-deleting principle is **deeply connected to the unitary structure** of quantum mechanics.

Definition 2: No-Deleting Principle

Suppose we want a quantum gate U that deletes a qubit state:

$$U |x\rangle = |0\rangle$$

For an arbitrary unknown state $|x\rangle$.

Equivalently, if we have two identical copies of a state, we may wish to delete one copy while leaving the other unchanged:

$$U |x\rangle |x\rangle = |x\rangle |0\rangle \quad (85)$$

The **No-Deleting Theorem** states that **no universal unitary gate** capable of performing this operation **can exist** for arbitrary unknown quantum states.

Intuitively, this means that **quantum information cannot simply disappear through reversible quantum evolution**.

As with the no-cloning theorem, we can prove the no-deleting principle by contradiction, using the fact that unitary operators preserve inner products (see page 177).

Proof. Suppose, by contradiction, that there exists a deleting gate U such that:

$$U |x\rangle = |0\rangle$$

For an arbitrary quantum state $|x\rangle$.

Since U is supposed to be universal, it must also delete another arbitrary state $|y\rangle$:

$$U |y\rangle = |0\rangle$$

At this point, we have two equations describing the action of U on two different input states.

Now, the only thing we know about unitary operators is that they **preserve inner products**. Therefore, if U is unitary, the inner product between the input states must be equal to the inner product between the corresponding output states.

- **Before deleting**, the inner product between the two input states is:

$$(U|x\rangle) \cdot (U|y\rangle)$$

Using the definition of inner product:

$$\begin{aligned} (U|x\rangle) \cdot (U|y\rangle) &= (U|x\rangle)^\dagger (U|y\rangle) \\ &= \langle x|U^\dagger U|y\rangle \\ &\downarrow \text{since } U \text{ is unitary, } U^\dagger U = I \\ &= \langle x|y\rangle \end{aligned}$$

- **After deleting**, both output states become $|0\rangle$, so the inner product is:

$$|0\rangle \cdot |0\rangle$$

Again:

$$\begin{aligned} |0\rangle \cdot |0\rangle &= |0\rangle^\dagger |0\rangle \\ &= \langle 0|0\rangle \\ &\downarrow \text{since } |0\rangle \text{ is a normalized state (page 16)} \\ &= 1 \end{aligned}$$

Since unitary operators preserve inner products:

$$\langle x|y\rangle = 1$$

But this is **not true in general for arbitrary quantum states**. It is true only if:

$$|x\rangle = |y\rangle$$

Therefore, such a **universal deleting gate U cannot exist.**

QED

5.3 No-Signaling Principle

Quantum entanglement creates strong correlations between distant particles (i.e., the measurement of one particle instantaneously affects the state of the other, regardless of the distance between them, page 121). However, these correlations cannot be used to transmit information instantaneously.

This idea is known as the **no-signaling principle**: although measurements on **entangled systems are correlated, they cannot be exploited for faster-than-light communication**.

⚠ The Communication Problem

Suppose Alice and Bob are at opposite ends of the universe. They each possess one qubit of the Bell pair (page 115):

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

The first qubit belongs to Alice, while the second belongs to Bob. More precisely, Alice owns the **first subsystem** of the bipartite state and Bob owns the **second subsystem**.

$$\begin{aligned} |00\rangle &= |0\rangle_A \otimes |0\rangle_B & |11\rangle &= |1\rangle_A \otimes |1\rangle_B \\ |\Phi^+\rangle &= \frac{|0\rangle_A \otimes |0\rangle_B + |1\rangle_A \otimes |1\rangle_B}{\sqrt{2}} \end{aligned}$$

Because the two qubits are **entangled**, measuring one qubit instantaneously determines the state of the other. For example:

- If Alice measures $|0\rangle$, Bob's qubit immediately collapses to $|0\rangle$;
- If Alice measures $|1\rangle$, Bob's qubit immediately collapses to $|1\rangle$.

⚠ At first sight, this **seems to violate relativity**. One may wonder:

“Can Bob detect instantly whether Alice has already measured her qubit?”

If the answer were yes, then **Alice could communicate information to Bob instantaneously**. For example, Alice could decide:

- If Alice measures her qubit $\Rightarrow$ transmit bit 1;
- If Alice does not measure her qubit $\Rightarrow$ transmit bit 0.

This **would allow faster-than-light communication**, contradicting special relativity.

The **no-signaling principle states that this is impossible**: although entanglement creates instantaneous correlations, **Bob cannot locally determine whether Alice has performed a measurement or not**.

Definition 3: No-Signaling Principle

The **No-Signaling Principle** states that quantum **entanglement cannot be used to transmit information instantaneously** between distant observers.

More precisely, local measurements performed on one subsystem of an entangled quantum system do not allow another observer to determine whether a measurement has already been performed on the distant subsystem.

Equivalently, although entangled measurements produce strong correlations, the local measurement statistics observed by one party are independent of the actions performed by the other party.

Proof. We want to understand whether Bob can locally detect whether Alice has already measured her qubit or not.

Alice and Bob share the Bell pair:

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{|0\rangle_A \otimes |0\rangle_B + |1\rangle_A \otimes |1\rangle_B}{\sqrt{2}}$$

The question is **whether Alice's measurement changes the probabilities observed by Bob**. If Bob's probabilities changed, then **Alice could use her measurement choice to send information instantaneously**.

Case 1: Alice does not measure.

If Alice does not measure her qubit, the global state remains:

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

If Bob measures his qubit in the computational basis, he obtains:

$$P_B(0) = \frac{1}{2}, \quad P_B(1) = \frac{1}{2}$$

So Bob sees a **completely random result**.

Case 2: Alice measures.

Now suppose Alice measures her qubit first in the computational basis. There are **two possible outcomes**:

- With probability $\frac{1}{2}$, Alice obtains $|0\rangle$, and the global state collapses to:

$$|00\rangle$$

In this case, Bob's qubit becomes $|0\rangle$.

- With probability $\frac{1}{2}$, Alice obtains $|1\rangle$, and the global state collapses to:

$$|11\rangle$$

In this case, Bob's qubit becomes $|1\rangle$.

Therefore, from Bob's point of view:

$$P_B(0) = \frac{1}{2}, \quad P_B(1) = \frac{1}{2}$$

So **Bob again sees a completely random result.**

Conclusion.

In both cases:

$$P_B(0) = P_B(1) = \frac{1}{2}$$

Therefore, **Bob's local measurement statistics are the same whether Alice has measured or not.** This means Bob cannot infer Alice's action by measuring only his own qubit. Thus, no usable information is transmitted instantaneously, and the no-signaling principle is preserved. QED

❓ What are we trying to prove? We start with the Bell state:

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{|0\rangle_A \otimes |0\rangle_B + |1\rangle_A \otimes |1\rangle_B}{\sqrt{2}}$$

Alice has the first qubit, while Bob has the second qubit. Now suppose Alice is billions of light-years away from Bob. The concern is that if Alice measures her qubit, the joint entangled state changes, and Bob's qubit can subsequently be described by a definite state correlated with Alice's outcome. Can Bob detect this change? If yes, then Alice could use her measurement choice to send information to Bob faster than light, for example:

- If Alice measures her qubit $\Rightarrow$ transmit bit 1;
- If Alice does not measure her qubit $\Rightarrow$ transmit bit 0.

This would violate the relativity principle.

❓ What is the relativity principle? It is not a topic of this course, but it is fundamental to understand the motivation behind the no-signaling principle. The **Einstein's Relativity Principle** core idea is: **no information, influence, or signal can travel faster than the speed of light.** This is not merely a technological limitation, but a fundamental law of spacetime.

Now, related to quantum mechanics, why is this principle relevant? Imagine Alice is on Earth and Bob is on Mars. Suppose Alice could send a message instantly. Then there would exist observers for whom: Bob receives the message before Alice sends it, which creates a paradox. In some reference frames, the **effect would occur before the cause**, but this contradicts our understanding of causality. Therefore, Einstein looked at this and thought⁹: “*wait, did*

⁹We will discuss the EPR paradox later, but it is interesting to note that Einstein historically disliked the idea that measuring one particle could apparently affect another distant particle instantaneously. He famously referred to this phenomenon as “spukhafte Fernwirkung” (spooky action at a distance), expressing his discomfort with the idea that quantum mechanics could allow for such non-local effects. As we will see later, Schrödinger coined the term “entanglement” to describe this phenomenon, and it has since become a fundamental aspect of quantum mechanics. Nevertheless, Einstein's concerns about entanglement's implications for locality and causality remain a topic of discussion and debate in the foundations of quantum mechanics.

something travel from Alice to Bob faster than light?”. This is the motivation behind the no-signaling principle: yes, the correlations are real, but Alice cannot choose the outcome she obtains, so she cannot force Bob’s qubit to collapse to a specific state. Therefore, Bob cannot detect whether Alice has measured her qubit or not, and no information is transmitted faster than light. In simple terms, **Entanglement creates correlations, but it does not allow for Communication**. Entanglement makes distant measurement outcomes correlated, but the no-signaling principle guarantees that these correlations cannot be controlled to transmit information.

5.4 EPR Paradox and Quantum Correlation

5.4.1 Definition of EPR Paradox

In 1935, Albert Einstein, Boris Podolsky, and Nathan Rosen published a paper entitled *Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?*. [2] Their goal was not to disprove quantum mechanics, but to **show** that the **theory appeared to be incomplete**.

The source of their concern was a peculiar phenomenon predicted by quantum mechanics: two particles could become so strongly correlated that measuring one particle would instantaneously determine the state of the other, regardless of the distance separating them (explained in the proof of the no-signaling principle, page 183).

Einstein found this deeply troubling. According to special relativity, **no information can travel faster than the speed of light**. Yet entanglement seemed to imply an instantaneous connection between distant particles, a phenomenon Einstein famously called “*spooky action at a distance*” (*spukhafte Fernwirkung*).

The EPR paper sparked one of the most important debates in the history of physics. Shortly after its publication, Erwin Schrödinger analyzed the EPR argument and realized that it revealed what he considered the most distinctive feature of quantum mechanics. In a series of papers published the same year, he introduced the term *Verschränkung* (“entanglement”), which is now known as quantum entanglement (page 114).

Interestingly, Schrödinger largely shared Einstein’s discomfort with the standard interpretation of quantum mechanics. However, while Einstein viewed the EPR argument as evidence that the theory was incomplete, **Schrödinger recognized entanglement as a fundamental property of quantum systems**.

The central **question raised by the EPR paradox** is therefore:

*If measuring Alice’s qubit immediately determines the state of Bob’s qubit, **does quantum mechanics allow faster-than-light communication?***

As we have already seen through the no-signaling principle (page 183), the **answer is no**. Entanglement creates correlations between distant measurements, but these correlations cannot be used to transmit information.

Precisely, Einstein’s concern was fundamentally:

*If measuring Alice’s qubit instantaneously determines the state of Bob’s qubit, **what is actually happening?***

In other words, **how can nature produce these instantaneous correlations?** Einstein was not primarily asking about the possibility of faster-than-light communication, but rather about the underlying mechanism that could explain the observed correlations without violating relativity. He suspected

that Bob's particle already possessed some hidden information before the measurement, which would determine the outcome of Alice's measurement without any need for instantaneous influence.

Modern Perception of the EPR Paradox

To understand the paradox in a modern quantum-computing setting, consider the Bell state:

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Suppose Alice and Bob each possess one qubit of this entangled pair and are separated by a very large distance.

If Alice measures her qubit in the computational basis and obtains:

$$|0\rangle$$

Then Bob's qubit immediately collapses to:

$$|0\rangle$$

Similarly, if Alice measures:

$$|1\rangle$$

Then Bob's qubit immediately collapses to:

$$|1\rangle$$

At first sight, this appears to imply an instantaneous influence between distant particles. How can Bob's qubit "know" which outcome Alice obtained if they may be separated by billions of light-years? This is the modern formulation of the EPR paradox.

Definition 4: Einstein-Podolsky-Rosen (EPR) Paradox

The **Einstein-Podolsky-Rosen (EPR) Paradox** is the **apparent contradiction** between quantum **entanglement** and the **classical principle of locality**.

More precisely, if two particles are prepared in an entangled state, a measurement performed on one particle appears to instantaneously determine the state of the other, regardless of the distance separating them.

This raises the question of whether **quantum mechanics is a complete description of physical reality**, or whether **additional hidden information is required to explain these correlations without instantaneous influence**.

5.4.2 Entanglement and Quantum Correlation

The EPR paradox highlights a fundamental feature of entangled systems: measurement outcomes may appear completely random to individual observers, yet exhibit strong correlations when compared afterward.

In simple terms, Alice and Bob each observe **random outcomes** when measuring their respective qubits. However, when they later compare their results, they discover that these outcomes are **correlated in a way that appears difficult to reconcile with classical intuition**.

❓ Alice and Bob obtain correlated results. Why is that surprising?

At first glance, it might seem that correlated results are not surprising at all. After all, **classical systems can also exhibit correlations**.

🎲 **Classical example.** Suppose two coins are prepared in such a way that they always show the same face. If one observer sees Heads, they immediately know that the other coin must also be Heads. Likewise, if one observer sees Tails, the other must see Tails. This is an example of a **classical correlation**.

In **classical physics**, such **correlations** are explained by assuming that the **system possessed predetermined properties** before any observation was made. In our example, the two coins were already both Heads or both Tails before anyone looked at them.

💡 **Einstein's intuition.** Einstein believed that the **correlations observed in entangled particles might be explained in exactly the same way**. According to this view, the particles could carry some form of **hidden information that predetermines the outcome of future measurements**. If such hidden information existed, the apparent mystery of entanglement would disappear: measuring one particle would simply reveal information that was already present from the beginning.

Quantum entanglement exhibits a superficially similar phenomenon. Individual measurement outcomes appear random, yet they become strongly correlated when compared afterward.

⚠️ **The difference.** Unlike classical correlations, which are completely determined by pre-existing properties of the system, **quantum correlations depend on the measurement basis chosen by the observers**. This means that **changing how Alice and Bob perform their measurements can change the observed correlations in a way that has no classical analogue**.

Therefore, the existence of correlations is not itself surprising. The **real challenge is determining whether quantum correlations can be explained using the same classical intuition** that explains correlated coins, cards, or dice, or whether they **represent an entirely new physical phenomenon**.

This question leads directly to the distinction between **classical correlation** and **quantum correlation**.

5.4.3 Classical vs. Quantum Correlation

🎲 Classical Correlation

A **Classical Correlation** arises from **information** that **already exists before any measurement** is performed.

In a classical system, correlated outcomes can be explained by assuming that the **system possesses predetermined properties**. These properties are often called **Hidden Variables**, because they **determine the outcome of future observations** even if the observers do not know their values.

Example 2: Classical Correlation

For example, consider again two coins prepared so that they always show the same face. Before any observation takes place, the system is already in one of two possible configurations:

- Heads - Heads;
- Tails - Tails.

The measurement does not create the correlation. It merely reveals information that was already present.

Therefore, knowing the outcome observed by one observer immediately allows the other outcome to be inferred. No instantaneous influence is required, because the **correlation existed before the measurement**.

⚙️ Quantum Correlation

A **Quantum Correlation** arises from **measurements performed on an entangled quantum system**.

As in the classical case, each observer sees random outcomes individually. However, the observed **correlations** possess a remarkable property: they **depend on the measurement basis chosen by the observers**.

Example 3: Quantum Correlation

For example, if Alice and Bob share the Bell state

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

And both measure in the same basis, their outcomes are perfectly correlated. However, if they choose different measurement bases, the observed correlations change.

The Main Difference

In a **classical system**, changing how an observer looks at the system does not alter the underlying correlation. The **correlation is determined entirely by the hidden variables** that existed **before the measurement**.

In contrast, **quantum correlations** depend on the **measurements that are actually performed**. The observed correlations cannot always be explained by assuming that the particles carried predetermined answers before the measurement.

This distinction is what makes quantum correlation fundamentally different from classical correlation and lies at the heart of the EPR debate.

Bell Pair Example

The distinction between classical and quantum correlation becomes clearer when we analyze an entangled Bell pair. Consider the Bell state:

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

This state can also be expressed in the Hadamard basis as (see the full derivation in section 4.7):

$$|\Phi^+\rangle = \frac{|++\rangle + |--\rangle}{\sqrt{2}}$$

Suppose Alice and Bob each possess one qubit of this entangled pair:

$$|\Phi^+\rangle_{AB} = \frac{|00\rangle_{AB} + |11\rangle_{AB}}{\sqrt{2}} = \frac{|+\rangle_A \otimes |+\rangle_B + |-\rangle_A \otimes |-\rangle_B}{\sqrt{2}} = \underbrace{\frac{|0\rangle_A \otimes |0\rangle_B + |1\rangle_A \otimes |1\rangle_B}{\sqrt{2}}}_{\text{classical correlation}}$$

Example 4: Same Basis Measurements

Suppose Alice and Bob both measure in the computational basis $\{|0\rangle, |1\rangle\}$.

Alice measures first and obtains:

$$|0\rangle \quad \text{or} \quad |1\rangle$$

With equal probability:

$$P(0) = P(1) = \frac{1}{2}$$

If Alice obtains:

$$|0\rangle$$

The Bell state collapses to:

$$|00\rangle$$

Therefore Bob's qubit becomes:

$$|0\rangle$$

Similarly, if Alice obtains:

$$|1\rangle$$

The Bell state collapses to:

$$|11\rangle$$

And Bob's qubit becomes:

$$|1\rangle$$

Consequently, **Bob always obtains the same outcome as Alice.**

Although each observer sees **random outcomes individually**, they discover **perfect correlation when they later compare their measurement results**.

Example 5: Different Basis Measurements

Now suppose Alice measures in the Hadamard basis:

$$\{|+\rangle, |-\rangle\}$$

Where

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Bob, however, continues measuring in the computational basis.

Alice obtains either:

$$|+\rangle \quad \text{or} \quad |-\rangle$$

With probability:

$$\frac{1}{2}$$

If Alice measures $|+\rangle$, Bob's qubit collapses to:

$$|+\rangle$$

If Alice measures $|-\rangle$, Bob's qubit collapses to:

$$|-\rangle$$

However, **Bob is measuring in the computational basis.** For both $|+\rangle$ and $|-\rangle$:

$$P(0) = P(1) = \frac{1}{2}$$

Therefore Bob obtains:

$$0 \quad \text{or} \quad 1$$

With equal probability, regardless of Alice's outcome.

When Alice and Bob later compare their results, they find that the **perfect correlation** observed in the previous experiment has **disappeared**.

💡 In a classical hidden-variable model, the correlation would be determined entirely by pre-existing information carried by the particles. In contrast, the Bell-pair example shows that the **observed quantum correlation depends on the measurement basis chosen by the observers** (same basis, perfect correlation; different basis, no correlation). This basis dependence is one of the fundamental signatures of quantum correlation.

5.4.4 Entanglement vs. Quantum Correlation

Throughout the discussion of the EPR paradox, we have repeatedly encountered two closely related concepts: **entanglement** and **quantum correlation**. Although they are strongly connected, they describe different aspects of a quantum system.

 **Entanglement.** We have already defined **entanglement** (page 114) as a property of the quantum state itself. A system is entangled when its state cannot be written as a tensor product of the states of its individual subsystems.

For example, the Bell state:

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Is entangled because it cannot be expressed as:

$$|\psi_A\rangle \otimes |\psi_B\rangle$$

The important point is that entanglement is an intrinsic property of the system. It exists independently of whether anybody performs a measurement (see the discussion on page 116).

 **Quantum Correlation.** **Quantum Correlation** refers to the **statistical relationship observed between measurement outcomes**. Unlike entanglement, quantum correlation is **not a property of the state** alone. It **depends on how the system is measured**. For instance, Alice and Bob may share the same entangled Bell pair, yet the observed correlations change depending on the measurement bases they choose.

Entanglement vs. Quantum Correlation

Entanglement does not depend on the measurement basis:

- Entanglement is a **property of the quantum system**.
- It exists before any measurement is performed.

Quantum correlation, on the other hand, depends on the measurement basis:

- Quantum correlation is a **property of the measurement outcomes**.
- Different measurement bases may produce different observed correlations.

In summary:

- **Entanglement** is a property of the quantum state and does not depend on how the system is measured.
- **Quantum correlation** is a property of the measurement outcomes and depends on the measurement basis chosen by the observers.

6 Transpiling

6.1 Quantum Fidelity

6.1.1 Why Quantum Computers Are Noisy

In all the previous chapters, we implicitly assumed that quantum gates exactly as their mathematical definitions. For example:

- Hadamard gate:

$$H|0\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Behaves as a perfect beam splitter, splitting the input state into an equal superposition of $|0\rangle$ and $|1\rangle$.

- Pauli-X gate:

$$X|0\rangle = |1\rangle$$

Behaves as a perfect NOT gate, flipping the state of the qubit from $|0\rangle$ to $|1\rangle$.

- CNOT gate:

$$\begin{aligned} \text{CNOT}|00\rangle &= |00\rangle, & \text{CNOT}|01\rangle &= |01\rangle \\ \text{CNOT}|10\rangle &= |11\rangle, & \text{CNOT}|11\rangle &= |10\rangle \end{aligned}$$

Behaves as a perfect Controlled-NOT gate, flipping the target qubit if the control qubit is in the state $|1\rangle$ and leaving it unchanged if the control qubit is in the state $|0\rangle$.

Mathematically these transformations **are exact**. However, in a real quantum computer, gates are implemented by physical hardware:

- Microwave pulses (superconducting qubits);
- Laser pulses (trapped ions);
- Electromagnetic fields;
- Control electronics;

Because the **hardware is imperfect** (e.g., due to manufacturing defects, environmental noise, or limitations in control precision), the **resulting operation is only an approximation of the ideal gate**. As a consequence:

$$U_{\text{physical}} \neq U_{\text{ideal}} \xrightarrow{\text{better}} U_{\text{physical}} \approx U_{\text{ideal}}$$

This discrepancy leads to **errors** in the quantum computation, which can accumulate over multiple gates and degrade the performance of quantum algorithms.

▲ The Two Main Sources of Noise

- **Gate Infidelity.** Gate infidelity means that the **gate physically applied by the hardware is not exactly the gate we intended to execute.**

Suppose we want to apply a Hadamard gate:

$$U_{\text{ideal}} = H$$

Ideally, we would like the physical implementation to match the ideal gate:

$$H|0\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

However, due to **imperfections** like inaccurate control pulses, calibration errors, electronic noise, manufacturing imperfections, the hardware may actually apply:

$$U_{\text{physical}} = H + \varepsilon$$

Where ε is a **small error**. The resulting state is therefore slightly different from the desired one. After thousands of gates, these small errors can **accumulate** and significantly affect the final result.

❓ **Why Multi-Qubit Gates Are Worse?** Intuitively, multi-qubit gates (like CNOT) are **more complex to implement** than single-qubit gates (like Hadamard) because they require precise control over interactions between multiple qubits. For this reason, the **fidelity of multi-qubit gates is typically lower** than that of single-qubit gates:

$$\text{Fidelity}_{\text{CNOT}} < \text{Fidelity}_{\text{Hadamard}}$$

This is one reason why transpilers try to minimize the number of CNOTs and other multi-qubit operations. However, we will understand this in more detail in the next chapters.

- **Decoherence.** A **qubit is never perfectly isolated** from the surrounding environment. Even if we do not intentionally measure it, the environment continuously interacts with it: electromagnetic radiation, thermal fluctuations, vibrations, surrounding particles, etc. These **interactions gradually destroy the quantum properties of the qubit**. In summary, decoherence is the process by which a quantum system progressively loses its “*quantumness*” and starts behaving more like a classical object (e.g., a bit instead of a qubit).

For example, imagine preparing a qubit in the state $|+\rangle$:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Initially the qubit exhibits quantum superposition, interference, and phase relationships. However, as time passes, interactions with the environment disturb the phase information. Eventually, the qubit behaves more like a classical probabilistic bit than a true quantum state. The information

is not necessarily destroyed instantly, but the quantum coherence gradually leaks into the environment, making it harder to maintain the desired quantum state over time.

⚠ Why is Decoherence a serious problem? A quantum algorithm often requires many sequential operations:

$$U_n U_{n-1} \cdots U_2 U_1$$

Where each operation takes a certain amount of time to execute. If the algorithm runs for too long, decoherence may occur before the computation finishes. **When decoherence becomes dominant**, the superposition is lost, the interference is lost, the entanglement is lost, and the **quantum advantage disappears**.

🔧 Hardware Limitations

Both gate infidelity and decoherence originate from the same fundamental fact: **quantum computers are physical devices**. Unlike an ideal mathematical model, qubits are not perfectly isolated, gates are not perfectly accurate, and measurements are not perfectly reliable. Therefore, every real quantum computer has a finite error rate, coherence time, and computational depth. These limitations ultimately restrict how deep a quantum circuit can be before errors become overwhelming.

In summary, quantum computers are noisy because the **gates are imperfectly implemented** and the **qubits decohere over time**.

6.1.2 Gate Infidelity

In the mathematical model of quantum computing, quantum gates are represented by ideal unitary matrices that transform quantum states exactly as predicted by theory.

For example, the Hadamard gate is defined as:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

And when applied to the $|0\rangle$ state, it always produces the $|+\rangle$ state:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$$

In theory, this transformation is **exact**.

Real quantum computers, however, do not manipulate abstract vectors and matrices. They manipulate physical systems through microwave pulses, laser pulses, electromagnetic fields, and control electronics. As a consequence, the operation implemented by the hardware is never perfectly identical to the ideal gate (as we have seen in the previous section).

Ideal Gate vs Physical Gate

It is useful to distinguish between:

- **Ideal gate:** the mathematical operation specified by the algorithm;
- **Physical gate:** the operation actually implemented by the hardware.

Ideally we would like:

$$U_{\text{physical}} = U_{\text{ideal}}$$

But in practice we have:

$$U_{\text{physical}} \approx U_{\text{ideal}}$$

But there is always a small discrepancy due to imperfections in the hardware. This **difference between the ideal and physical implementation** is called **Gate Infidelity**.

References

- [1] Bryan Eastin and Emanuel Knill. Restrictions on transversal encoded quantum gate sets. *Physical review letters*, 102(11):110502, 2009.
- [2] Albert Einstein, Boris Podolsky, and Nathan Rosen. Can quantum-mechanical description of physical reality be considered complete? *Physical review*, 47(10):777, 1935.
- [3] Cremonesi Paolo. Quantum computing. Slides from the HPC-E master's degree course on Politecnico di Milano, 2024.

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